

Selected Works on Conditionals

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Preface

This document was produced as part of the project “Research on conditional and modal language”, funded by the National Science Foundation (Grant #2116972; M. Kaufmann, PI; S. Kaufmann, Co-PI), 2021-23. The project included a comparative analysis of a variety of modal and conditional constructions in English in Japanese. This part included a survey of the Japanese linguistic literature on the topic, which produced a collection of English-language summaries. Teruyuki Mizuno and Muyi Yang deserve most of the credit for this work. The summaries have subsequently proven useful to researchers outside of our project, so we decided to make the publicly available.

These summaries are not authorized by the authors or publishers of the original works. We do not guarantee their accuracy or completeness. Their intended use is as a study guide, not as a substitute for the original works. Feedback welcome.

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Japanese Conditionals and Tensedness

Setsuko Arita

Chapter 1 Introduction

September 9, 2021

Everyday inference based on uncertain knowledge:

- E.g., On the day before the sports day, you say to yourself ‘I’m going to get the first place!’. But this requires the prerequisite that the sports day takes place as scheduled, which you do not know yet.
- E.g., You’re late for a date. While rushing to the place that you’re supposed to meet up, you think ‘I can finally meet him in just a second!’. But this requires the prerequisite that your date has already arrived on time, which you do not know yet.

Two types of uncertainty

1. Uncertainty of unsettled propositions: the states of affairs described by the propositions have not obtained yet (e.g., the sports day example)
2. Uncertainty of settled propositions: the states of affairs described by the propositions have obtained, and are unknown to the speaker (e.g., the date example)

This study assumes that conditionals are a construction that expresses inferences based on uncertain knowledge. So, based on the two types of uncertainty, we can classify conditionals into three types

1. Predictive conditionals: Antecedent propositions are unsettled
2. Epistemic conditionals: Antecedent propositions are settled but unknown to the speaker
3. Counterfactual conditionals: Antecedent propositions are settled, and the speaker knows they are false

(Un)settledness in Japanese is marked by the (in)complete tensedness (complete tensedness for settledness). This is manifest in conditionals: some conditional markers take complete tensed forms, and some take incomplete tensed forms. The goal of this study is to explore the hypothesis that there is a correspondence between the three types of conditionals classified above and conditional markers that mark (in)complete tensedness.

Chapter 2 Modality and tensedness in Japanese

September 22, 2021

Main point of this chapter: there is a correspondence between modality and tense on the predicate preceding the modal.

Japanese modality: the classification follows Palmer (2001):

- (1) a. Propositional modality: About the speaker's judgment regarding a proposition
 - (i) Epistemic modality
In Jp: *darou hazu da, ni chigainai*, etc.
 - (ii) Evidential modality
In Jp: *you da, rashii, sou da*, etc.
- b. Event modality: About the speaker's attitude regarding the realization of a state of affairs
 - (i) Deontic modality
In Jp: *beki da, nakereba naranai, temo ii*, etc.
 - (ii) Dynamic modality
In Jp: *u, you, koto ga dekiru*, etc.

Basic tense morphology 1: *Ta* form

- In main clauses, *ta* expresses that a relevant event occurred prior to the speech time. Apart from the basic use to express past tense, *ta* can also express perfect aspect (2) and mood (3).

- (2) *mou hanako-ga ki-ta.*
already Hanako-NOM come-PST
'Hanako has already come.' ((28), p. 17)

- (3) *nanda, konna tokoro-ni at-ta.*
oh such place-DAT be-PST
'Oh, it is here (I had been looking for it).' ((29), p. 17)

- Complexity in subordinate clauses: *ta* can mark an event whose event time follows the speech time.

- (4) a. Subordinate-clause event to be verified before the main-clause event (but after the speech time):

- b. [*hanako-ga kaet-ta baai*], *hoka-no hito-ga kawari-o*
 Hanako-NOM return-PST case other-GEN person-NOM replacement-ACC
tsutomeru koto-ni nat-tei-ru.
 work thing-DAT become-ASP-NPST
 ‘In case Hanako goes back, other people are supposed to perform her duty instead.’ ((31), p. 17)

It might be tempting to assume that *ta* introduces temporal interpretation relative to the time of the main-clause event, but this would not be able to explain the following:

- (5) a. Subordinate-clause event is verified later than the main-clause event:
 b. [*kesa kotoba-o kawashi-ta josei*]-o *kinou eigakan-de*
 this.morning word-ACC exchange-PST woman-ACC yesterday cinema-LOC
mikake-ta.
 see-PST
 ‘Yesterday at the cinema, I saw the woman who I spoke to this morning.’ ((32), p. 28)
- (6) a. Subordinate-clause event occurs simultaneously with the main-clause event:
 b. *tarou-wa [toukyou-ni i-ta toki], apaato-ni sun-dei-ta.*
 Taro-TOP Tokyo-DAT be-PST when apartment-DAT live-ASP-PST
 ‘When Taro was in Tokyo, he lived in an apartment.’ ((34), p. 28)

- Generalization: Japanese *-ta* introduces (i) an event time *E* and (ii) **a reference time *R* that is independent of the speech time *S*.** (X, p. 19)

Basic tense morphology 2: *Ru* form

- The basic interpretation of *ru* is dependent on whether the predicate is eventive or stative:
 - Stative predicates: the speech time falls within the interval of the event, i.e., *ru* expresses present tense.

- (7) *tanaka-ga heya-ni i-ru.*
 Tanaka-NOM room-DAT be-NPST
 ‘Tanaka is in the room.’ ((35), p. 29)
 (Fn. 11: The temporal interpretation can be relativized to a time that is different from the speech time. E.g., with the adverb *goji koro* ‘at about 5’, (7) will be interpreted in the future tense.)

- Eventive predicates:

- * Speech time precedes event time, i.e., *ru* expresses future tense

- (8) *tanaka-ga heya-o de-ru.*
 Tanaka-NOM room-ACC go.out-NPST
 ‘Tanaka will get out of the room.’ ((36), p. 19)

- * With eventive predicates expressing feeling or thinking, *ru* expresses present tense.

- (9) *doumo mi-gi-kata-ga itam-u.*
 somehow right-shoulder-NOM hurt-NPST
 ‘My right shoulder hurts (and I can’t help it).’ ((38), p. 20)

Side-note (Fn. 12): Performative verbs marked with *ru* also express present tense:

- nidoto uso-o tsuk-anai to yakusoku su-ru yo.*
 second.time lie-ACC say-NEG COMP promise do-NPST SFP
 ‘I promise not to lie again.’ ((f), p. 20)

- Generalization: *Ru* signals that the relevant event occurs within a time interval that includes the speech time.
- In (8), *ru* is interpreted as future tense because the truth-values of events like *go out* need to be determined based on at least two ‘scenes’ (cf. Takubo 1993): one where Tanaka is in the room, and one where Tanaka is not. When the speech time is included in the interval of the former scene, the change of state (going out) will have to be realized in the future.
- In (9), *ru* is interpreted as present tense because the thinking/feeling concerns a scene whose interval includes the speech time.

Complete and incomplete tensed clauses in Japanese

- Complete tensed clauses:
 - allow both *ru* form and *ta* form; and
 - *ru* form expresses the temporal relation s.t. speech time < event time, and
 - *ta* form expresses the temporal relation s.t. event time < speech time
- Incomplete tensed clauses: all the other clauses, namely the clauses that
 - allow only *ru* form or *ta* form; or
 - allow neither form; or
 - allow both forms but do not reflect different temporal relations between the event time and the speech time.

Correspondence between tense and modality (cf. the classification in (1))

- **Propositional modality selects complete tensed clauses.** (XI, p. 23)

- (10) *tarou-ga {a. ku-ru / b. ki-ta} {darou / hazu da / ni chigainai}.*
 Taro-NOM come-NPST come-PST be.probable should no.doubt
 a. ‘{It’s probable / It should be the case / There is no doubt} that Taro will come.’
 b. ‘{It’s probable / It should be the case / There is no doubt} that Taro came.’ ((43), p. 22)

- (11) *tarou-ga* {*a. ku-ru* / *b. ki-ta*} {*youda* / *rashii* / *souda*}.
 Taro-NOM come-NPST come-PST INFER REPORT REPORT
 a. ‘{It seems / I heard} that Taro will come.’
 b. ‘{It seems / I heard} that Taro came.’ ((44), p. 22)

Temporal relation expressed in (10a) and (11a): speech time < event time;

Temporal relation expressed in (10b) and (11b): event time < speech time

• **Event modality selects incomplete tensed clauses.** (XII, p. 23)

Temporal relation expressed (even with *ta* form in (14)): speech time < event time

- (12) *ashita eiga-ni* {*a. ik-ou* / *b. ik-itai*}.
 tomorrow movie-DAT go-VOL go-want
 a. ‘Let’s go to see a movie tomorrow.’
 b. ‘I want to go to see a movie tomorrow.’ ((45), (46), p. 22)
- (13) *shinsenna yasai-o* {*tabe-ru* / **tabe-ta*} {*a. beki da* / *b. koto-ga dekiru*}.
 fresh vegetable-ACC eat-NPST eat-PST should thing-NOM can
 a. ‘One/You/I/etc. should eat fresh vegetables.’
 b. ‘One is/You are/I am/etc. able to eat fresh vegetables.’ ((47), (48), p. 22)
- (14) *tarou-ga* {*ku-ru* / *ki-ta*} *hou-ga ii*.
 Taro-NOM come-NPST come-PST direction-NOM good
 ‘Taro should come.’ ((49), p. 23)

With stative predicates – Temporal relation expressed: speech time $\subseteq$ event interval

- (15) *kyou-wa hima da kara boku-ga kimi-no kawari-ni koko-ni iru*
 today-TOP free COP since I-NOM you-GEN replace-DAT here-DAT be
koto-ga dekiru yo.
 thing-NOM can SFP
 ‘Since I’m not busy today, I can be here in place of you.’ ((62), p. 25)
- (16) *reji-ni tenchou-ga i-nake-reba nar-anai*.
 cashier-DAT shop.manager-NOM be-NEG-COND become-NEG
 ‘The shop manager must be at the cashier.’ ((64), p. 26)

Interaction between stativity and modality (cf. Table 1)

• *aru* ‘be (inanimate)’

With deontic modality:

- (17) *konnani tenki-ga ii n da kara, hanabitaikai-wa aru beki da*.
 such weather-NOM good FIN COP since fireworks-TOP be should
 ‘Since the weather is so good, there should be fireworks.’ ((69), p. 26)

	Eventive predicate	Stative predicates		
		<i>iru</i> ‘be (animate)’	<i>aru</i> ‘be (inanimate)’	adjectives
Propositional modality	✓	✓	✓	✓
Eventive modality (deontic)	✓	✓	✓	w. restriction
Eventive modality (dynamic)	✓	✓	*	*

Table 1: Interaction between eventivity/stativity and modality (p. 29)

With dynamic modality:

- (18) *ashita-wa hanabitaikai-ga ar-ou.*
tomorrow-TOP fireworks-NOM be-VOL
OK as epistemic modality: ‘There might be fireworks tomorrow’ (sounds archaic)
Cannot be dynamic modality: ‘Let there be fireworks tomorrow.’ ((67), p. 26)

- Adjectives

With deontic modality: Some modals can directly take adjectives (e.g., *te(mo) ii* ‘can’), some cannot (e.g., *koto-ga dekiru* ‘can’).

- (19) *machiawase-jikan-wa motto {hayaku-te ii / *hayai koto-ga dekiru} yo.*
meeting-time-TOP more early-CONT good early thing-NOM can SFP
‘We can meet earlier.’ ((78), p. 28)

- The reason that *iru* ‘be (animate)’ is fully compatible with eventive modality has to do with its property regarding settledness, which will be discussed later.

Chapter 3 Adverbial clauses and incompletely-tensed clauses in Japanese

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This chapter discusses two diagnoses for the (un)tensedness of adverbial clauses—realization of agentive subjects and temporal order between subordinate and matrix clauses. Within tensed adverbial clauses, the tests offer evidence for a distinction between completely-tensed clauses and incompletely-tensed clauses. The findings are summarized in Table 1:

	Untensed	Tensed (incomplete)	Tensed (complete)
Realization of agentive subjects	*	✓	✓
One-directional temporal order	✓	✓	✓
Two-directional temporal order	*	w. restriction	✓

Table 1: (p. 40)

(Un)tensedness and realization of agentive subjects: Tensed and untensed adverbial clauses can be distinguished based on whether an agentive subject (a transitive subject or an unergative subject) independent of the matrix subject can be realized. (Unaccusative subjects are non-agentive subjects.)

- Untensed clauses can't contain agentive subjects. Ex. *youni*-clauses, which express purposes

- (1) with unaccusative subjects

[*otoshiyori-ga korob-ana-i youni*] *senaka-kara sasae-ta.*

elderly-NOM fall-NEG-NPST YOUNI back-from support-PST

'In order for the elderly not to fall, I held him at his back.'

((13), p. 34)

- (2) with transitive subjects

*[*haha-ga sandoicchi-o tsukur-u youni*] *watashi-wa shokupan-o kai-ni*

mother-NOM sandwich-ACC make-NPST YOUNI I-TOP

bread-ACC buy-DAT

it-ta.

go-PST

Intended: 'In order for my mother to make sandwiches, I went to buy bread.' ((14), p. 34)

- (3) with unergative subjects

*[*musuko-ga daigaku-ni shingaku su-ru youni*] *shigoto-o fuyashi-ta.*
 son-NOM university-DAT enter do-NPST YOUNI job-ACC increase-PST
 Intended: ‘In order for my son to enter college, I increased my amount of work.’
 ((12), p. 34)

Note: Another type of untensed adverbial clauses that express purpose—*tameni*-clauses—differ from *youni*-clauses in terms of the distribution of different types of subjects.

- Tensed clauses can contain agentive subjects. Ex. *node*-clauses, which express reasons.

(4) with unaccusative subjects
 [*okane-ga amat-ta node*] *toushishintaku-ni te-o dash-ita.*
 money-NOM be.left.over-PST because investment.trust-DAT hand-ACC send-PST
 ‘Because I had money left over, I tried investment trust.’ ((17), p. 35)

(5) with transitive subjects
 [*haha-ga sandoicchi-o tsukut-ta node*] *chuushoku-dai-ga setsuyaku deki-ta.*
 mother-NOM sandwich-ACC make-PST because lunch-money-NOM save
 can-PST
 ‘Because my mother made sandwiches, I could save the money for lunch.’ ((19), p. 35)

(6) with unergative subjects
 [*musuko-ga daigaku-ni shingaku shi-ta node*] *seikatsu-ga kurushiku nat-ta.*
 son-NOM university-DAT enter do-PST because life-NOM tough
 become-PST
 ‘Because my son entered college, our life became harder (financially).’ ((18), p. 35)

(Un)tensedness and temporal order: Unlike tensed clauses, untensed clauses are restricted in terms of the temporal order between the subordinate event and the matrix event.

- Untensed clauses: in (7), the subordinate event must follow the matrix event; the reverse order is impossible.

(7) [*daigaku-ni shingaku su-ru youni*] *juku-ni kayot-ta.*
 university-DAT enter do-NPST YOUNI cram.school go-PST
 ‘In order to get in college, I went to a cram school.’ ((20), p. 35)

- Tensed clauses: the subordinate event can precede (e.g. (8)) or follow (e.g. (9)) the matrix event.

(8) [*kinou yasun-da kara*], *kyou-wa zangyou shi-you.*
 yesterday rest-PST because today-TOP work.overtime do-VOL
 ‘Because I took a day off yesterday, I should work overtime today.’ ((25), p. 36)

- (9) *[ashita yasum-u kara], kyou-wa zangyou shi-you.*
 tomorrow rest-NPST because today-TOP work.overtime do-VOL
 ‘Because I will take a day off tomorrow, I should work overtime today.’ ((27), p. 36)

Evidence for incompleteness of tensed clauses: *Ba*-clauses, *tara*-clauses and *temo*-clauses are morphologically similar to untensed clauses in that they cannot take the past or the non-past suffixes. However, in terms of the properties examined above, they behave in part similarly to untensed clauses (such as *youni*-clauses) and in part similarly to tensed clauses (such as *kara*-clauses). This suggests that it is necessary to recognize a category that sits in between untensed and (what have been traditionally called) tensed clauses. We will identify this category as incompletely-tensed clauses; what have been traditionally called tensed clauses will be identified as completely-tensed clauses.

- *Ba/tara/temo*-clauses allow agentive subjects:

- (10) with unaccusative subjects
nedan-ga sagar-eba, kat-temo ii kana.
 price-NOM drop-BA buy-TEMO good Q
 ‘If the price drops, maybe I can buy it.’ ((32), p. 37)

- (11) with unergative subjects
tarou-ga shusseki su-reba, minna-ga yorokob-u darou.
 Taro-NOM attend do-BA everyone-NOM be.glad-NPST be.probable
 ‘If Taro comes, people will probably be happy.’ ((29), p. 37)

- The event described by a *ba/tara/temo*-clause normally precedes the matrix event.

- (12) *kono kusuri-o {nom-eba / non-dara} kibun-ga yoku nar-u*
 this medicine-ACC drink-BA drink-TARA feeling-NOM good become-NPST
darou.
 be.probable
 ‘If you take this medicine, you will probably feel better.’ ((35)/(36), p. 37)

- (13) **ashita {yasum-eba / yasun-dara} kyou-wa zangyou shi-nake-reba*
 tomorrow rest-BA rest-TARA today-TOP work.overtime do-NEG-BA
nar-ana-i.
 become-NEG-NPST
 Intended: ‘If I take a day off tomorrow, I will have to work overtime today.’ ((38)/(39), p. 38)

- Restriction regarding temporal order: The subordinate event can temporally follow the matrix event when the consequent contains modals that allow backward inferences (from effects to causes), such as *no da* and *koto-ni naru*.

- (14) *asatte tarou-ga {ki-tara / ku-reba}, ashita hanako-ga*
 day.after.tomorrow Taro-NOM come-TARA come-BA tomorrow Hanako-NOM

yoppodo settoku shi-ta {no da /no darou /koto-ni nar-u /
 very.much persuade do-PST FIN COP FIN be.probable thing-DAT become-NPST
 *darou / *hazu da}.
 be.probable should COP
 ‘If Taro comes the day after tomorrow, it will have to be the case that he was deeply
 persuaded by Hanako tomorrow.’ ((43), p. 38)

(Note: the modal *darou* is a conjugated form of the copula *da* or *dearu*.)

Evidence that *no da* and *koto-ni naru* allows backward inferences:

- (15) *douro-ga nure-tei-ru. (to iu koto-wa) ame-ga fut-ta {no da /no*
 street-NOM be.wet-ASP-NPST C say thing-TOP rain-NOM fall-PST FIN COP FIN
*darou /koto-ni nar-u / *darou}*.
 be.probable thing-DAT become-NPST be.probable
 ‘The streets are wet. It must be the case that it rained.’ ((45), (46), p. 39)

Contra completely-tensed clauses like *kara*-clauses and *node*-clauses, which does not shown the correlation between the ‘matrix event < subordinate event’ temporal order and modality:

- (16) *asatte tarou-ga ku-ru kara, ashita hanako-ga*
 tomorrow Taro-NOM come-NPST because tomorrow Hanako-NOM
motenashi-no junbi-o su-ru {darou /no darou /no
 hospitality-GEN preparation-ACC do-NPST be.probable FIN be.probable FIN
da /koto-ni nar-u}.
 COP thing-DAT become-NPST
 ‘Because Taro will come the day after tomorrow, it will {probably / have to} be the
 case tomorrow that Hanako will do lots of preparation to welcome him.’ ((47), p. 39)

Chapter 4 Complete tensed clauses and settledness in Japanese

October 7, 2021

Recall the definitions of complete and incomplete tensed clauses:

- Complete tensed clauses are clauses where
 - both the plain form and the *ta*-form are allowed, and
 - the plain form corresponds to the temporal order ‘Utterance Time < Event Time; or ‘Utterance Time $\subseteq$ Event Time’, and
 - the *ta*-form corresponds to the temporal order ‘Event Time < Utterance Time’.
- Incomplete tensed clauses are clauses that
 - allow only the plain form or only the *ta*-form, or
 - allow neither the plain form nor the *ta*-form, or
 - allow both the plain form and the *ta*-form, but the two forms do not give rise to different temporal orders.

Asymmetry between past and future

Intuitively, the states of affairs that have already happened cannot be changed, but the states of affairs that are yet to happen in the future may come about in various different ways. Suppose the utterance time is t_s , this intuition is visualized below:

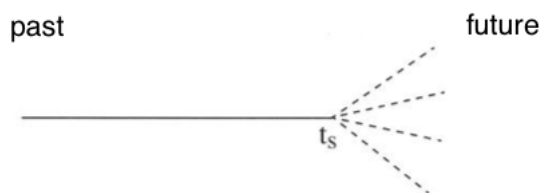


Figure 1: p. 42

The asymmetry between the past and the future can also be reflected by the fact that the truth-values of propositions are time-sensitive:

- (1) a. The coin landed heads.
 b. The coin has landed heads.
 c. The coin will land heads. (Kaufmann 2001:763, cited in (1)-(3), p. 43)

Regardless of whether we know, the truth-values of the propositions expressed by (1a) and (1b) are already settled at the utterance time. In contrast, the truth-value of (1c) needs to be determined by what turns out to be the case in the future.

Technicalities

To capture the time-sensitivity of the truth-values of propositions, we will assume indices I , which reflect time T , worlds W and their combination.

Definition I:

T is a non-empty set of moments. W is a non-empty set of worlds. I is a non-empty set of indices, which are the combinations of moments and worlds.

$$\begin{aligned} T &= \{t_1, t_2, t_3, \dots\} \\ W &= \{w_1, w_2, w_3, \dots\} \\ I &= \{\langle w_1, t_1 \rangle, \langle w_1, t_2 \rangle, \dots, \langle w_2, t_1 \rangle, \langle w_2, t_2 \rangle, \dots\} \end{aligned}$$

Definition II:

The elements in T are ordered by a transitive, irreflexive and linear relation $<$. $t' < t''$ means that t' temporally precedes t'' .

Definition III:

Propositions are written as the letters $p, q, r, \dots$

An interval i is a set of moments.

A proposition expresses that a state of affairs (state or event) holds at an interval i .

The interval inherently expressed by a proposition p is called ‘the time specified by the proposition’, and is written as p_i .

Definition IV:

There are three possible temporal relations between an interval i and a moment.

- a. $i < t$: All moments in i precede t .
 b. $t < i$: All moments in i follow t .
 c. $t \in i$: i contains t .

Definition V:

The interpretation of a proposition p at an index $\langle w, t \rangle$ is written as $\llbracket p \rrbracket \langle w, t \rangle$, and has the following three possibilities:

- a. $\llbracket p \rrbracket \langle w, t \rangle = 1$ only if p holds at the index $\langle w, t \rangle$.
 In such cases, we say that p is true at $\langle w, t \rangle$.

- b. $\llbracket p \rrbracket \langle w, t \rangle = 0$ only if the negation of p holds at the index $\langle w, t \rangle$.
In such cases, we say that p is false at $\langle w, t \rangle$.
- c. Otherwise, $\llbracket p \rrbracket \langle w, t \rangle = *$.
In such cases, we say that p is undetermined at $\langle w, t \rangle$.

We assume that when a proposition is interpreted with respect to a time that follows the interval specified by the proposition, the truth-value is either 1 or 0, but not $*$.

Assumption:

For all propositions p and the interval i specified by p , the following holds:

$$\forall w \forall t [(i < t \vee t \in i) \rightarrow (\llbracket p_i \rrbracket \langle w, t \rangle = 1 \vee \llbracket p_i \rrbracket \langle w, t \rangle = 0)]$$

Definition VI: Set of basic propositions $\Phi \langle w, t \rangle$

$$\Phi \langle w, t \rangle = \{p \mid \llbracket p \rrbracket \langle w, t \rangle = 1 \vee \llbracket p \rrbracket \langle w, t \rangle = 0\}$$

The following theorem follows from Assumption and Definition VI. It reflects informational growth as time passes.

Theorem:

For all indices $\langle w, t \rangle$, for all t' such that $t < t'$, $\Phi \langle w, t \rangle \subseteq \Phi \langle w, t' \rangle$.

Settledness

Settled proposition are propositions that are either true or false with respect to all indices $\langle w, t_s \rangle$; e.g., ‘*It rained yesterday*’. Propositions that are not settled propositions are unsettled propositions; e.g., ‘*It will rain tomorrow*’.

Definition IX: Settled propositions

Suppose t_s is the utterance time. A proposition p is a settled proposition if $p \in \Phi \langle w, t_s \rangle$.

Definition X: Unsettled propositions

A proposition p is an unsettled proposition if $p \notin \Phi \langle w, t_s \rangle$.

Settled propositions typically talk about the truthfulness/falsity of past events or current states/properties. Suppose that the utterance time is January 30, 2004, the following examples express settled propositions:

- (2) a. Japan defeated Turkey on June 19, 2002.
- b. Hanako Yamada is male.
- c. There is a succulent on the table. ((5)-(7), p. 49)

The proposition expressed by (3) does not talk about past events, states or properties, and its truth-value is not determined at the utterance time. However, given the schedule of the 2006 World Cup that is available at the utterance time in the actual world, it can also be seen as settled in the sense that it is **presumed to be settled**.

- (3) Japan plays Singapore on March 31, 2004. ((9), p. 50)

More specifically, sentences like (3) express propositions that are presumed to be settled, with the presumption being based on the propositions that are settled in the actual world at the utterance time (e.g., the World Cup schedule). Such propositions can be considered as Kratzer's (1981) conversational backgrounds (fn. 5, p. 51).

Definition XI: Propositions presumed to be settled

With respect to the actual world w_R and the utterance time t_s , p is a proposition that is presumed to be settled at the index $\langle w_R, t_s \rangle$ if for a set of propositions $\Psi\langle w_R, t_s \rangle \subseteq \Phi\langle w_R, t_s \rangle$, $\Psi\langle w_R, t_s \rangle$ entails p .

Henceforth, propositions that are settled as defined in Definition IX will be called '**narrow-sense settled propositions**'. Those defined in Definition XI will be called '**propositions presumed to be settled**'.

Expressing settledness in Japanese

To express settledness for past events, the *ta*-form is used:

- (4) *tanaka-ga nai-ta.*
Tanaka-NOM cry-PST
'Tanaka cried.' ((12), p. 52)
- (5) *kono atari-ni-wa kura-ga at-ta.*
this area-DAT-TOP warehouse-NOM be-PST
'There was a warehouse around here.' ((15), p. 52)

To express settledness for states of affairs that hold at the utterance time, the plain form of stative predicates or the stative form of eventive predicates (*te*-form of predicates + the stative verbs *iru*, *aru* 'be') is used:

- (6) *tanaka-ga heya-no naka-ni i-ru.*
Tanaka-NOM room-GEN inside-DAT be-NPST
'Tanaka is in the room.' ((17), p. 52)
- (7) *tanaka-ga ima undoujou-o hashit-te-i-ru.*
Tanaka-NOM now playground-ACC run-TE-be-NPST
'Tanaka is running at the playground now.' ((18), p. 52)
- (8) *tsukue-no ue-ni shinbun-ga oi-te-a-ru.*
table-GEN up-DAT newspaper-NOM put-TE-be-NPST
'There are newspapers on the table.' ((19), p. 52)

Propositions presumed to be settled can be expressed by the plain form and the *te* + *iru* form. In particular, in (10), the sentence expresses that the speaker's state of sitting will continue to hold until some point after the utterance time.

- (9) *ashita toukyou-ni shucchou su-ru.*
tomorrow Tokyo-DAT business.trip do-NPST

- ‘Tomorrow, I will go to Tokyo for work.’ ((20), p. 52)
- (10) *mou sukoshi, koko-ni suwat-te-i-ru.*
 more a.bit here-DAT sit-TE-be-NPST
 ‘I will sit here for a bit longer.’ ((24), p. 53)

When the agent of the state described by a proposition is not the speaker, in order to interpret the proposition as one that is presumed to be settled, epistemic modals are needed.

- (11) *tanaka-wa sanji-made benchi-ni suwat-te-i-ru hazu da.*
 Tanaka-TOP 3o'clock-until bench-DAT sit-TE-be-NPST should DAT
 ‘Tanaka should be sitting here until 3 o'clock.’ ((26), p. 53)
- (12) *kono furu-shinbun-wa asatte-made koko-ni oi-te-a-ru hazu da.*
 this old-newspaper-TOP day.after.tomorrow-until here-DAT put-TE-be-NPST should COP
 ‘These old newspapers should be here until the day after tomorrow.’ ((25), p. 53)

Settledness and speaker knowledge

We also introduce another important concept: speaker knowledge (K).

Definition XII: Speaker knowledge

A speaker’s knowledge at the current world and the utterance time is written as $K\langle w, t_s \rangle$, for which the following holds: $K\langle w, t_s \rangle \subseteq \Phi\langle w, t_s \rangle$.

Definition XIII: Settledness with respect to speaker knowledge

For all indices $\langle w, t_s \rangle$, a proposition p is settled with respect to a speaker’s knowledge if $p \in K\langle w, t_s \rangle$. Likewise, a proposition p is unsettled with respect to a speaker’s knowledge if $p \notin K\langle w, t_s \rangle$.

Suppose a proposition p expressed by the sentence ‘Luis came to the office yesterday’. Since the sentence is settled, $p \in \Phi\langle w, t_s \rangle$. In the declarative with the *ta*-form (13), $p \in K\langle w, t_s \rangle$. In the sentence that contains the modal *kamoshirenai* ‘might’ (14), $p \notin K\langle w, t_s \rangle$.

- (13) *ruisu-ga kinou kenkyuushitsu-ni ki-ta.*
 Luis-NOM yesterday office-DAT come-PST
 ‘Luis came to the office yesterday.’ ((30), p. 55)
- (14) *ruisu-ga kinou kenkyuushitsu-ni ki-ta kamoshirena-i.*
 Luis-NOM yesterday office-DAT come-PST might-NPST
 ‘Luis might have come to the office yesterday.’ ((31), p. 55)

Suppose a proposition q expressed by the sentence ‘Luis comes to the office tomorrow’. q is narrow-sense unsettled. In the declarative with the plain form (15), the speaker’s knowledge contains a set of settled propositions that entails q . With the modal *kamoshirenai* ‘might’ (16), the speaker’s knowledge does not contain such a set.

- (15) *ruisu-ga kinou kenkyuushitsu-ni ku-ru.*
 Luis-NOM yesterday office-DAT come-NPST

‘Luis will come to the office tomorrow.’ ((32), p. 55)

- (16) *ruisu-ga kinou kenkyuushitsu-ni ku-ru kamoshirena-i.*
 Luis-NOM yesterday office-DAT come-NPST might-NPST
 ‘Luis might come to the office tomorrow.’ ((33), p. 55)

The *ta*-form can also express that the speaker obtains the relevant knowledge at a point preceding the utterance time:

- (17) *ashita-no kaigi-wa niji-kara dat-ta.*
 tomorrow-GEN meeting-TOP 2o'clock-from COP-PST
 ‘Tomorrow’s meeting is at 2 (I have just found out / remembered /etc.).’ ((27), p. 54)

Evaluative adverbs and tensedness

The notions concerning settledness and speaker knowledge introduced above can help us capture the distribution of two types of evaluative adverbs, as summarized in Table 1.

	narrow-sense settled or presumed to be settled		unsettled
	in speaker knowledge	not in speaker knowledge	
Type A <i>shosen</i> ‘after all’ <i>douse</i> ‘anyway’	*	✓	*
Type B <i>saiwai</i> ‘fortunately’ <i>ainiku</i> ‘unfortunately’	✓	?	*

Table 1: p. 64

Saiwai ‘fortunately’ and *ainiku* ‘unfortunately’ are allowed with propositions that the speaker presupposes to be true. To use the notions introduced above, this means that these adverbs are allowed with settled propositions that are in the speaker’s knowledge. In contrast, *shosen* ‘after all’ and *douse* ‘anyway’ are allowed with propositions that the speaker assigns considerably high credence to (but she must not know the truth-values). Using the notions introduced above, we can say that these adverbs are allowed with settled propositions that are not in the speaker’s knowledge.

First, *shosen* ‘after all’ and *douse* ‘anyway’ are degraded in declaratives with the *ta*-form, but not in declaratives with the plain form. *Saiwai* ‘unfortunately’ and *ainiku* ‘unfortunately’ do not display this contrast. We assume that declaratives with the *ta*-form describe states of affairs that actually obtain at the actual world, and that declaratives with the plain form convey a ‘sense of future’ and are easy to be associated with speaker inference (Morimoto 1994: 66). Hence, the data can be taken to show that *saiwai* ‘fortunately’ and *ainiku* ‘unfortunately’ presuppose that the propositions are true, whereas *shosen* ‘after all’ and *douse* ‘anyway’ do not.

- (18) Narrow-sense settled propositions (past events)
 {*saiwai* / *ainiku* / ??*shosen* / ??*douse*}, *tanaka-ga iinchou-ni*
 fortunately unfortunately after.all anyway Tanaka-NOM committee.chair-DAT

nat-ta.

become-PST

‘Fortunately / Unfortunately / After all / Anyway, Tanaka became the committee chair.’
((51), (59), p. 59)

- (19) Narrow-sense settled propositions (states of affairs that hold at the utterance time)

{*saiwai* / *ainiku* / *shosen* / *douse*}, *kono hon-wa taka-i.*

fortunately unfortunately after.all anyway this book-TOP good-NPST

‘Fortunately / Unfortunately / After all / Anyway, the book is expensive.’ ((57), p. 59, (65), p. 60)

- (20) Propositions presumed to be settled

{*saiwai* / *ainiku* / *shosen* / *douse*}, *ashita toukyou-ni shucchou*

fortunately unfortunately after.all anyway tomorrow Tokyo-DAT business.trip

su-ru.

do-NPST

‘Fortunately / Unfortunately / After all / Anyway, I will travel to Tokyo for work.’ ((55), p. 59, (63), p. 60)

There is further evidence for the contrast between the two types of adverbs. The *no da* construction signals that the speaker has just come to recognize a state of affairs that she did not recognize until the utterance time. Hence, the speaker of a *no da* sentence believes the embedded proposition, but does not know it. When a declarative with the *ta*-form is embedded under *no da*, *shosen* and *douse* become acceptable. Compare (21) with (19).

- (21) {*saiwai* / *ainiku* / *shosen* / *douse*}, *tanaka-ga iinchou-ni*
fortunately unfortunately after.all anyway Tanaka-NOM committee.chair-DAT
nat-ta n da.

become-PST FIN COP.NPST

‘Fortunately / Unfortunately / After all / Anyway, it is that Tanaka became the committee chair.’ ((59’), p. 60, (67) p. 61)

Shosen and *douse* can also appear on propositions that are embedded under the epistemic modal *darou* ‘be probable’, whereas *saiwai* and *ainiku* are degraded, as in (22). However, *saiwai* and *ainiku* become acceptable if the speaker assigns high credence to the prejacent proposition, as in (23).

- (22) {*??saiwai* / *??ainiku* / *shosen* / *douse*}, *kondo-no shichou-sen-de-wa*
fortunately unfortunately after.all anyway next.time-GEN mayor-election-LOC-TOP
sakagamishi-ga kahansuu-o kakutoku su-ru darou.

Sakagami-NOM over.half-ACC get do-NPST be.probable

‘Fortunately / Unfortunately / After all / Anyway, Sakagami will probably get over half of the votes in the next election.’ ((71), p. 61)

- (23) {*saiwai* / *ainiku* / *shosen* / *douse*}, *sakagamishi-ga kahansuu-o*
fortunately unfortunately after.all anyway Sakagami-NOM over.half-ACC
war-u koto-wa man-ni hitotsu-mo na-i darou.
break-NPST thing-TOP ten.thousand-DAT one-even NEG-NPST be.probable

‘Fortunately / Unfortunately / After all / Anyway, Sakagami will not have a single chance of getting over half of the votes.’ ((72)-(74), p. 62)

In cleft constructions, the predicates are presupposed to be true. As shown in (24), *saiwai* and *ainiku*, but not *shosen* and *douse*, can appear in the predicate position of cleft constructions.

- (24) *kare-ni aribai-ga a-ru no-wa {saiwai / ainiku / *shosen / *douse}*
 he-DAT alibi-NOM be-NPST FIN-TOP fortunately unfortunately after.all anyway
da.
 COP.NPST
 lit. ‘It is fortunate / unfortunate / after all / anyway that he had an alibi.’ ((78), p. 62)

The incompatibility of the four adverbs with event modalities shows that the adverbs are not allowed with unsettled propositions.

- (25) *{*saiwai / *ainiku / *shosen / *douse} ashita shucchou shi-you.*
 fortunately unfortunately after.all anyway tomorrow business.trip do-VOL
 lit. ‘Fortunately / Unfortunately / After all / Anyway, let’s travel for work tomorrow.’
 ((79), p. 63)

Settledness and incomplete tensedness

Complete tensed clauses allow both narrow-sense settled propositions and propositions that are presumed to be settled. Incomplete tensed clauses allow narrow-sense settled propositions when the predicates are stative; they do not allow propositions that are presumed to be settled. The results are summarized in Table 2 and illustrated in more details below.

		Narrow-sense settled	Presumed to be settled	Type A <i>shosen</i> <i>douse</i>	Type B <i>saiwai</i> <i>ainiku</i>
Complete tensed		✓	✓	✓	✓
Incomplete tensed	Non-stative	*	*	*	*
	Stative	✓	*	*	?

Table 2: p. 68

States of affairs that hold at the utterance time are expressed by stative forms (stative verbs *aru*, *iru* ‘be’ or the stative forms of eventive predicates), and are allowed in both complete and incomplete tensed clauses. In such cases, Type B adverbs like *ainiku* are allowed in complete tensed clauses, as in (26). The adverbs are slightly degraded in incomplete tensed clauses, as in (27).

- (26) Narrow-sense settled (current states of affairs) + complete tensed clause
(ainiku) jikan-ga na-i {kara / node}, doko-ni-mo yor-azu-ni
 unfortunately time-NOM NEG-NPST because because where-DAT-ADD go-NEG-DAT

kaer-ou.

go.home-VOL

‘Since we don’t have time, (unfortunately,) let’s go home without going anywhere else.’

((87’), p. 65)

- (27) Narrow-sense settled (current states of affairs) + incomplete tensed clause

(?ainiku) jikan-ga {nak-ereba / nak-attara}, sono mune shir-ase-te

unfortunately time-NOM NEG-BA NEG-TARA that idea know-CAUSE-TE

hoshi-i.

want-NPST

lit. ‘If you don’t have time, (unfortunately,) please let me know.’

((88’), p. 65)

Settled propositions that express past events are allowed in complete tensed clauses, as in (28). In such cases, Type B adverbs are also allowed. In incomplete tensed clause, however, such propositions are acceptable only with non-stative predicates, as in (29). Similarly to (27), Type B adverbs are slightly degraded in such cases, as in (30).

- (28) Narrow-sense settled (past event) + complete tensed clause

(saiwai) kinou-no kaigi-de-wa sekkyokuteki-ni hatsugen shi-ta {kara /

fortunately yesterday-GEN meeting-LOC-TOP active-DAT speak.up do-PST because

node}, ashita-mo kono choushi-de ganbar-u tsumori da.

because tomorrow-ADD this way-LOC make.effort-NPST plan COP.NPST

‘Because I spoke up actively at the meeting yesterday, (fortunately,) I’ll do it like this tomorrow, too.’

((82’), p. 65)

- (29) Narrow-sense settled (past event) + incomplete tensed clause

moshi kinou yamada-ga kaisha-o {*yasun-eba / *yasun-dara /

MOSHI yesterday Yamada-NOM company-ACC rest-BA rest-TARA

yasun-de-i-reba / yasun-de-i-tara}, ashita-wa hayame-ni it-te, yamada-no

rest-TE-be-BA rest-TE-be-TARA tomorrow-TOP early-DAT go-TE Yamada-GEN

bun-o anaume shi-nak-ereba nar-ana-i.

part-ACC make.up do-NEG-BA become-NEG-NPST

‘If Yamada took a day off yesterday, tomorrow I will have to go to the company and make up his part.’

((83) p. 65, (83’), p. 66)

- (30) ?moshi ainiku kinou yamada-ga kaisha-o {yasun-de-i-reba /

MOSHI unfortunately yesterday Yamada-NOM company-ACC rest-TE-be-BA

yasun-de-i-tara}, ashita-wa hayame-ni it-te, yamada-no bun-o anaume

rest-TE-be-TARA tomorrow-TOP early-DAT go-TE Yamada-GEN part-ACC make.up

shi-nak-ereba nar-ana-i.

do-NEG-BA become-NEG-NPST

lit. ‘If Yamada took a day off yesterday, unfortunately, tomorrow I will have to go to the company and make up his part.’

((83’), p. 66)

Propositions that are presumed to be settled (e.g., according to a schedule) can appear in complete tensed clauses. In such cases, Type A adverbs like *douse* ‘anyway’ are allowed, as in (31). Note that it is the subordinate clause, instead of the whole sentence, that is modified by *douse*.

- (31) (*douse*) *kanojo-to-wa wakare-ru {kara /node}, hanashi nado*
 anyway girlfriend-and-TOP break.up-NPST because because talk etc.
shi-taku-na-i.
 do-want-NEG-NPST
 ‘Since I will break up with my girlfriend (anyway), I don’t want to talk to her.’ ((89’) p. 66)

Propositions in incomplete tensed clauses cannot be interpreted as presumed to be settled, regardless of the (non-)stativity of the predicate in the clause, as in (32) and (33). Note that *douse* are allowed in both (32) and (33), but it is the whole sentence, instead of the subordinate clause, that is modified by *douse*.

- (32) (*douse*) *kanojo-to {wakare-reba / wakare-tara}, hanashi-o su-ru koto nado*
 anyway girlfriend-and break.up-BA break.up-TARA talk-ACC do-NPST thing etc
na-i sa.
 NEG-NPST SFP
 ‘If I break up with my girlfriend, I won’t have anything to talk with her (anyway).’ ((91) p. 67)
- (33) (*douse*) *kanojo-to {wakare-te-i-reba / wakare-te-i-tara}, hanashi-o*
 anyway girlfriend-and break.up-BA break.up-TE-be-TARA talk-TE-be-ACC
su-ru koto nado na-i sa.
 do-NPST thing etc NEG-NPST SFP
 ‘If I have broken up with my girlfriend, I won’t have anything to talk with her (anyway).’ ((91’) p. 67)

Chapter 5 The system of logical sentences

October 15, 2021

Hypotheticality and factuality

The term *ronribun* ‘logical sentences’ is coined by Koizumi (1987) to cover conditional sentences, concessive sentences and causal sentences. Researchers have adopted the classification of logical sentences based on two parameters, harmony vs. dissonance, and hypothetical vs. factual (Maeda 1991, König and Siemund 2000).

	Hypothetical	Factual
Harmony	Conditional	Causal
Dissonance	Concessive conditional	Concessive causal

Table 1: König and Siemund 2000, cited in Table 2, p. 72

For a logical sentence that describe two states of affairs, ‘harmony’ refers to the situation that one state of affairs happens incidentally to the other or in the same chain with the other; otherwise, the meaning expressed by the sentence can be taken to express ‘dissonance’. This parameter distinguishes conditional sentences and causal sentences on the one hand, and concessive conditionals and concessive causal sentences on the other.

To distinguish between conditional sentences and concessive conditionals, as well as between causal sentences from concessive causal sentences, the relevant parameter is hypothetical vs. factual. Specifically, it refers to whether the subordinate clause of the logical sentence is true. Examples for the four types of logical sentences are provided below.

(1) Conditional

doryoku su-reba, seikou su-ru darou.

work.hard do-BA success do-NPST be.probably

‘If I work hard, it is probable that I succeed.’

(Maeda 1991: 55 cited in (1), p. 72)

(2) Causal

doryoku shi-ta node, seikou shi-ta.

work.hard do-PST because success do-PST

‘Because I worked hard, I succeeded.’

(Maeda 1991: 55 cited in (2), p. 72)

(3) Concessive conditional

doryoku shi-te-mo, seikou shi-na-i darou.

work.hard do-TE-even success do-NEG-NPST be.probably

‘Even if I work hard, it is probable that I don’t succeed.’ (Maeda 1991: 55 cited in (3), p. 72)

(4) Concessive causal

doryoku shi-ta noni, seikou shi-nakat-ta.

work.hard do-PST NONI success do-NEG-PST

‘Even though I worked hard, I didn’t succeed.’ (Maeda 1991: 55 cited in (4), p. 72)

However, separation of conditionals and causal sentences based on hypotheticality/factuality is problematic, because (1) can be uttered when the speaker has confirmed the truthfulness of the subordinate clause, or the truthfulness of both the subordinate and the main clause. What sentences like (1) expresses is not specific events, but rather the causal relation between the two relevant states of affairs. The relation might involve instances where the subordinate-clause state of affairs (and hence the main-clause state of affairs) has not yet obtained.

Implicit prerequisite

Apart from ‘harmony’ and ‘dissonance’, another way to make the distinction between conditionals and concessive conditionals, as well as between causal sentences and concessive causal sentences, is to resort to Sakahara’s (1985) notion of ‘implicit prerequisites’. As shown in the following figure, in the worlds where the relevant implicit requisite is satisfied, the speaker will utter conditionals, causal sentences, or counterfactuals. Otherwise, concessive conditionals, concessive causal sentences, or concessive counterfactuals will be chosen.

Implicit prerequisite is satisfied		Implicit prerequisite is not satisfied
Conditionals	→	Concessive conditionals
Causals		Concessive (causal)
Counterfactuals		Concessive counterfactuals

Figure 1: p. 77

For instance, (5) is a true statement. However, in a non-stereotypical world where the speaker wears heat-resistant glove, it is possible that she does not get burnt when she puts her hands into boiling water. Hence, utterances of conditionals like (5) comes with the implicit assumption that the non-stereotypical state of affairs that the speaker wears heat-resistance gloves does not hold.

(5) Conditional

nitat-ta oyu-ni te-o ire-reba yakedo su-ru.

come.to.boil-PST hot.water-DAT hand-ACC put.in-BA get.burnt do-NPST

‘If you put your hand into boiling water, you will get burnt.’ ((12), p. 72)

Meanwhile, Sakahara (1985) treats concessive conditionals as the form that is used when the implicit prerequisite is not satisfied. Sentences like (6) talk about non-stereotypical worlds, for instance, worlds where the speaker wears heat-resistant gloves. In such worlds, the causal relation between the states of affairs described by the two clauses no longer holds.

(6) Concessive conditional

nitat-ta oyu-ni te-o ire-te-mo yakedo shi-na-i.
 come.to.boil-PST hot.water-DAT hand-ACC put.in-TE-even get.burnt do-NEG-NPST
 ‘Even if you put your hand into boiling water, you won’t get burnt.’ ((16), p. 73)

Causal sentences are like conditionals in relying on the assumption that the implicit prerequisite is satisfied. (7) thus ignores non-stereotypical events such as ‘Inoue got invited to have a drink with his colleagues’, in which case the causal relation between finishing work early and going to the English school would hold. When the implicit prerequisite is not satisfied, concessive causal sentences are used, as in (8).

(7) Causal
inoue-wa shigoto-ga hayaku owat-ta kara, eikaiwa gakkou-ni it-ta.
 Inoue-TOP work-NOM early end-PST because English.conversation school-DAT go-PST
 ‘Because Inoue finished work early, he went to the English school.’ ((17), p. 76)

(8) Concessive causal
inoue-wa shigoto-ga hayaku owat-ta noni, eikaiwa gakkou-ni
 Inoue-TOP work-NOM early end-PST NONI English.conversation school-DAT
ik-anakat-ta.
 go-NEG-PST
 ‘Even though Inoue finished work early, he didn’t go to the English school.’ ((17), p. 76)

The same can be said for counterfactual and concessive counterfactuals, like (9) and (10).

(9) Counterfactual
inoue-wa shigoto-ga hayaku owat-te i-tara, eikaiwa gakkou-ni it-ta
 Inoue-TOP work-NOM early end-TE be-TARA English.conversation school-DAT go-PST
darou.
 be.probable
 ‘If Inoue had finished work early, he probably would have gone to the English school.’
 ((18), p. 76)

(10) Concessive counterfactual
inoue-wa shigoto-ga hayaku owat-te i-te-mo, eikaiwa gakkou-ni
 Inoue-TOP work-NOM early end-TE be-TE-even English.conversation school-DAT
ik-anakat-ta darou.
 go-NEG-PST be.probable
 ‘Even if Inoue had finished work early, he probably would not have gone to the English school.’
 ((18), p. 76)

As for the distinction between conditionals and causal sentences, we have seen above that hypotheticality/factivity is insufficient to draw the line. Rather, the two types of sentences can be distinguished based on the inference that they give rise to. As noted by Sakahara, conditionals ‘if *p*, then *q*’ often come with the pragmatic inference that ‘if not *p*, then not *q*’. Causal sentences also give rise to such inference as shown below, although it is counterfactual conditionals that get inferred from causal sentences.

- (11) Because you called the police, they killed the kid. (Sakahara 1985: 121, in Japanese, cited in (14), p. 75)
- (12) If you had not called the police, they would not have killed the kid. (Sakahara 1985: 121, in Japanese, cited in (15), p. 75)

The rest of this study assumes that conditionals are linguistic expressions that express inferences based on unsettled epistemic states. For a sentence ‘if p , then q ’, p is a proposition whose truth value is unsettled, and q is the consequence that follows from p and the implicit prerequisite.

Chapter 6 Conditional meaning and two types of uncertainty

December 26, 2021

Classification of conditionals

Conditionals can be classified based on **indicative vs. subjunctive mood** such as in Spanish, with the subjunctive mood signaling counterfactual conditionals. The mood-based classification is not suitable for languages whose mood system is less robust, such as English. For English, the most widely accepted way of classifying conditionals is the one that separates **open conditionals** and **hypothetical conditionals**, with the latter being signaled by **backshift**.

- (1) Open conditionals (no backshift)
If Collins is in London, he is undoubtedly staying at the Hilton.
(Quirk et al. 1985: 1091, (6), p. 81)
- (2) Hypothetical conditionals (with backshift)
 - a. If he changed his opinions, he'd be a more likable person.
 - b. If you had listened to me, you wouldn't have made so many mistakes.
 (Quirk et al. 1985: 1091, cited in (7), (9), p. 81)

In both the mood-based system and the open vs. hypothetical system, the key to the classification is whether the antecedent is true, which is encoded explicitly (by mood selection or backshift).

Another way to classify conditionals is based on predictivity. Dancygier (1998) suggests to separate (3a) and (4) into different groups. Note that in the open vs. hypothetical system, the two sentences would both qualify as open conditionals.

- (3) Dancygier's (1998) predictive conditionals
 - a. If it rains, the match will be canceled.
 - b. If it rained, the match would be canceled.
 - c. If it had rained, the match would have been canceled.
- (4) Dancygier's (1998) non-predictive conditionals
If he had a cold last week, he couldn't come to your talk.

Specifically, he identifies two types of backshift in conditionals:

- if-backshift (e.g., (3a)): the use of the simple present form of verbs instead of 'will + V' in the antecedent

- Dancygier's classification is problematic in two respects. First, by grouping (3a)-(3c) under the same type of conditionals, we would lose the intuitive distinction between counterfactual and non-counterfactual conditionals. Second, as argued by Kaufmann (2005), the lack of '*will*' in the antecedent of (3a) should in fact be attributed to an independent syntactic reason, namely that '*will*' is not required in non-matrix environments. Kaufmann's reasoning is as follows. Matrix sentences with unmodalized verbs must be interpreted with Certainty Condition, namely that the speaker is certain that the state of affairs will obtain. This is exemplified in (5b):

- Kaufmann proposes that Certainty Condition is required in all matrix non-modalized environments. For matrix sentences with non-modalized verbs in their past tense, Certainty Condition holds trivially, since such sentences describe states of affairs that have already obtained. Conditional antecedents are not matrix; accordingly, even with non-modalized verbs in their simple present tense like in (3a), they do not need to be interpreted with Certainty Condition. Consequently, there is no motivation to group (3a) together with the other conditionals that attest backshift.

Funk (1985) proposes to classify conditionals based on two types of uncertainty,

- (Funk 1985, cited in (18)-(19), p. 85)

- (Funk 1985, cited in (20), p. 85)

This book will adopt a classification of conditionals based on settledness as well as the two types of uncertainty of the antecedent proposition. The classification distinguishes three types of conditionals, as shown in the table below:

Settledness	Speaker's attitude towards antecedent	Type of conditional
Unsettled	N/A	Predictive conditionals
Settled	Speaker doesn't know the antecedent	Epistemic conditionals
	Speaker knows the antecedent is false	Counterfactual conditionals

Table 1: p. 90

Overall, conditionals in the form of '*if p, q*' express the speaker's inference of *q* based on the implicit prerequisite (represented as *E* below) in her knowledge (represented as *K* applied to a world-time pair below) and the antecedent proposition *p*. (Note that in conditionals, the speaker must not know *p*.) The definitions for the three types of conditionals are given below. Predictive conditionals are conditionals with unsettled antecedents (note that the term 'predictive' is used in a different sense from the one used in Dancygier's classification mentioned above):

Definition (Predictive conditionals):

Suppose that the propositions expressed by *P* and *Q* are *p* and *q*, respectively. A sentence '*if P, Q*' is a predictive conditional if $p \notin \Phi\langle w, t_s \rangle$ and $E, \{p\} \models q$, where $\Phi\langle w, t_s \rangle$ is the set of propositions that are settled at *w* and *t_s*.

Examples include the ones in (6), and also the following:

- (8) If it rains tomorrow, the match will be canceled. ((21), p. 88)

Epistemic conditionals refer to conditionals whose antecedents are settled, but are not known by the speaker:

Definition (Epistemic conditionals)

Suppose that the propositions expressed by *P* and *Q* are *p* and *q*, respectively. A sentence '*if P, Q*' is an epistemic conditional if $p \in \Phi\langle w, t_s \rangle$, $p \notin K\langle w, t_s \rangle$, and $E, \{p\} \models q$.

Examples include (7), as well as the following:

- (9) If it rained today, the match would be canceled. ((22), p. 88)

Counterfactual conditionals refer to conditionals whose antecedents are known by the speaker to be false.

Definition (Counterfactual conditionals)

Suppose that the propositions expressed by *P* and *Q* are *p* and *q*, respectively. A sentence '*if P, Q*' is a counterfactual conditional if $p \in \Phi\langle w, t_s \rangle$, $\neg p \in K\langle w, t_s \rangle$ and $E, \{p\} \models q$.

The interaction between the implicit prerequisite *E* and the antecedent proposition *p* will be discussed in Ch. 10. Below is an example of counterfactual conditionals:

- (10) If it had rained yesterday, the match would have been canceled. ((23), p. 88)

Chapter 7 Japanese conditionals

November 8, 2021

Conditional expressions

Conditional expressions in Japanese are illustrated in the following.

- (1) Basics types:
 - a. *ba*: verb stem + (*r*)*eba*
 - b. *tara*: verb stem + PAST + *ra*
 - c. *to*: verb stem + NPST + *to*
 - d. *nara*: verb stem + NPST/PST + *nara*
- (2) *no da* series
 - a. *no nara*: verb stem + NPST/PST + *no* + *nara*
 - b. *no de ar-eba*: verb stem + NPST/PST + *no* + *dea* ('be') + *reba*
 - c. *no dat-tara*: verb stem + NPST/PST + *no* + *dat-ta* ('be'-PST) + *ra*
 - d. *no da to*: verb stem + NPST/PST + *no* + *da* ('be'.NPST) + *to*
- (3) *to suru* series
 - a. *to su-reba*: verb stem + NPST/PST + *to* + *su* + *reba*
 - b. *to shi-tara*: verb stem + NPST/PST + *to* + *shi-ta* + *ra*
 - c. *to su-ru nara*: verb stem + NPST/PST + *to* + *su-ru* + *nara*
 - d. *to shi-ta nara*: verb stem + NPST/PST + *to* + *shi-ta* + *nara*
- (4) *te wa* series
 - a. *te wa*: verb stem + *te* + *wa*
 - b. *no de wa*: verb stem + NPST/PST + *no* + *de* + *wa*

There are two things to note about (3) and (4). The phrase *to suru* means 'to hypothesize', 'to suppose', and should thus be differed from the functional items that signal conditionals. We will thus gloss over the expressions in (3). In addition, *te wa* in (4) usually gives rise to the speaker's negative feeling towards the consequent. In addition, *te wa* has a non-conditional, repetitive use. Ch. 11 will provide a case study of *te wa*.

We will also not discuss *to*, because it is unclear to what extent it should be treated as a conditional marker. Like *tewa*, *to* also has a non-conditional, repetitive use. It is also more constrained than the other conditional markers. For instance, its consequent cannot express commands or event modality:

- (5) **taifuu-ga ku-ru to ie-ni {kaer-inasai / kaer-itai / kaer-ou}*.
 typhoon-NOM come-NPST TO home-DAT return-IMP return-want return-VOL
 ‘If the typhoon comes, go home / I want to go home / let’s go home.’ (p. 95, (1))

It is also degraded with *moshi* ‘supposedly’:

- (6) ?*moshi taifuu-ga ku-ru to kono ie-wa taorete shima-u*
 supposedly typhoon-NOM come-NPST TO this home-TOP fall end-NPST
kamoshirena-i.
 might-NPST
 ‘If the typhoon comes, this house might fall.’ (p. 95, (2))

It also cannot be used to seek information about the hearer’s intention:

- (7) *futtou su-ru to {dou nar-imas-u ka? / *dou shi-mas-u ka?}*
 boil do-NPST TO how become-POL-NPST Q how do-POL-NPST Q
 ‘If the water boils, what happens next? / what will you do?’ (p. 95, (3), (4))

The discussions will thus focus on the following four conditional markers:

- (8) a. *ba*: verb stem + (r)*eba*
 b. *tara*: verb stem + PAST + *ra*
 c. *nara*: verb stem + NPST/PST + *nara*
 d. *no nara*: verb stem + NPST/PST + *no* + *nara*

Complete tensedness of *nara*- and *no nara*-clauses

We have seen in Ch. 3 and Ch. 4 that *ba*-clauses and *tara*-clauses are incomplete tensed clauses. Recall in *ba*-conditionals and *tara*-conditionals, the state of affairs described by the antecedent must temporally precede the state of affairs described by the consequent, unless the consequent contains auxiliaries that express backward inference (e.g. *no da*, *koto ni naru*):

- (9) **ashita {yasum-eba / yasun-dara} kyou-wa zangyou shi-nak-ereba*
 tomorrow rest-BA rest-TARA today-TOP work.overtime do-NEG-BA
nar-ana-i.
 become-NEG-NPST
 ‘If I take a day off tomorrow, I will have to work overtime today.’ (p. 96, (13), (14))
- (10) *asatte tarou-ga {ki-tara / ku-reba}, ashita hanako-ga*
 day.after.tomorrow Taro-NOM come-TARA come-BA tomorrow Hanako-NOM
yoppodo settoku shi-ta {no da / no darou / koto-ni nar-u /
 very.much persuade do-PST FIN COP FIN be.probable thing-DAT become-NPST
 **darou / *hazu da}*.
 be.probable should COP
 ‘If Taro comes the day after tomorrow, it will have to be the case that he was deeply persuaded by Hanako tomorrow.’ (p. 96, (16))

They are also incompatible with adverbs that require clauses that are presumed to be settled:

- (11) **douse kanojo-to {wakare-reba / wakare-tara}, hanashi-o su-ru koto nado*
 after.all girlfriend-with break.up-BA break.up-TARA word-ACC do-NPST thing etc.
na-i sa.
 NEG-NPST SFP
 ‘If I will break up with my girlfriend after all, I have nothing to talk about with her.’ (p. 96, (17))

Nara-clauses and *no nara*-clauses behave differently. Like reason clauses (*kara*, *node*), they are complete tensed clauses:

- (12) *ashita yasum-u {nara / no nara} kyou-wa zangyou shi-nak-ereba*
 tomorrow rest-NPST NARA FIN NARA today-TOP work.overtime do-NEG-BA
nar-ana-i.
 become-NEG-NPST
 ‘If I take a day off tomorrow, I will have to work overtime today.’ (p. 97, (20))
- (13) *hontouni aitsu-ga korosh-ita {nara / no nara}, nanika shouko-o*
 really that.guy-NOM kill-PST NARA FIN NARA something evidence-ACC
nokosh-itei-ru hazu da.
 leave-ASP-NPST should COP.NPST
 ‘If he is really the killer, there must be some evidence for it.’ (p. 97, (19))
- (14) *douse wakare-ru {nara / no nara}, tsukiaw-anak-ereba yokat-ta.*
 after.all break.up-NPST NARA FIN NARA date-NEG-BA good-PST
 ‘If we will break up after all, we should not have dated.’ (p. 97, (22))

However, unlike reason clauses, *nara* and *no nara* are incompatible with adverbs that presuppose the truth of the antecedent:

- (15) ??*saiwai itsumo yori oome-ni mot-tei-ru {nara / no nara}, harat-te*
 fortunately always than much-DAT hold-ASP-NPST NARA FIN NARA pay-CONT
hoshi-i.
 want-NPST
 ‘If fortunately you have more (money) with you than you usually do, I want you to pay for it.’ (p. 98, (25))
- (16) *saiwai itsumo yori oome-ni mot-tei-ru {kara / node}, boku-ga hara-u*
 fortunately always than much-DAT hold-ASP-NPST because because I-NOM pay-NPST
yo.
 SFP
 ‘Since fortunately I have more (money) with me than I usually do, let me pay for it.’ (p. 98, (27))

Chapter 8 Predictive conditionals in Japanese

November 19, 2021

Main point of this chapter: In Japanese, predictive conditionals require the antecedent clauses to be incomplete-tensed clauses, whereas complete-tensed clauses are not allowed as the antecedents of predictive conditionals.

Antecedents of predictive conditionals

The distribution of conditional expressions in predictive conditionals is illustrated in (1):

- (1) *ashita ame-ga {fur-eba / fut-tara / *fur-u no nara / *fut-ta no nara /*
 tomorrow rain-NOM fall-BA fall-TARA fall-NPST FIN NARA fall-PST FIN NARA
??fur-u nara / fut-ta nara}, *shiai-wa chuushi-ni nar-u darou.*
 fall-NPST NARA fall-PST NARA game-TOP cancel-DAT become-NPST be.probable
 ‘If it rains tomorrow, the game will probably be cancelled.’ (p. 102, (8)-(13))

According to the definition in Ch. 6, the antecedents of predictive conditionals express unsettled propositions. We thus predict that in Japanese, *ba* and *tara*, which require incomplete tensedness, can appear in the antecedents of predictive conditionals; in contrast, *no nara*, which requires complete tensedness, cannot appear in such examples. These predictions are borne out, as in (1). In particular, with *furu no nara*, the sentence is acceptable under the interpretation that raining is a settled state of affairs; in other words, *no nara* forces the conditional to be an epistemic conditional instead of a predictive conditional.

It is possible for *nara*, which requires complete tensedness, to appear in predictive conditionals. In such cases, past + *nara* is more natural than non-past + *nara*. However, with past + *nara* in predictive conditionals, the antecedent clauses do *not* count as complete tensed clauses, because *ta* does not indicate the temporal order event time < utterance time. For instance, in (1), the event time of the antecedent state of affairs clearly follows the utterance time. As discussed in Ch. 2, Japanese *ta* can introduce a reference time that is independent of the utterance time and express the temporal order event time < reference time, which is the case in (1).

Consequents of predictive conditionals

Apart from epistemic modality, the consequents of predictive conditionals can also contain evidential modality (2) and event modality (3):

- (2) *ashita ame-ga {fur-eba / fut-tara}, shiai-wa chuushi-ni nar-u*
 tomorrow rain-NOM fall-BA fall-TARA game-TOP cancel-DAT become-NPST
{rashi-i / you da}.
 seem-NPST seem COP.NPST
 ‘If it rains tomorrow, it seems that the game will be cancelled.’ (p. 106, (21))
- (3) *ashita ame-ga {fur-eba / fut-tara}, shiai-wa chuushi-ni {shi-you / shi-ta-i*
 tomorrow rain-NOM fall-BA fall-TARA game-TOP cancel-DAT do-VOL do-want-NPST
/ shi-nak-ereba nar-ana-i}.
 do-NEG-BA become-NEG-NPST
 ‘If it rains tomorrow, {let’s / I want to / I have to} cancel the game.’ (p. 106, (22), (23))

Note that modals differ in terms of the scope that they can take in conditionals. Evidential modalities scope over the whole conditionals:

- (4) *ashita ame-ga fut-tara, shiai-wa chuushi-ni nar-u you da.*
 tomorrow rain-NOM fall-TARA game-TOP cancel-DAT become-NPST seem COP.NPST
 ‘If it rains tomorrow, it seems that the game will be cancelled.’
 a. [[If it rains, the games gets canceled] *youda*]
 b. *[If it rains, [[the games gets canceled] *youda*]] (fn. 4, p. 106)

In contrast, epistemic modalities and event modalities scope over the consequents only:

- (5) *ashita ame-ga fut-tara, shiai-wa chuushi-ni nar-u darou.*
 tomorrow rain-NOM fall-TARA game-TOP cancel-DAT become-NPST be.probable
 ‘If it rains tomorrow, the game will probably be cancelled.’
 a. *[[If it rains, the games gets canceled] *darou*]
 b. [If it rains, [[the games gets canceled] *darou*]] (fn. 4, p. 106)
- (6) *ashita ame-ga fut-tara, shiai-wa chuushi-ni shi-nak-ereba nar-ana-i.*
 tomorrow rain-NOM fall-TARA game-TOP cancel-DAT do-NEG-BA become-NEG-NPST
 ‘If it rains tomorrow, I have to cancel the game.’
 a. *[[If it rains, the games gets canceled] *nakereba naranai*]
 b. [If it rains, [[the games gets canceled] *nakereba naranai*]] (fn. 4, p. 106)

More discussions for the scope properties of evidentials in conditionals can be found in Takubo (2006).

As discussed in Ch. 3, the consequents of predictive conditionals can also contain modalities that license backward inferences, such as *koto-ni naru* or *no da*. In such cases, the states of affairs described by the antecedents can temporally follow the state of affairs described by the consequents. Without these modalities, this type of temporal order is not allowed in Japanese predictive conditionals:

- (7) *kondo-no shiken-ni musuko-ga {uk-areba / ukat-tara}, watashi-wa kamisama-kara*
 next.time-GEN test-DAT son-NOM pass-BA pass-TARA I-TOP god-from
*yuushuuna kodomo-o sazukat-ta {koto-ni nar-u / *darou}*.
 gifted child-ACC give-PST thing-DAT become-NPST be.probable

‘If my son passes the upcoming exam, {it means that / it’s probably the case that} god gave me a gifted child.’ (p. 108, (26), (28))

This observation does not carry over to English predictive conditionals:

- (8) If the leaves wither in a day or two, you added too much fertilizer. (Dancygier 1998: 48, cited on p. 107, (32))

The contrast between English and Japanese follow from the difference between the interpretation of past tense in the two languages. In English, backward inferences can be supported by unmodalized simple past sentences, whereas Japanese requires explicit modals:

- (9) a. The road is wet. It rained.
 b. *douro-ga nure-tei-ru. ame-ga fut-ta no da.*
 road-NOM be.wet-ASP-NPST rain-NOM fall-PST FIN COP.NPST
 ‘The road is wet. It rained.’ (p. 107, (33))

Generic conditionals

What we call generic conditionals in this study refer to the conditionals whose antecedents and consequents are in a general or habitual relation, such as:

- (10) I I drink too much milk, I get a rash. (Dancygier 1998: 63, cited in p. 108, (34))

We classify generic conditionals as a type of predictive conditionals, because e.g. in (10), the antecedent proposition may still occur in the future, and is thus unsettled at the utterance time. A seeming counterexample comes from generic conditionals that describe states of affairs happened in the past:

- (11) If he said yes, she said no.

However, such examples are allowed even if the antecedent has only occurred once in the past:

- (12) When she was young, if he said yes, though I only observed this once. (p. 109, (37))

This means that even with generic conditionals describing past events, the antecedent refer to states of affairs that will obtain in the future. Hence, past-tense generic conditionals also have antecedents that express unsettled propositions.

Classifying generic conditionals as predictive conditionals is consistent with the fact that in Japanese, generic conditionals allow *ba*- and *tara*-clauses (which are incomplete-tensed), but not *nara*- and *no nara*-clauses (which are complete-tensed):

- (13) *kare-ga {tanom-eba / tanon-dara / *tanom-u nara / *tanon-da nara / *tanom-u no*
 he-NOM ask-BA ask-TARA ask-NPST NARA ask-PAST NARA ask-NPST FIN
*nara / *tanon-da no nara}, (futsuu) kanojo-wa shoudaku su-ru.*
 NARA ask-PST FIN NARA usually she-TOP agree do-NPST
 ‘If he asks, usually she agrees.’ (p. 110, (38), (39))

Chapter 9 Epistemic conditionals in Japanese

December 9, 2021

Main idea of this chapter: Epistemic conditionals in Japanese require antecedents that are complete-tensed clauses. This means that the antecedent of epistemic conditionals can be led by *nara* and *no nara* clauses, which take tensed clauses. The conditional connectives that take incomplete tensed clauses, namely *tara* and *ba*, are also allowed in epistemic conditionals, but are subject to extra constraints. The constraints on epistemic conditionals are the following:

1. When the predicate in the antecedent is eventive, *tara* and *ba* require their stative forms.
2. *tara* and *ba* cannot appear in epistemic conditionals whose antecedents express propositions that are presumed to be settled.
3. Epistemic conditionals cannot have consequents that contain evidential modalities.

Antecedents of epistemic conditionals in Japanese

Table 1: p. 118

		<i>nara/no nara</i>	Statives + <i>tara/ba</i>
Narrow-sense settled propositions	Supposition by the speaker	✓	✓
	Introduced by the hearer	✓	*
Propositions presumed to be settled	Supposition by the speaker	✓	*
	Introduced by the hearer	✓	*

When the antecedent expresses narrow-sense settled propositions

When the antecedent expresses narrow-sense settled propositions, *nara* and *no nara* are allowed, as in (1). The fact that the speaker does not know the truth/falsity of the antecedent proposition can be shown by the adverb *moshi* ‘supposedly’:

- (1) (*moshi*) *kinou kin'ippuu-ga {de-ta nara / de-ta no nara}*, *kyou-wa*
 MOSHI yesterday bonus-NOM release-PST NARA release-PST FIN NARA today-TOP
minna nomi-ni ik-u darou.
 everyone drink-DAT go-NPST be.probbale
 ‘If they got their bonus yesterday, they will probably go to have a drink together today.’
 ((5), (10), p. 113)

With *ba* and *tara*, the verb must be in its stative form:

- (2) *kinou kin'ippuu-ga {*de-reba / *de-tara}, kyou-wa minna nomi-ni ik-u darou.*
 yesterday bonus-NOM release-BA release-TARA today-TOP everyone drink-DAT
 go-NPST be.probbale
 'If they got their bonus yesterday, they will probably go to have a drink together today.'
 ((16), p. 114)
- (3) (*moshi*) *kinou kin'ippuu-ga {de-te i-reba / de-te i-tara}, kyou-wa MOSHI yesterday bonus-NOM release-CONT be-BA release-CONT be-TARA today-TOP minna nomi-ni ik-u darou.*
 everyone drink-DAT go-NPST be.probbale
 'If they got their bonus yesterday, they will probably go to have a drink together today.'
 ((17), (28) p. 114-116)

The antecedent can also be introduced by the hearer in the discourse; in such cases, *nara* and *no nara*, but not *ba* and *tara* (even with stative verbs), are allowed:

- (4) A: *kinou kin'ippuu-ga de-ta yo.*
 yesterday bonus-NOM release-PST SFP
 'I got my bonus yesterday.'
- B: *sou ka. kinou kin'ippuu-ga {de-ta nara / de-ta no nara / *de-reba / *de-tara / *de-te i-reba / *de-te i-tara}, ashita kaimono-ni ik-ou.*
 that Q yesterday bonus-NOM release-PST NARA release-PST FIN NARA
 release-BA release-TARA release-CONT be-BA release-TE be-TARA tomorrow
 shopping-DAT go-VOL
 'I see. If you got their bonus yesterday, let's go shopping tomorrow.' ((15), (35), p. 114, p. 116)

Also, *moshi* is not allowed in such cases:

- (5) A: *kinou kin'ippuu-ga de-ta yo.*
 yesterday bonus-NOM release-PST SFP
 'I got my bonus yesterday.'
- B: *sou ka. *moshi kinou kin'ippuu-ga {de-ta nara / de-ta no nara}, ashita kaimono ik-ou.*
 that Q MOSHI yesterday bonus-NOM release-PST NARA release-PST FIN NARA
 tomorrow shopping-DAT go-VOL
 'I see. If you got their bonus yesterday, let's go shopping tomorrow.' ((15), p. 114)

Another difference between *nara/no nara* vs. *ba/tara* in epistemic conditionals is that *ba/tara* are degraded when the consequent contains event modalities:

- (6) *ima densha-ni {not-te i-reba / not-te i-tara / no-te iru nara}*
 now train-DAT ride-CONT be-BA ride-CONT be-TARA ride-CONT be NARA

- go-fun-go-ni Umeda-ni touchaku su-ru hazu da.*
 five-minutes-later-DAT Umeda-DAT arrive do-NPST should COP.NPST
 ‘If you’re on the train now, you should arrive at Umeda in five minutes.’ ((24), p. 115)
- (7) *ima densha-ni {??not-te i-reba / ??not-te i-tara / no-te iru nara} ato-de*
 now train-DAT ride-CONT be-BA ride-CONT be-TARA ride-CONT be NARA later-LOC
kakenaos-u yo.
 call.again-NPST NPST
 ‘If you’re on the train now, I’ll call you again later.’ ((25), p. 115)

When the antecedent expresses propositions that are presumed to be settled

When the antecedent expresses propositions that are presumed to be settled, *nara* and *no nara* are allowed. In contrast, *tara* and *ba* are not allowed:

- (8) *(moshi) raishuu-no suiyoubi-ni shucchou su-ru {nara / no nara},*
 MOSHI next.week-GEN Wednesday-DAT business.trip do-NPST NARA FIN NARA
konshuujuu-ni shorui-o junbi shi-te ok-anak-ereba nar-ana-i.
 this.week-DAT paperwork-ACC prepare do-CONT prepare-NEG-BA become-NEG-NPST
 ‘If I (am scheduled to) travel for work next Wednesday, I have to prepare the paper work this week.’ ((36), p. 116, (38), p. 117)
- (9) *(moshi) raishuu-no suiyoubi-ni shucchou {*su-reba / *shi-tara / *shi-te*
 MOSHI next.week-GEN Wednesday-DAT business.trip do-BA do-TARA do-CONT
*i-reba / *shi-te i-tara}, konshuujuu-ni shorui-o junbi shi-te*
 be-BA do-CONT be-TARA this.week-DAT paperwork-ACC prepare do-CONT
ok-anak-ereba nar-ana-i.
 prepare-NEG-BA become-NEG-NPST
 ‘If I (am scheduled to) travel for work next Wednesday, I have to prepare the paper work this week.’ ((42), (43), p. 117)
- (10) A: *raishuu-no suiyoubi-ni shucchou su-ru yo.*
 next.week-GEN Wednesday-DAT business.trip do-NPST SFP
 ‘I’ll travel for work next Wednesday.’
 B: *raishuu-no suiyoubi-ni shucchou {su-ru nara / su-ru no nara*
 next.week-GEN Wednesday-DAT business.trip do-NPST NARA do-NPST FIN NARA
*/ *shi-te i-reba / *shi-te i-tara}, sono shuumtasu-wa ie-de yukkuri*
 do-CONT be-BA do-CONT be-TARA that weekend-TOP home-LOC slowly
shi-ta hou-ga i-i.
 do-PST way-NOM good-NPST
 ‘If you (are scheduled to) travel for work next Wednesday, you should relax at home on the weekend of that week.’ ((41), (46), p. 117)

Consequents of epistemic conditionals in Japanese

We have seen in the previous chapter that in the consequents of predictive conditionals, both epistemic and evidential modals are allowed. In epistemic conditionals, however, evidential modalities

are not allowed:

- (11) *kinou kin'ippuu-ga {de-ta nara / de-ta no nara / de-te i-reba}*
 yesterday bonus-NOM release-PST NARA release-PST FIN NARA release-CONT be-BA
/ de-te i-tara}, *kyou-wa minna-de nomi-ni ik-u {darou /}*
 release-TE be-TARA today-TOP everyone-LOC drink-DAT go-NPST be.probable
**youda}*.

seem

'If they got their bonus yesterday, they would probably they will go to have a drink together today.' /

'It seems that if they got their bonus yesterday, they will go to have a drink together today.'

((64) p. 124)

- (12) *raishuu-no suiyoubi-ni shucchou {su-ru nara / su-ru no nara}*,
 next.week-GEN Wednesday-DAT business.trip do-NPST NARA do-NPST FIN NARA
konshuujuu-ni jimusho-no mono-ga shucchou-ni hitsuyouna shorui-o
 this.week-DAT office-GEN people-NOM business.trip-DAT necessary paperwork-ACC
*soroe-te kure-ru {darou / *rashii}*.

prepare-CONT give-NPST be.probable seem

'If you (are scheduled to) travel for work next Wednesday, the people at the office will probably prepare all the paperworks for you this week.' /

'It seems that if you (are scheduled to) travel for work next Wednesday, the people at the office will prepare all the paperworks for you this week.'

((65), p. 124)

The contrast can be explained by Takubo's (2001) observation that epistemic and evidential modals in conditional consequents lexically encode different types of inferences (below are from Takubo 2001: 1003-1006, cited on pp. 123-124):

- (13) *darou*-type: inferences based on evidence; in conditionals, they express inferences based on conditional knowledge. Hence, the antecedent clause is outside the scope of the *darou*-type modals.
- (14) *youda*-type: in conditionals, they express a conclusion that is itself a conditional statement. Hence the antecedent clause is within the scope of the *youda*-type modals.

The idea is illustrated in the bracketing of the following examples:

- (15) a. *(moshi) kouteibuai-ga 1%-mo ag-areba, tousan su-ru kigyuu-ga*
 MOSHI discount.rate-NOM 1%-ADD rise-BA bankrupt do-NPST company-NOM
de-ru {kamoshirenai / hazu da / darou}.
 come.out-NPST might should NPST.COP be.probable
 'If the discount rate rises by even 1%, {there might be / there should be / there will probably be} companies that go bankrupt.' ((51), p. 121, (55), p. 123)
- b. *[[(moshi) kouteibuai-ga 1%-mo ag-areba,] tousan su-ru kigyuu-ga de-ru {kamoshirenai / hazu da / darou} .]*
- (16) a. *(moshi) kouteibuai-ga 1%-mo ag-areba, tousan su-ru kigyuu-ga*
 MOSHI discount.rate-NOM 1%-ADD rise-BA bankrupt do-NPST company-NOM

de-ru {*youda* / *rashii* / *souda*}.

come.out-NPST seem seem seem

‘It seems that if the discount rate rises by even 1%, there will be companies that go bankrupt.’ ((52), p. 121, (56), p. 123)

- b. [[(*moshi*) *kouteibuai-ga* 1%-*mo* *ag-areba*, *tousan* *su-ru* *kigyō-ga de-ru*] {***youda*** / ***rashii*** / ***souda***}].

The contrast between epistemic and evidential modals in epistemic conditionals can be explained based on Takubo’s idea. In (12), the antecedent signals a supposition that the hearer is scheduled to travel for work next Wednesday. From this supposition, one is able to infer what would follow, as expressed by the epistemic modal *darou* in the consequent. However, the evidential modal *rashii* would require the supposition to be part of the conclusion, specifically, as the supporting evidence for the concluded statement that people will prepare the documents this week; but this cannot be the case, since the truthfulness/falseness of the supposition is not known.

Special case: Past predictive conditionals

There are conditionals that have an antecedent describing a past state of affairs, yet have the conditional connective *ba* with the verb in its eventive form:

- (17) My father sent me quite a lot of money to pay for tuition and books, so...

sorera-o ryōkūhiyō-ni ate-reba, gakusei toshite-wa zeitakuna ryōkō-ga deki-ta.
those-ACC travel.fee-DAT use-BA student as-TOP luxurious travel-NOM can-PST

‘If I use the money for travel, I could have a luxurious trip.’ ((66), p. 125)

The fact that the antecedent describes a settled proposition seems to suggest that the sentence is an epistemic conditional, but the acceptability of *ba* with an eventive verb seems to suggest that the sentence is a predictive conditional.

We analyze examples like (17) as predictive conditionals. (18) shows that (17) shares with predictive conditionals in terms of the unacceptability of evaluative adverbs *saiwai* ‘fortunately’ and *douse* ‘after all’. Recall that these adverbs require complete tensed clauses:

- (18) {**saiwai* / **douse*} *sorera-o ryōkūhiyō-ni ate-reba, gakusei toshite-wa zeitakuna*
fortunately after.all those-ACC travel.fee-DAT use-BA student as-TOP luxurious
ryōkō-ga deki-ta.
travel-NOM can-PST
‘If, {fortunately / after all}, I use the money for travel, I could have a luxurious trip.’ ((68), p. 125)

The acceptability of evidential modality in the consequent also suggest that (17) patterns with predictive conditionals rather than epistemic conditionals:

- (19) *sorera-o ryōkūhiyō-ni ate-reba, gakusei toshite-wa zeitakuna ryōkō-ga deki-ta*
those-ACC travel.fee-DAT use-BA student as-TOP luxurious travel-NOM can-PST

youda.

seem

‘It seems that if I use the money for travel, I could have a luxurious trip.’ ((69), p. 125)

Subsequently, the antecedent of (17) expresses an *unsettled* proposition, despite describing a past state of affairs. This can be analyzed as the result of an operation that shifts the evaluation time from the actual time to a past time.

Chapter 10 Counterfactual conditionals in Japanese

December 26, 2021

Antecedents of counterfactual conditionals

As argued in earlier chapters, counterfactual conditionals share with epistemic conditionals the property of having antecedents that express settled propositions. Hence, we expect the antecedents of counterfactuals to be complete tensed clauses, which is borne out.

Antecedent propositions that are presumed to be settled (and false)

We have seen that in epistemic conditionals, when the antecedent proposition is presumed to be settled, incomplete tensed clauses are not allowed. In contrast, counterfactuals seem to be less restricted. As shown in (1), both complete tensed clauses (led by *nara*) and incomplete tensed clauses (led by *ba* and *tara*) are allowed.

- (1) [Nakata has retired. So, he will not play in the next game with Australia.]

nakata-ga shutsujou {*a. su-ru nara* / *b. shi-ta nara* / *c. su-reba* / *d. shi-te i-reba* /
Nakata-NOM join do-NPST NARA do-PST NARA do-BA do-CONT be-BA
e. shi-tara / *f. shi-te i-tara*}, *zensen-e-no pasu-ga sumuuzu-ni toor-u*
do-TARA do-CONT be-TARA front-DAT-GEN pass-NOM smooth-DAT go-NPST
darou ga, ima-no chuuban-no menbaa dake-de-wa muzukashi-i na.
be.probable but now-GEN midfield-GEN member only-LOC-TOP difficult-NPST SFP

‘If Nakata had played, the pass to the front might have go smoothly, but that’s quite difficult given the current members of the midfield.’ (p. 132, (9)-(11))

In particular, between (1a) vs. (1b). (1c) vs. (1d) and (1e) vs. (1f), there is no difference in terms of how probable the antecedent proposition is. In other words, they do not attest the difference between English (2a) vs. (2b).

- (2) a. Used either when it is unsettled whether Boris will come or when it is settled that he will not come:
If Boris came tomorrow, Olga would be happy.
b. Used only when it is settled that Boris will not come:
If Boris had come tomorrow, Olga would be happy. (p. 131, (3)-(4))

No nara cannot be used in such environments. (3a) is allowed only if it is still possible that Nakata plays in the game (i.e. if the sentence is an epistemic conditional). (3b) is not allowed.

- (3) [Nakata has retired. So, he will not play in the next game with Australia.]

nakata-ga shutsujou {a. *#su-ru no nara* / b. **shi-ta no nara*}, *zensen-e-no*
 Nakata-NOM join do-NPST FIN NARA do-PST FIN NARA front-DAT-GEN
pasu-ga sumuuzu-ni toor-u darou ga, ima-no chuuban-no menbaa
 pass-NOM smooth-DAT go-NPST be.probable but now-GEN midfield-GEN member
dake-de-wa muzukashi-i na.
 only-LOC-TOP difficult-NPST SFP

‘If Nakata had played, the pass to the front might have go smoothly, but that’s quite difficult given the current members of the midfield.’ (p. 133, (12))

Antecedents propositions that are narrow-sense settled (to be false)

When the antecedent proposition is a narrow-sensed settled proposition, *nara* is allowed regardless of the (non-)stativity of the predicate in the antecedent. In contrast, *no nara* is not allowed.

- (4) [Why didn’t I give money to my son?...]

ano toki okane-o {a. *watashi-ta nara* / b. *watashi-te i-ta nara* /
 that time money-ACC give-PST NARA give-CONT be-PST NARA
 {c. **watashi-ta no nara* / d. **watashi-te i-ta no nara* / e. *a-ta nara* / f. **a-ta*
 give-PST FIN NARA give-CONT be-PST FIN NARA be-PST NARA be-PST
no nara}, *musuko-ga manbiki su-ru koto-wa nakat-ta darouni.*
 FIN NARA son-NOM shoplift do-NPST thing-TOP NEG-PST be.probable.CF

(a)-(d): ‘If I had given my son money back then, he probably would not have ended up shoplifting.’

(e)-(f): ‘If my son had had money back then, he probably would not have ended up shoplifting.’ (p. 134, (16)-(17), p. 135, (19)-(20))

Recall that in Ch. 8, we have seen that *no nara* also cannot be used in predictive conditionals, unlike *nara*. We thus conclude that *no nara* is an epistemic marker; specifically, it marks the status of the antecedent proposition as a settled proposition whose truth value is not known by the speaker. Note that apart from *no nara*, epistemic markers with similar functions have been identified in Japanese. Takubo (1989) argues that the topic marker *tte* indicates referents that (i) are introduced by the addressee and (ii) cannot be identified by the speaker:

- (5) A: *kinou tanaka-ni at-ta yo.*
 yesterday Tanaka-DAT meet-PST PST
 ‘I met Tanaka yesterday.’
 B: *tanaka-tte doitsu.*
 Tanaka-TTE who
 ‘Who’s Tanaka?’

(p. 137)

A similar phenomenon comes from the the *so*-series and *a*-series of demonstratives. For identifiable referents, *a*-series demonstratives or bare proper names like *Tanaka* can be used. For unidentifiable referents, *so*-series demonstratives are used, and bare proper names are not allowed:

- (6) A: *keiribu-no tanaka da yo.*
 accounting.department-GEN Tanaka COP SFP
 ‘It’s the Tanaka from the accounting department.’
 B: *aa, ano hito ne. wakat-ta. de, tanaka-ga dou shi-ta no?*
 ah that person SFP see-PST then Tanaka-NOM how do-PST FIN
 ‘Ah, that Tanaka, I see. What about him?’
 B’: *chotto wakar-ana-i naa. keiribu-to-wa amari*
 a.little understand-NEG-NPST SFP accounting.department-and-TOP not.so.much
*kouryuu-ga na-i shi. de, {sono tanaka / *tanaka}-ba dou shi-ta*
 communication-NOM NEG-NPST and so that Tanaka Tanaka-NOM how do-PST
no?
 FIN
 ‘I don’t know. I haven’t talked much with the people from the accounting department.
 So, what about him?’ (p. 137)

Similarly to what we have seen for epistemic conditionals, when the antecedent is a narrow-sensed settled proposition, *ba* and *tara* are allowed only with stative predicates:

- (7) [Why didn’t I give money to my son?...]
ano toki okane-o {a. ??watas-eba / b. ??watashi-tara / c. watashi-te i-reba /
 that time money-ACC give-BA give-TARA give-CONT be-BA
d. watashi-te i-tara / e. a-reba / f. at-tara}, musuko-ga manbiki su-ru koto-wa
 give-CONT be-TARA be-BA be-TARA son-NOM shoplift do-NPST thing-TOP
nakat-ta darouni.
 NEG-PST be.probable.CF
 (a)-(d): ‘If I had given my son money back then, he probably would not have ended up shoplifting.’
 (e)-(f): ‘If my son had had money back then, he probably would not have ended up shoplifting.’
 (13)-(15)) (p. 134,

As we have argued repeatedly, both epistemic conditionals and counterfactual conditionals have antecedents that express settled propositions. The fact that *ba* and *tara* are allowed in epistemic and counterfactual conditionals with the stative *-tei*-forms suggests that *-tei* carries settledness itself. The reason of this has to do with the semantics of *-tei*, or more specifically, the semantics of *-te*. Ogiwara (1998) argues that the continuative form *-te* has a [+perfect] and a [-perfect] variant. The [+perfect] variant can be disambiguated in cases like (8b), where a specific order restriction is required regarding the two actions described in the sentence:

- (8) a. No order restriction between the two actions:

tarou-wa uwagi-o nugi, hangaa-ni kake-ta.

Taro-TOP coat-ACC take.off hanger-DAT hang-PST

‘Taro took off the coat and hung it on a hanger.’

- b. Taking off the coat must be finished before hanging the coat:

tarou-wa uwagi-o nugi-de, hangaa-ni kake-ta.

Taro-TOP coat-ACC take.off-DAT hanger-DAT hang-PST

‘Taro took off the coat and hung it on a hanger.’ (Kuno 1973: 122, cited on p. 143, (42))

The [-perfect] variant is exemplified in (9):

- (9) *tarou-ga toukyou-ni it-te, hanako-ga oosaka-ni it-ta.*

Taro-NOM Tokyo-DAT go-CONT Hanako-NOM Osaka-DAT go-PST

‘Taro went to Tokyo and Hanako went to Osaka.’ (Kuno 1973, cited on p. 143, (43))

Ogihara (1998) further argues that the distinction between the two types of *-te* is supported by the distribution of time adverbials. Specifically, [-perfect] *-te* does not co-occur with time adverbials, whereas the [+perfect] *-te* does:

- (10) a. (*sakunen*) *koutsuu-jiko-de hyakunin-ga shin-de i-ru.*

last.year traffic.accident-LOC 100.people-NOM die-CONT be-NPST

‘Last year, 100 people have died of traffic accidents.’

(p. 142, (41))

- b. (**sakunen*) *hito-ga asoko-de taore-te i-ru.*

last.year person-NOM there-LOC fall-CONT be-NPST

‘Last year, someone (fainted and) was lying over there.’

(p. 142 (39))

We argue that in incomplete tensed antecedent clauses that express settled propositions like (7), the *-te* in *-tei* is what Ogihara (1998) analyzes as the [+perfect] *-te*.

Consequents of counterfactual conditionals

Expressions like *darouni* or *noni* can indicate the counterfactual status of the sentence unambiguously, but they are not obligatory:

- (11) *ano toki iw-are-ru mama-ni okane-o watashi-te i-tara, musuko-wa*

that time say-PASS-NPST just.as-DAT money-ACC give-CONT be-TARA son-TOP

manbiki nante shi-nakat-ta {a. Ø / b. noni / c. darou / d. darouni}.

shoplift such.thing do-NEG-PST CF be.probable be.probable.CF

(a)-(b): ‘If I had given my son money as I was told back then, he would not have ended up shoplifting.’

(c)-(d): ‘If I had given my son money as I was told back then, he probably would not have ended up shoplifting.’ (p. 145, (53))

In counterfactuals without modal auxiliaries that indicate counterfactuality like (11a) and (11c), it has been pointed out by Ogihara (2005) that the *-ta*-form of the verb in the consequent indicates the counterfactual status of the sentence unambiguously. See below for one more example:

- (12) *moshi tarou-ga ashita ki-tara, hanako-wa yorokon-da darou.*
 supposedly Taro-NOM tomorrow come-TARA Hanako-TOP be.glad-PST be.probable
 ‘If Taro came tomorrow, Hanako would have been glad.’ (p. 146, (55))

However, consequents of counterfactuals do not have to contain *-ta*-forms:

- (13) *kimi-ga ashita-no jugyou-ga kyuukou da tte koto*
 you-NOM tomorrow-GEN class-GEN cancel COP.NPST COMPL thing
shir-ase-te kure-nakat-tara, boku ashita {it-te i-ru yo / ik-u
 inform-CAUS-CONT give-NEG-TARA I tomorrow go-CONT be-NPST SFP go-NPST
tokoro da yo / it-ta yo}.
 TOKORO COP.NPST SFP go-PST SFP
 ‘If you hadn’t let me know of the cancellation of the class tomorrow, I would have gone to the class.’ (p. 146-147, (57)-(59))

Another marker that can indicate counterfactuality unambiguously is *tokoro da*. It will be discussed in more details in the next chapter.

Chapter 11 Settledness in conditional antecedents and modality in conditional consequents

December 29, 2021

Topic of this section: How the settledness of antecedent propositions relates to the modality in the consequents.

Event modality in consequents

The restriction on event modalities in conditional consequents is that in counterfactuals, the modals must appear in the past *-ta*-form. The volitional modal *-you* does not have a *-ta*-form, and is not allowed in counterfactuals.

Dynamic modality

When the consequent contains a volitional modal, the antecedent provides the condition under which the volitional speech act expressed by the modal comes into effect. So, in the predictive conditional (1a) and the epistemic conditional (1b), the speech act comes into effect when the antecedent proposition gets verified. However, in the counterfactual conditional (1c), the antecedent proposition is known by the speaker to be false, and is thus unable to provide the condition for the speech act expressed by the volitional modal.

(1) a. Predictive conditionals

ashita ooyuki-de basu-ga futsuu-ni nat-tara kenkyuukai-wa
tomorrow heavy.snow-LOC bus-NOM stop-DAT become-TARA workshop-TOP
chuushi-ni shi-you.
cancel-DAT do-VOL

‘If the bus stops because of the heavy snow, let’s cancel the workshop.’

b. Epistemic conditionals

ooyuki-de basu-ga futsuu-ni nat-ta no nara kenkyuukai-wa
heavy.snow-LOC bus-NOM stop-DAT become-PST FIN NARA workshop-TOP
chuushi-ni shi-you.
cancel-DAT do-VOL

‘If the bus stopped because of the heavy snow, let’s cancel the workshop.’

c. Counterfactual conditionals

#ooyuki-de basu-ga futsuu-ni nat-te i-tara kenkyuukai-wa
 heavy.snow-LOC bus-NOM stop-DAT become-CONT be-TARA workshop-TOP
 chuushi-ni shi-you.
 cancel-DAT do-VOL

‘If the bus had stopped because of the heavy snow, let’s cancel the workshop.’

(p. 150, (1)-(3))

When the consequent of a counterfactual conditional contains an ability modal, the modal needs to appear in the past *-ta*-form. This can be explained as follows. Event modals, including ability modals, express the speaker’s attitude towards some future state of affairs. Such attitudes cannot be supported by propositions that are known to be false at the utterance time, as we have shown in (1c). However, at some point earlier than the utterance time, it was possible for the attitude to come into effect. This is why the past form *-ta* is required in (2c). Predictive and epistemic conditionals are not subject to such restriction.

(2) a. Predictive conditionals

ashita basu-ga ugoi-tara, yotei-doori kenteishiken-o kaisai su-ru
 tomorrow bus-NOM move-TARA plan-just.as certificate.exam-ACC hold do-NPST
 koto-ga **deki-ru**.
 thing-NOM can-NPST

‘If the bus runs tomorrow, we will be able to have the certification exam as scheduled.’

b. Epistemic conditionals

[I wonder if the bus still running now. Let’s check with the bus company...]

basu-ga ugoi-te i-tara, yotei-doori kenteishiken-o kaisai su-ru
 bus-NOM move-CONT be-TARA plan-just.as certificate.exam-ACC hold do-NPST
 koto-ga **deki-ru**.
 thing-NOM can-NPST

‘If the bus is still running, we are able to have the certification exam as scheduled.’

c. Counterfactual conditionals

[The bus stopped operating because of the heavy snow...]

basu-ga ugoi-te i-tara, kenteishiken-o kaisai su-ru koto-ga
 bus-NOM move-CONT be-TARA certificate.exam-ACC hold do-NPST thing-NOM
 {#**deki-ru** / **deki-ta**}.
 can-NPST can-PST

‘If the bus had still been running, we would have been able to have the certification exam.’

(p. 151 (4), (6), (7))

Deontic modality

Similarly, when the consequent of a counterfactual conditional contains a deontic modal, the modal needs to appear in the past *-ta*-form. Predictive and epistemic conditionals are not subject to such constraint.

(3) a. Predictive conditionals

basu-ga futsuu-ni nat-tara, takushii-o tsukat-te-mo i-i.
 bus-NOM stop-DAT become-TARA taxi-ACC use-CONT-ADD good-NPST
 ‘If the bus stops, we are allowed to take the taxi.’

b. Epistemic conditionals

[That’s a lot of snow. Let’s ask the bus company whether the bus is still running...]

moshi basu-ga futsuu-ni nat-te i-tara, takushii-o tsukat-te-mo
 supposedly bus-NOM stop-DAT become-CONT be-TARA taxi-ACC use-CONT-ADD
i-i des-u.
 good-NPST COP.POL-NPST

‘If the bus has stopped, we are allowed to take the taxi.’

c. Counterfactual conditionals

[Please don’t worry. The snow is heavy, but the bus is still operating...]

*moshi basu-ga futsuu-ni nat-te i-tara, takushii-o {*tsukat-te-mo*
 supposedly bus-NOM stop-DAT become-CONT be-TARA taxi-ACC use-CONT-ADD
i-i / tsukat-te-mo yokat-ta n} des-u ga.
 good-NPST use-CONT-ADD good-NPST FIN COP.POL-NPST but

‘If the bus had stopped, we would still be allowed to take the taxi, though.’

(p. 152-153, (9), (13), (15))

Propositional modality in consequents

Epistemic modality

Epistemic modals are allowed in predictive and epistemic conditionals. In counterfactual conditionals, either the verb in the consequent proposition or the epistemic modal itself needs to appear in their *-ta*-form.

(4) a. Predictive conditionals

go-satsu ijou chuumon shi-tara, ichiwari-biki-ni nar-u kamoshirena-i.
 5-CL over order do-TARA 10%-off-DAT become-NPST might-NPST
 ‘If they order more than five copies, they might get 10% off.’

b. Epistemic conditionals

[I wonder whether the students have ordered the textbooks on their own. Let’s check...]

moshi tomodachi-to soudan shi-te go-satsu ijou chuumon shi-tara,
 supposedly friend-with discuss do-CONT 5-CL over order do-TARA
ichiwari-biki-ni nar-u kamoshirena-i.
 10%-off-DAT become-NPST might-NPST

‘If they asked their friends and ordered more than five copies, they might get 10% off.’

c. Counterfactual conditionals

[I didn’t ask my friends and ordered only one copy of the textbook for myself...]

moshi go-satsu ijou chuumon shi-te i-tara, ichiwari-biki-ni
 supposedly 5-CL over order do-CONT be-TARA 10%-off-DAT

{**nar-u* *kamoshirena-i* / *nat-ta* *kamoshirena-i* / *nar-u*
 become-NPST might-NPST become-PST might-NPST become-NPST
kamoshirenakat-ta / *nat-ta* *kamoshirenakat-ta* }.
 might-PST become-PST might-PST

‘If I had ordered more than five copies, I might have gotten 10% off.’

(p. 154, (16), (19), (21), (23)).

Evidential modality

Evidential modals are allowed in predictive conditionals. Epistemic conditionals and counterfactual conditionals both have settled antecedent propositions, and are both degraded when the consequents contain evidential modals.

(5) a. Predictive conditionals

ashita ame-ga fur-eba, shiai-wa chuushi-ni nar-u {*you da* /
 tomorrow rain-NOM fall-BA game-TOP cancel-DAT become-NPST EVID COP.NPST
sou da / *rashi-i*}.
 EVID COP.NPST EVID-NPST

‘It seems that if it rains tomorrow, the game will be canceled.’

b. Epistemic conditionals

[I wonder how the bargaining went yesterday. Let’s ask the organizers...]

**moshi kinou kin’ippuu-ga de-te i-tara, kyou-wa minna*
 supposedly yesterday bonus-NOM release-CONT be-TARA today-TOP everyone
nomi-ni ik-u sou da.
 drink-DAT go-NPST EVID COP.NPST

‘It seems that if the bonus was released yesterday, everyone will go for a drink today.’

c. Counterfactual conditionals

[The bargaining yesterday didn’t work out, and we didn’t manage to make them release the bonus..]

kinou kin’ippuu-ga de-te i-tara, kyou-wa minna nomi-ni
 yesterday bonus-NOM release-CONT be-TARA today-TOP everyone drink-DAT
it-ta {??*you da* / ?*sou da*}.
 go-PST EVID COP.NPST EVID COP.NPST

‘It seems that if the bonus had been released yesterday, everyone would have gone for a drink today.’

(p. 156, (30), (32)-(34))

As discussed in Ch. 8, predictive conditionals ‘*p ba, q*’ express the speaker’s inference about the relation between *p* and *q* based on the implicit prerequisite constituted by part of her actual knowledge at the utterance time (see Ch. 6 for implicit prerequisite). The evidential modal in the consequent indicates that the speaker has evidence in the actual world that supports such inference. By contrast, in epistemic conditionals ‘*p ba, q*’, the implicit prerequisite is updated with the addition of the antecedent proposition; such conditionals thus express the speaker’s inference about whether *q* holds based on the *updated* implicit prerequisite. But evidential modals are supposed to indicate inferences based on the speaker’s knowledge *in the actual world*; this is impossible if the

modal is in an epistemic conditional, since the conditional will have to express inferences about *q* based on the *updated* implicit prerequisite instead of the speaker's actual knowledge. This is why evidential modals cannot be used in epistemic conditionals. The degradedness of evidential modals in counterfactuals can be explained similarly.

However, (5c) shows that evidential modals are more acceptable in counterfactuals than in epistemic conditionals. The reason of this is that counterfactuals make statements about the actual world by describing states of affairs in counterfactual worlds; e.g. (5c) can be interpreted as 'if the bonus was released yesterday in the actual world, then we would have evidence that everyone goes for a drink'.

The counterfactual interpretation of *tokoro da* in consequents

As pointed out by Takubo (1993) and Takubo and Sasaguri (2002), the expression *tokoro da* (lit. place COP) in conditional consequents must be interpreted counterfactually. This is shown in (6a)-(6c).

- (6) a. Predictive conditionals
**ashita ame-ga fur-eba, shiai-wa chuushi-ni nar-u tokoro da.*
 tomorrow rain-NOM fall-BA game-TOP cancel-DAT become-NPST place COP.NPST
 'If it rains tomorrow, the game will be canceled.'
- b. Epistemic conditionals
**kinou ame-ga fut-ta no nara, shiai-wa chuushi-ni nat-ta tokoro da.*
 yesterday rain-NOM fall-PST FIN NARA game-TOP cancel-DAT become-PST place
 COP.NPST
 'If it rained yesterday, the game was canceled.'
- c. Counterfactual conditionals
 [It was sunny yesterday...]
kinou ame-ga fut-te i-reba, shiai-wa chuushi-ni nat-ta tokoro da.
 yesterday rain-NOM fall-CONT be-BA game-TOP cancel-DAT become-PST place
 COP.NPST
 'If it had rained yesterday, the game would have been canceled.' (p. 159, (37)-(39))

Tokoro da has an aspectual use:

- (7) a. *ima-kara kaer-u tokoro da.*
 now-from go.back-NPST place COP.NPST
 'I am just about to leave now.'
- b. *ima densha-ni not-te i-ru tokoro da.*
 now train-DAT ride-CONT be-NPST place COP.NPST
 'I am on the train now.'
- c. *ima kaet-ta tokoro da.*
 now go.back-PST place COP.NPST

‘I have just returned.’

(p. 160, (40)-(42))

In these sentences, *tokoro* expresses different stages of the relevant action—the beginning of the action (7a), the middle of the action (7b), and the end of the action (7c). The copula *da* expresses that the utterance time (the ‘now’) has the property expressed by *tokoro*; e.g. in (7a), that the utterance time has the property that it is at the beginning of a leaving action, and in (7b), that the utterance time has the property that it is in the middle of a being-on-the-train action.

Similarly, in conditional consequents, *tokoro da* expresses that the utterance time at a hypothetical world has the property described by the consequent proposition. For instance, (6c) says that in a counterfactual world where it rained yesterday, the utterance time has the property that the game was canceled. In contrast, the property of the hypothetical world described in (6a)—that the game gets canceled—is not a property of the utterance time. Hence, *tokoro da* cannot be used. The infelicity of *tokoro da* in (6b) can be explained in the same way.

Chapter 12 The compositionality of the meaning of *tewa*-sentences

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Main point of this chapter: Japanese *tewa*-sentences mainly have two uses, the conditional use (1) and the non-conditional use (2). The two uses differ in terms of the tensed status of the *te*-clauses and the scope of *wa*, as summarized in the table below. The main argument of this chapter is that the meaning of the two uses of *tewa* can be derived compositionally based on the meaning of *te*, the scope of *wa* and the meaning of *wa* (i.e. *X-wa Y* expresses the application of the predicate *Y* to *X*).

- (1) Conditional *tewa*
tomodachi-o uragit-te-wa shin'you sare-naku nar-u.
 friend-ACC betray-CONT-TOP trust PASS-NEG become-NPST
 'If you betray a friend, you will no longer be trusted.' (p. 164, (1))
- (2) Non-conditional (repetitive) *tewa*
tomodachi-o uragit-te-wa sute-ta.
 friend-ACC betray-CONT-TOP throw.away-NPST
 'I betrayed and left my friends (repetitively).' (p. 164, (2))

	Conditional <i>tewa</i>	Non-conditional <i>tewa</i>
Syntactic property of <i>te</i> -clauses	incomplete tensed	untensed
Scope of <i>wa</i>	only <i>te</i> -clauses	<i>te</i> -clause + main clause

Table 1: p. 184

Properties of *tewa*-sentences to be captured

Observation A: Negative connotation

It has been noted that under the conditional use, *tewa*-clauses give rise to the negative connotation such that the state of affairs expressed by the *tewa*-clause is something that one has tried to avoid (Hasunuma 1987 a.o.). However, this does not have to be the case. In (3), the sunset being beautiful does not need to be a negative state of affairs:

- (3) *konnani yuuyake-ga utsukushiku-te-wa, ashita-mo ii o-tenki darou.*
 such sunset-NOM beautiful-CONT-TOP tomorrow-ADD good HON-weather be.probable
 ‘If the sunset is so beautiful, the weather will probably be great tomorrow.’
 (Kawabata 1958: 51, cited in p. 165, (9))

Observation B: Negative connotation outside *tewa*-sentences

It is also possible for conditional forms other than *tewa* to give rise to negative connotation.

- (4) [You should not go...]
kimigaga it-tara minnaga meiwaku suru.
 you-NOM go-TARA everyone-NOM bother do-NPST
 ‘If you go, people will feel bothered.’ (p. 165, (10))

Observation C: Repetitive meaning

As shown in (2), the non-conditional use can express that the relevant action occurs repetitively. But this also does not have to be the case. The actions expressed by the *tewa*-clauses in (5) (going to the reception desk, going on the elevator) do not need to have occurred repetitively:

- (5) *uketsukejou-no tokoro-e it-te-wa maigo-no furi-o shi-te*
 reception.lady-GEN place-DAT go-CONT-TOP lost-GEN pretend-ACC do-CONT
karakai... erebeetaa-ni not-te-wa kaku-kai-no botan-o oshi...
 make.fun.of elevator-DAT ride-CONT-TOP each-floor-GEN button-ACC press
 ‘He went to the reception desk, and pretended that he was lost to make fun of the reception ladies... And he went on the elevator and pressed the buttons of all floors...’
 (modified from p. 166, (12))

Observation D: Lack of generic use

The non-conditional *tewa* cannot receive generic interpretations:

- (6) *tomodachi-o uragit-te-wa sute-ru.*
 friend-ACC betray-CONT-TOP throw.away-NPST
 ‘He/She betrays and leaves his/her friends (repetitively).’
 Not ‘One betrays and leaves one’s friends (repetitively).’ (p. 166, (17))

Note that *to* contrasts with *tewa* in this respect:

- (7) *tomodachi-o uragir-u to, itsuka jibun-mo uragir-are-ru.*
 friend-ACC betray-NPST TO some.day self-ADD betray-PASS-NPST
 ‘If one betrays a friend, one will be betrayed some day, too.’ (p. 166, fn. 2 (b))

Dichotomy of *tewa*-sentences in terms of syntactic structures

We will use four tests to show that conditional *tewa*-clauses are incomplete tensed clauses and non-conditional *tewa*-clauses are untensed clauses.

	Untensed clauses	Non-conditional <i>tewa</i>	Incomplete tensed clauses	Conditional <i>tewa</i>
Independent subject	*	*	✓	✓
Scope of <i>mo</i>	inside	inside	outside	outside
<i>teiru</i> -form	*	*	✓	✓
Restricted temporal order	*	(*)	✓	✓

Table 2: p. 172

Test 1: Scope of *mo*

Untensed and incomplete tensed clauses exhibit different scope interactions with the focus particle *mo* ‘also’. Consider the interaction between untensed *nagara*-clauses with *mo*:

(8) Untensed *nagara*-clauses

wa-ga ko-no me-o mitsume-nagara naki-mo shi-ta.

I-GEN child-GEN eye-ACC look.into-while cry-ADD do-PST

‘While I was looking into my child’s eyes, I also cried.’

(p. 168 (20b))

In this sentence, *mo* can associate with either the verb phrase *naki* ‘cry’, or the constituent including the *nagara*-clause.

- (9) a. *wa-ga ko-no me-o mitsume-nagara [naki]-mo shi-ta.*
 implies: Besides crying, I also did things like being angry while looking into my child’s eyes.
- b. *[wa-ga ko-no me-o mitsume-nagara naki]-mo shi-ta.*
 implies: Besides crying while looking into my child’s eyes, I also did things like sighing deeply while thinking about my child’s future.

In contrast, incomplete tensed *ba*-clauses cannot be in the scope of *mo*:

(10) Incomplete tensed *ba*-clauses

kiyomi-wa chichioya-ni shika-rare-reba naki-mo su-ru.

Kiyomi-TOP father-DAT scold-PASS-BA cry-ADD do-NPST

‘If Kiyomi is scolded by her father, she also cries.’

(p. 168 (21b))

- (11) a. *kiyomi-wa chichioya-ni shika-rare-reba [naki]-mo suru.*
 implies: Besides crying, Kiyomi also gets upset if her father scolds her.
- b. **[kiyomi-wa chichioya-ni shika-rare-reba naki]-mo suru.*
 implication unavailable: Besides crying when her father scolds her, Kiyomi also cries when she’s in trouble.

The following test shows that non-conditional *tewa*-clauses pattern with untensed clauses, while conditional *tewa*-clauses pattern with incomplete tensed clauses.

- (12) Non-conditional *tewa*
kiyomi-wa shorui-no taba-o yabut-te-wa horinage-mo shi-ta.
 Kiyomi-TOP paper-GEN bundle-ACC tear-CONT-TOP throw-ADD do-NPST
 ‘Kiyomi tore up the documents and also threw them everywhere.’ (p. 169 (24b))
- (13) a. *kiyomi-wa shorui-no taba-o yabut-te-wa [horinage]-mo shi-ta.*
 implies: Besides throwing them everywhere, Kiyomi also stepped on the documents after tearing them up.
 b. *[kiyomi-wa shorui-no taba-o yabut-te-wa horinage]-mo shi-ta.*
 implies: Besides tearing up the documents and throwing them everyone, Kiyomi also cried out loud and spoke loudly.
- (14) Conditional *tewa*
daijina shigoto-o sabot-te-wa, risutora sare-mo su-ru.
 important job-ACC skip-CONT-TOP be.fired PASS-ADD do-NPST
 ‘If you slack off when you have important work, you’ll also get fired.’ (p. 169 (25b))
- (15) a. *daijina shigoto-o sabot-te-wa, [risutora sare]-mo su-ru.*
 implies: Besides getting fired, your pay will also be cut when you slack off.
 b. **[daijina shigoto-o sabot-te-wa, risutora sare]-mo su-ru.*

Test 2: Independent subjects

We have seen in Sec 3 that untensed clauses do not allow the realization of a subject that is independent of the subject of the main clause:

- (16) Untensed *nagara*-clauses
**watashi-wa musuko-ga terebi-o mi-nagara gohan-o tabe-ta.*
 I-TOP son-NOM TV-ACC see-while meal-ACC eat-PST
 Intended: ‘I ate while my son was watching TV.’ (p. 169 (26))
- (17) Incomplete tensed *ba*-clauses
watashi-wa musuko-ga monku-o i-eba daigaku-o yasum-u tsumori
 I-TOP son-NOM complaint-ACC say-BA university-ACC rest-NPST plan
da.
 COP.NPST
 ‘I plan to have a day off (from the university) if my son complains.’ (p. 169 (27))

Again, non-conditional *tewa* parallels with untensed clauses, while conditional *tewa* parallels with incomplete tensed clauses:

- (18) Non-conditional *tewa*
**tarou-wa hanako-ga shorui-o yabut-te-wa hourinage-ta.*
 Taro-TOP Hanako-NOM paper-ACC tear-CONT-TOP throw-PST
 Intended: ‘Hanako tore up the documents and Taro threw them everywhere.’ (p. 170 (30b))

- (19) Conditional *tewa*
hoka-no menbaa-wa, kimi-ga sossen shi-te yar-anaku-te-wa, kyouryoku
 other-GEN member-TOP you-NOM take.the.lead do-CONT do-NEG-CONT-TOP cooperate
shi-na-i yo.
 do-NEG-NPST SFP
 ‘If you don’t take the lead, other members will not cooperate.’ (p. 170 (31b))

Test 3: Aspect *teiru*

Untensed clauses do not allow the aspectual marker *teiru*, whereas incomplete tensed clauses do:

- (20) Untensed *nagara*-clauses
**terebi-o mi-te i-nagara gohan-o tabe-ta.*
 TV-ACC see-CONT be-while meal-ACC eat-PST
 ‘I ate while I was watching the TV.’ (p. 170 (28))
- (21) Incomplete tensed *ba*-clauses
soko-de mat-te i-reba, dare-ka-ga ku-ru hazu des-u.
 there-LOC wait-CONT be-BA who-Q-NOM come-NPST should COP.POL-NPST
 ‘If you keep waiting there, someone should come to you.’ (p. 170 (29))

In this respect, too, non-conditional *tewa* parallels with untensed clauses, while conditional *tewa* parallels with incomplete tensed clauses:

- (22) Non-conditional *tewa*
**hanako-wa shorui-o yabut-te i-te-wa hourinage-ta.*
 Hanako-TOP paper-ACC tear-CONT be-CONT-TOP throw-PST
 Intended: ‘Hanako was tearing up the documents and threw them everywhere.’ (p. 171 (32))
- (23) Conditional *tewa*
kimi bakari yat-te i-te-wa, hoka-no menbaa-wa nani-mo yar-ana-i
 you only do-CONT be-CONT-TOP other-GEN member-TOP what-ADD do-NEG-NPST
yo.
 SFP
 ‘If it’s only you that has been doing it, other members won’t do it.’ (p. 171 (33))

Test 4: Temporal order

Similarly to incomplete tensed clauses (cf. Ch. 3), conditional *tewa*-clauses describe states of affairs that temporally precede the states of affairs described by the main clauses:

- (24) Conditional *tewa*
 a. *konna kusuri-o non-de-wa kibun-wa yoku nar-ana-i yo.*
 such medicine-ACC drink-CONT-TOP feeling-TOP good become-NEG-NPST SFP
 ‘If you take this kind of medicine, you won’t get better.’

- b. **ashita yasun-de-wa kyou-wa hayaku kaer-e-na-i.*
 tomorrow rest-CONT-TOP today-TOP quickly return-can-NEG-NPST
 ‘If you take a day of tomorrow, you can’t leave early today.’ (p. 171 (34)-(35))

With unconditional *tewa*, the sentence often expresses that the states of affairs described by the *tewa*-clause and the main clause happen repetitively. The fact that (25a) and (25b) have the same interpretation and can be uttered in a sequence suggests that there is no restriction regarding the temporal order between the states of affairs described by the *tewa*-clause and the main clause:

- (25) Unconditional *tewa*
- a. *chigit-te-wa nage...*
 tear-CONT-TOP throw
 ‘I keep tearing them up and throwing them away...’
- b. *nage-te-wa chigiri...*
 throw-CONT-TOP tear
 ‘I keep throwing them away and tearing them up...’ (p. 171 (36)-(37))

The *te*-construction

This section shows that there are two types of *te*-clauses, one that expresses an action preceding the action described by the main clause, and one that expresses the causes, reasons or the setting of the state of affairs described by the main clause. As summarized in the table below, the former type of *te*-clauses resemble untensed clauses, thus corresponding to the *te* in non-conditional *tewa*-clauses; the latter type resemble incomplete tensed clauses, thus corresponding to the *te* in conditional *tewa*-clauses.

	Actions that precede the main-clause actions	Causes, reasons, settings
Independent subject	*	✓
Scope of <i>mo</i>	inside	outside
<i>teiru</i> -form	*	✓

Table 3: p. 175

Te-clauses can express an action that precedes the action described by the main clause, the reason, cause, or the setting for the state of affairs described by the main clause, and a state of affairs that contrasts with the main clause:

- (26) Actions that precede the main-clause actions
jon-wa kuruma-o azuke-te sugu-ni kaet-ta.
 John-TOP car-ACC leave-CONT soon-DAT return-PST
 ‘John asked someone to take care of his car and left right away.’ (p. 172 (38))
- (27) Reasons
jon-ga ki-te biru-ga kaet-ta.
 John-NOM come Bill-NOM leave-PST
 ‘Because John came, Bill left.’ (p. 172 (39))

- (28) Causes
jon-ga kuruma-o kat-te minna yorokon-da.
 John-NOM car-ACC buy-CONT everyone be.happy-PST
 ‘Because John bought a car, everyone was pleased.’ (p. 173 (47))
- (29) Setting
hi-ga kure-te ame-ga fut-te ki-ta.
 sun-NOM sunset-CONT rain-NOM fall-CONT come-PST
 ‘When the sun set, it started to rain.’ (p. 173 (49))
- (30) Contrast
ao-shingou-de hito-ya kuruma-wa susun-de, aka-de teishi su-ru.
 green-traffic.light-LOC people-and car-TOP go-CONT red-LOC stop do-NPST
 ‘People and cars go at the green lights, stop at the red lights.’ (p. 173 (48))

The following tests show that *te*-clause behave differently when they describe actions that precede the main-clause actions and when they describe reasons, causes and settings. First, consider the test of whether *te*-clauses allow independent subjects:

- (31) Actions that precede the main-clause actions
**takuya-wa ootoo-ga doa-o tatai-te oogoe-de saken-da.*
 Takuya-TOP brother-NOM door-ACC knock-CONT loud-LOC scream-PST
 Intended: ‘Takuya’s brother knocked on the door loudly and Takuya screamed out loud.’ (p. 173 (50b))

In contrast, we have seen in (27)-(29) that independent subjects are possible in *te*-clauses that describe reasons, causes and settings.

Te-clauses can be in the scope of the focus particle *mo* when they describe actions that precede the main-clause actions, but not when they describe causes, reasons, or settings.

- (32) Actions that precede the main-clause actions
- takuya-wa doa-o tatai-te oogoe-de sakebi-mo shi-ta.*
 Takuya-TOP door-ACC knock-CONT loud-LOC scream-ADD do-PST
 ‘Takuya knocked on the door loudly and also screamed out loud.’ (p. 173 (51a))
 - takuya-wa [doa-o tatai-te oogoe-de sakebi]-mo shi-ta.*
 implies: Besides knocking on the door loudly and screaming out loud, Takuya also went to the back door to see what happened inside.
- (33) Cause
- chichi-ga koukyuusha-o kat-te minna odoroki-mo shi-ta.*
 father-NOM luxurious.car-ACC buy-CONT everyone be.surprised-ADD do-PST
 ‘Because my father bought a fancy car, everyone was also surprised.’ (p. 173 (52a))
 - *[chichi-ga koukyuusha-o kat-te minna odoroki]-mo shi-ta.*
 implication unavailable: Besides being surprised when my father bought a fancy car, everyone also felt bothered when my father borrowed money from unreliable places.
- (34) Reason

- a. *tomodachi-ga keisatsu-ni tsukamat-te, keisatsu-ni mimoto-o hikiuke-ni*
 friend-NOM police-DAT be.caught-CONT police-DAT identity-ACC guarantee-DAT
iki-mo shi-ta.
 go-ADD go-PST
 ‘Because my friend was caught by the police, I also went to the police as a guarantor.’
- b. *[*tomodachi-ga keisatsu-ni tsukamat-te, keisatsu-ni mimoto-o hikiuke-ni iki*]-mo
 shi-ta. (p. 174 (53a))
- (35) Setting
- a. *kuraku nat-te, ame-ga furi-mo shi-ta.*
 dark become-CONT rain-NOM fall-ADD do-PST
 ‘When it got dark, it also rained.’ (p. 174 (54a))
- b. *[*kuraku nat-te, ame-ga furi*]-mo shi-ta.

Te-clauses do not allow the aspectual *teiru*-form when they describe actions that precede the main-clause actions, in contrast to when they describe causes, reasons and settings:

- (36) Actions that precede the main-clause actions
 **takuya-wa doa-o tatai-te i-te oogoe-de saken-da.*
 Takuya-TOP door-ACC knock-CONT be-CONT loud-LOC scream-PST
 Intended: ‘Takuya was knocking on the door loudly and screamed out loud.’
 (p. 173 (51b))
- (37) Cause
chichi-ga koukyuusha-o kat-te i-te minna odoroi-ta.
 father-NOM luxurious.car-ACC buy-CONT be-CONT everyone be.surprised-PST
 ‘Because my father has bought a fancy car, everyone was surprised.’ (p. 174 (52c))
- (38) Reason
tomodachi-ga keisatsu-ni tsukamat-te i-te, keisatsu-ni mimoto-o
 friend-NOM police-DAT be.caught-CONT be-CONT police-DAT identity-ACC
hikiuke-ni it-ta.
 guarantee-DAT go-PST
 ‘Because my friend has been caught by the police, I went to the police as a guarantor.’
 (p. 174 (53b))
- (39) Setting
kuraku nat-te i-te, hitokko hitori mie-nakat-ta.
 dark become-CONT be-CONT people one.person can.see-NEG-PST
 ‘When it has gotten dark, I didn’t see one single person.’ (p. 174 (54b))

As a side note, *te*-clauses that indicate contrast appears to differ from *te*-clauses that express reasons, causes and setting. For instance, contrastive *te*-clauses cannot contain interrogative words:

- (40) a. Contrast
 **dare-ga nagasaki-ni it-te, jirou-wa fukuoka-ni nokot-ta no?*
 who-NOM Nagasaki-DAT go-CONT Jiro-TOP Fukuoka-DAT remain-PST Q
 Intended: ‘Who went to Nagasaki and Jiro wen to Fukuoka?’

b. Cause

dare-ga jiko-ni at-te yamada-ga byouin-ni tsure-te
 who-NOM accident-DAT meet-CONT Yamada-NOM hospital-DAT take-CONT
it-ta no?
 go-PST Q

lit. ‘Who is it such that because he got involved in an accident, Yamada took him to the hospital?’

c. Reason

dare-ni shikar-are-te besoo kai-te i-ru no?
 who-DAT scold-PASS-CONT crying.face-ACC do-CONT be-NPST Q
 ‘Who is it such that because he was mad at you, you were about to cry?’

d. Setting

nanji-ni nat-te modot-ta no?
 what.time-DAT become-CONT go.back-PST Q
 ‘What time did you leave?’ (lit. When it became what time did you leave?)

(p. 175 fn. 7)

An analysis of these uses of *te* can be found in Arita (1999b).

The scope of *wa*

The marker *wa* typically expresses contrast. But the constituent under contrast does not have to be the one that *wa* attaches to. For instance, in (41), *wa* attaches to the adverb *hayaku* ‘fast’, but what is under contrast is the whole verb phrase *hayaku hashiru* ‘run fast’. In other words, the scope of *wa* in this sentence is the whole VP, as shown in (42).

- (41) *hayaku-wa hashit-ta ga, taimu-wa tot-te i-na-i.*
 fast-TOP run-PST but time-TOP take be-NEG-NPST
 ‘I did run fast, but did not measure the time.’ (p. 177 (57))

- (42) [_{XP} [_{VP} **hayaku hashiru**] *wa*] *ta*

Now, we can understand the difference between conditional and non-conditional *tewa* based on the scope of *wa*. That is, in non-conditional *tewa*-sentences, where the *tewa*-clauses are untensed, *wa* takes the whole VP of the main clause as its scope. In conditional *tewa*-sentences, where the *tewa*-clauses are incomplete tensed, *wa* takes its surface scope.

(43) Non-conditional *tewa*

- a. *tomodachi-o uragit-te-wa sute-ta.*
 friend-ACC betray-CONT-TOP throw.away-NPST
 ‘I betrayed and left my friends (repeatedly).’
 b. [_{IP} [_{XP} [_{VP} **tomodachi-o uragit-te sute**] *wa*] *ta*] (p. 177 (60))

(44) Conditional *tewa*

- a. *tarou-ga it-te-wa minna-ga komar-u.*
 Taro-NOM go-CONT-TOP everyone-NOM bother-NPST

- ‘If Taro goes, people will feel bothered.’
 b. [_{XP} [_{IP} **tarou-ga it-te**] *wa*] *minna-ga komar-u.* (p. 177 (61))

The meaning of *wa*

In a sentence ‘*X-wa Y*’, the predicate *Y* can be either an event predicate (45a) or a property predicate (45b).

- (45) a. *tarou-wa ki-ta.*
 Taro-TOP come-PST
 ‘Taro came.’
 b. *tarou-wa shinsetsu da.*
 Taro-TOP kind COP.NPST
 ‘Taro is kind.’ (p. 178, (62b), (63b))

However, *wa* is not allowed when the referent of *X* is not determined.

- (46) a. **dare-ka-wa i-ru.*
 who-Q-TOP be-NPST
 Intended: ‘Someone is here.’
 b. **dare-wa ki-ta no?*
 who-TOP come-PST Q
 Intended: ‘Who came?’ (p. 178, (65)-(68))

To felicitously use *wa* on an expression *X* whose referent is undetermined, it is necessary to explicitly indicate a set of individuals that include the referent of *X*. In such cases, *wa* is interpreted contrastively. For instance, (46a) is acceptable under the interpretation ‘someone among the people that we are considering came’. (46b) is acceptable if formed as follows (Miyagawa 1987):

- (47) Among the people that we are considering...
 dare-wa ki-te, dare-wa ko-nakat-ta no?
 who-TOP come-CONT who-TOP come-NEG-PST Q
 ‘Who came and who didn’t?’ (p. 179 (70))

We thus suggest that to use ‘*X-wa*’, the speaker must know the definitional property of *X*, i.e. the property that can help determine the referent of *X*:

- (48) The meaning of ‘*X-wa Y*’
 a. Apply the predicate *Y* to *X*.
 b. The definitional property of *X* is known in discourse.

Hence, the sentence (43) means roughly that the action chain ‘betraying and leaving a friend’ has the property expressed by *ta*, i.e. that the action chain occurs in some time interval in the past. Whether the action chain occurred only once or repetitively depends on the nature of the actions and the discourse. This is how the non-conditional interpretation of *tewa* arises.

In contrast, in (44), the scope of *wa* is *tarou-ga itte* ‘Taro goes’, i.e. the part of the action chain that indicates the reason of the following action, i.e. that everyone feels bothered. The sentence

roughly means that the event that Taro goes has the property of leading to the result that people feel bothered. In other words, the sentence expresses a generic causal relation between Taro's going and everyone's feeling bothered. This is how the conditional interpretation of *tewa* arises.

Explaining (the lack of) repetitive meanings

Recall that we have seen in (5) that the actions described by *tewa*-clauses do not need to have occurred repetitively. We can explain this with the condition on the use of *wa* given in (48b). In (5), (48b) requires the context to provide a definitional property for the actions described by the scope of *wa*, i.e. going to the reception desk and pretending to be lost, and going on the elevator and pressing the buttons of all floors. We can assume that the definition property is 'being a prank that one does at a department store'. The actions that satisfy this property are not required to occur repetitively.

Explaining (the lack of) negative connotation

We have seen in (3) that conditional *tewa*-clauses do not always give rise to negative connotation. We suggest that what conditional *tewa* expresses is not negative connotation, but rather contrastiveness, as part of the meaning of *wa*. The *tewa*-clause in (3) describes an unusual state of affairs, i.e. the sunset being so beautiful, thus indicating a contrast with usual states of affairs, where sunset is not as beautiful as today's. This point can also be shown by the contrast between (49a) and (49b). In (49a), the state of affairs described by the *tewa*-clause cannot be understood as something unusual; in (49b), we can easily contrast the state of affairs described by the *tewa*-clause with usual states of affairs, where one does not move to new places hurriedly.

- (49) a. **sengetsu hikkoshi-o nasat-te-wa, ooisogashi deshou.*
 last.month move-ACC do.HON-CONT-TOP very.busy be.probable.POL
 'If you have just moved to a new place last month, you must be very busy.'
- b. *annani awate-te hikkoshi shi-te-wa, ie-no naka-wa muchakucha*
 that hurry-CONT move do-CONT-TOP house-GEN inside-TOP messy
deshou.
 be.probable.POL
 'If you have moved to a new place so hurriedly, the place must be very messy.'
 (p. 182, (81), (82))

The reason that conditional *tewa* tends to give rise to negative connotation comes from the interaction between the meaning of *te* and the meaning of *wa*. First, *te*-clauses indicate states of affairs that *inevitably/directly* lead to the states of affairs described by the main clause. Second, we have said above that *wa* expresses contrastiveness; e.g. *A-wa* typically indicates a contrast of *A* with $\neg A$. Now, in a sentence '*A-te-wa B*', an easy way to contrast *A* with $\neg A$ is to express that the inevitable consequence of *A*, namely *B*, is undesirable; in such cases, the speaker suggests that we should avoid *A*, which is the cause of the undesirable consequence *B*. This is how the negative connotation arises.

When contrastiveness (expressed by *wa* in *tewa*-conditionals) and the inevitability of the main-clause states of affairs (expressed by *te* in *tewa*-sentences) are provided by the context, conditional constructions other than *tewa* may also give rise to negative connotation. This is what we saw in (4).

The category of *tewa*-conditionals

Since *tewa*-conditional clauses are incomplete tensed clauses, they can appear in predictive conditionals:

- (50) *kimi-ga it-te-wa minna-ga meiwaku su-ru.*
 you-NOM go-CONT-TOP everyone-NOM bother do-NPST
 ‘If you go, people will feel bothered.’ (p. 184, (83))

However, unlike incomplete tensed *ba*-clauses and *tara*-clauses, in epistemic conditionals, *tewa* does not require the stative forms of the verbs:

- (51) *annani awate-te hikkoshi shi-te-wa, ie-no-naka-wa muchakucha*
 that hurry-CONT move do-CONT-TOP house-GEN-inside-TOP messy
deshou.
 be.probable.POL
 ‘If you have moved to a new place so hurriedly, the place must be very messy.’
 (p. 182, (81), (82))

As argued earlier, for *ba* and *tara*, the [+perfect] *te* in the stative *teiru*-form is needed to turn the antecedent propositions into settled propositions, so as to serve as the antecedents of epistemic conditionals. *Tewa* does not require this extra [+perfect] *te*, because a [+perfect] *te* is contained in *tewa* already.

Another difference between *tewa* vs. *baltara* is that in epistemic conditionals like (51), *tewa*-antecedents always express propositions that the speaker knows to be true. Also, conditional *tewa* cannot be used in counterfactuals.

Hanazono, Satoru. 1999. ‘Zyōkenkei fukugō yōgen no nintei [Identifying complex conditional predicates]’. *Kokugogaku [Studies in the Japanese Language]*, vol. 197, 39–53.

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This paper explores ‘complex conditional predicates’ (or conditional predicates), which refer to modal expressions that appear in the form of ‘conditional form + evaluative predicates’, such as *shinakereba naranai* (lit. ‘if you don’t do it, it doesn’t become’, roughly translated to ‘you must do it’) and *suru to ii* (lit. ‘if you do it, it’s good’, roughly translated to ‘you can do it’). We show that the elements that make up conditional predicates have lost their own properties, and argue that conditional predicates are independent, fossilized expressions. In addition, we show that depending on the meaning and the environments where they appear, different conditional predicates exhibit different degrees of fossilization.

1 A case study of *-nakereba naranai* ‘must’

1.1 The conditional clause

The first evidence comes from co-occurrences of multiple conditional clauses. In principle, a conditional form cannot appear twice in a sentence, as shown in (1):

- (1) *?ashita ame-ga fut-tara, gakkou-ga yasumi dat-tara, kimi-no ie-ni ik-u yo.*
 tomorrow rain-NOM fall-TARA school-NOM day.off be-TARA you-GEN home-DAT
 go-NPST SFP
 ‘If it rains tomorrow, if there’s no school, I’ll go to your place.’ (p. 41 (4))

Multiple conditional clauses are allowed if different conditional expressions are used (e.g. (2)), or if two conditional clauses marked by the same conditional expression share the same main clause (e.g. (3)).

- (2) *pasokon-o kai-takat-tara, yamadasan-ni i-eba, waribikiken-o kure-ru yo.*
 computer-ACC buy-want-TARA Yamada-DAT say-BA coupon-ACC give-NPST SFP
 ‘If you want to buy a computer, if you talk to Yamada, he will give you coupons.’ (p. 102 (5))
- (3) *ippanshoku josei-no naka-ni-wa, kekkon shi-tara, aruiwa kodomo-ga umare-tara shigoto-wa yame-ru beki da to kangae-ru hito-ga sukunaku-na-i.*
 general.job woman-GEN inside-DAT-TOP marry do-TARA or child-NOM be.born-TARA job-TOP quit-NPST should COP.NPST C think-NPST people-NOM few-NEG-NPST
 ‘Quite a lot of people think that women who do general office work should quit their jobs if they get married or if they have children.’ (p. 41 (6))

However, *-nakereba naranai* ‘must’ can co-occur with another *reba*-clause, as in (4):

- (4) *mukou-de yar-eba, touzen hoomutaun deshijon-wa kakugo shi-nake-reba*
 there-LOC do-BA of.course hometown decision-TOP be.prepared do-NEG-BA
nar-ana-i.
 become-NEG-NPST
 'If we do it over there, of course we will have to be prepared for hometown decisions.'
 (p. 42 (7c))

Second, if a conditional clause retains its conditional meaning, it should be able to co-occur with adverbs that express hypotheticality, such as *moshi*, *moshimo* and *man'ichi*.¹ However, *-nakereba naranai* 'must' cannot co-occur with these adverbs:

- (5) ?{*moshi / karini / man'ichi*} *kimi-wa ik-anake-reba nar-ana-i.*
 supposedly supposedly in.case you-TOP go-NEG-BA become-NEG-NPST
 'You must go.'
 (p. 42 (10))

Note that *-nakereba dame da* 'if not...it's bad' and *-nakereba komaru* 'if not...I'm troubled', whose meaning is similar to that of *-nakereba naranai*, are allowed to co-occur with the above adverbs:

- (6) *moshi, kimi-ga ik-anake-reba {dame da / komar-u}.*
 supposedly you-NOM go-NEG-BA bad COP.NPST be.troubled-NPST
 'If you don't go, {it's bad / I'll be troubled}.'
 (p. 42 (9))

Third, as a grammatical category, conditional forms exhibit the opposition between the positive and the negative, as in (7):

- (7) *moshi ame-ga {fur-eba / fur-anake-reba}, ashita-no undoukai-wa*
 supposedly rain-NOM fall-BA fall-NEG-BA tomorrow-GEN sports.day-TOP
chuushi da.
 cancel COP.NPST
 'If it {rains / doesn't rain}, the sports day will be canceled tomorrow.'
 (p. 42 (11))

However, the conditional clause in *-nakereba naranai* 'must' only has a negative form:

- (8) *gakkou-e {?ik-eba / ik-anake-reba} nar-ana-i.*
 school-DAT go-BA go-NEG-BA become-NEG-NPST
 'If you {go to school / don't go to school}, it won't work.'
 (p. 42 (12))

1.2 The consequent clause

First, elements that can stand alone as phrases themselves can be omitted in coordination constructions, such as the consequent clause of (9). This is not allowed in *-nakereba naranai*, as in

¹Of course, there are circumstances when *reba* and *moshi* cannot co-occur, for instance, in conditionals that express habits or universal truths:

- (i) ?*moshi haru-ga ku-reba, hana-ga sak-u.*
 supposedly spring-NOM come-BA flower-NOM bloom-NPST
 'Flowers bloom if spring comes.'
 (p. 51, footnote 6)

(10).

- (9) *ashita ame-ga fur-eba gakkou-ga yasumi-ni nar-u.* *moshikuwa*
 tomorrow rain-NOM fall-BA school-NOM day.off-DAT become-NPST or
suto-ga a-reba gakkou-ga yasumi-ni nar-u.
 strike-NOM be-BA school-NOM day.off-DAT become-NPST
 ‘If it rains, or if there is a strike, there will be no school tomorrow.’ (p. 43 (14a))
- (10) *?shigotojou-de misu-o shi-ta baai, jibun-de kaiketsusaku-o kangae-nake-reba*
 work-LOC miss-ACC do-PST case self-LOC solution-ACC think-NEG-BA
nar-ana-i. *moshikuwa tadachini joushi-ni houkoku shi-nake-reba*
 become-NEG-NPST or immediately superior-DAT report do-NEG-BA
nar-ana-i.
 become-NEG-NPST
 lit. ‘In case one makes a mistake at work, if one doesn’t think of a solution himself or if
 one does not report to his superior immediately, it’s bad.’ (p. 43 (14b))

Second, in epistemic modals like *nichigainai* ‘must’ and *kamoshirenai* ‘might’ (in which the elements that make up them have lost their own properties), the negated part cannot be replaced with their positive counterpart:

- (11) a. *ni chigai na-i* / **ni chigai a-ru*
 DAT mistake NEG-NPST DAT mistake be-NPST
 b. *ka mo shir-e-na-i* / *ka mo shir-e-ru*
 Q all know-can-NEG-NPST Q all know-can-NPST

Similarly, the consequent clause of *-nakereba naranai* cannot be replaced with its positive counterpart:

- (12) *gakkou-e ik-anake-reba {?nar-u / nar-ana-i}.*
 school-DAT go-NEG-BA become-NPST become-NEG-NPST
 lit. ‘If you don’t go to school, {it becomes / it doesn’t become}.’ (p. 43 (15))

A note on the historical development of *naranai*: *naranai* in *-nakereba naranai* is not the negative form of the verb *naru* ‘become’, but rather expresses means prohibition. In the historical development of this use of *naranai*, the word had never had a positive counterpart (Yuzawa 1936², etc.).

1.3 *-nakereba naranai* as a fossilized expression

We have shown above that both *-nakereba* and *naranai* in *-nakereba naranai* have lost their own morphological and syntactic properties. We now show that *-nakereba naranai* has become a fossilized expression. First, the order of a conditional and a main clause can be reversed:

²Yuzawa, Kōkichi. 1936. *Tokugawa Jidai Gengo no Kenkyū* [A Study on the Language of the Tokugawa Era]. Tōkō Shoin.

- (13) *undoukai-wa chuushi da ne, ashita ame-ga fur-eba.*
 sports.day-TOP cancel COP.NPST SFP tomorrow rain-NOM fall-BA
 ‘The sports day will be canceled, if it rains tomorrow.’ (p. 44 (17b))

However, this is not possible with *-nakereba naranai*:

- (14) *?nar-ana-i, ashita made-ni kono shigoto-o shiage-nak-ereba.*
 become-NEG-NPST tomorrow until-DAT this job-ACC finish-NEG-BA
 lit. ‘It’s bad, if I don’t finish this work today.’ (p. 44 (18b))

Second, it is impossible to insert elements in between *-nakereba* and *naranai*:

- (15) *?gakkou-e ik-anak-ereba zettaini nar-ana-i.*
 school-DAT go-NEG-BA definitely become-NEG-NPST
 lit. ‘If you don’t go to school, it’s definitely bad.’ (p. 44 (19))

Third, as a fossilized predicate, *-nakereba naranai* expresses obligation and necessity. These meanings cannot be derived based on the meaning of the conditional and the consequent clause. In particular, *-nakereba naranai* can co-occur with expressive adverbs like *zahi* and *zhitomo* ‘at all costs’:

- (16) *zhitomo shimekiri made-ni kansei sase-nak-ereba nar-ana-i.*
 at.all.costs deadline until-DAT finish do.CAUS-NEG-BA become-NEG-NPST
 ‘You must finish it before the deadline at all costs.’ (p. 44 (20))

Importantly, neither *-nakereba* nor *naranai* can co-occur with *zhitomo* ‘at all costs’:

- (17) a. *?zhitomo kenkyuushitsu-ni it-te mi-reba, sensei-ni a-e-ru yo.*
 at.all.costs office-DAT go-CONT try-BA teacher-DAT meet-can-NPST SFP
 lit. ‘At all costs, if you go to the office, you can meet the teacher.’
 b. *?zhitomo nar-ana-i.*
 at.all.costs become-NEG-NPST
 ‘At all costs, it’s bad.’ (p. 44 (21))

2 Applications to other forms

The above tests can be applied to the other complex modals, such as *-nakereba ikenai* ‘must’ (lit. if not...it doesn’t work), *-tara ii* ‘should’ (lit. if...it’s good) and *-reba ii* ‘should’ (lit. if...it’s good). The results are summarized in Table 1 on the last page.

2.1 The conditional clause

First, the above modals allow the co-occurrence of the same conditional form:

- (18) *moshi oomiya-ni man'ichi-no koto-de-mo a-reba, tamakazura-mo*
 supposedly Omiya-DAT the.worst-GEN thing-FIN-ADD be-BA Tamakazura-ADD
fukumo shi-nak-ereba ik-e-na-i.
 mourning do-NEG-BA go-can-NEG-NPST
 'If anything should happen to Omiya, Tamakazura has to go into mourning, too.'
 (p. 44 (22a))
- (19) [I'm not sure if people will understand what I say...]
wakar-anaku nat-te kyoushitsu-ga konran shi-tara dou shuushuu
 understand-NEG become-CONT classroom-NOM chaos do-TARA how handle
shi-tara yo-i no ka.
 do-TARA good-NPST FIN Q
 'If they don't understand and the whole class gets confused, what should I do?'
 (p. 44 (22b))
- (20) [If they have a spy on us, they should just find us...]
tsuke-tak-ereba tsuk-ereba i-i.
 follow-want-BA follow-BA good-NPST
 'If they want to follow us, they should do so.'
 (p. 44 (22c))

However, this test is not applicable to *-to ii* 'should' (lit. if...it's good), because the conditional form *-to* does not allow the main clause to express a command, a request, or a wish:

- (21) *okane-ga {a-reba / at-tara / ?a-ru to}, ka-inasai.*
 money-NOM be-BA be-TARA be-NPST TO buy-IMP
 'If you have money, buy it.'

Second, similarly to *-reba*, the conditional forms *-tara* and *-to* are allowed to co-occur with conditional adverbs. However, the complex modals that contain these conditional forms do not allow conditional adverbs:

- (22) a. ?{*moshi / karini / man'ichi*} *ik-anak-ereba ik-e-na-i.*
 supposedly supposedly in.case go-NEG-BA go-can-NEG-NPST
 'You have to go.'
- b. ?{*moshi / karini / man'ichi*} *ik-u to i-i.*
 supposedly supposedly in.case go-NPST TO good-NPST
 'You should go.'
- c. ?{*moshi / karini / man'ichi*} *it-tara i-i.*
 supposedly supposedly in.case go-TARA good-NPST
 'You should go.'
- d. ?{*moshi / karini / man'ichi*} *ik-eba i-i.*
 supposedly supposedly in.case go-BA good-NPST
 'You should go.'
 (p. 44 (24))

Third, just like *-nakereba naranai*, *-nakereba ikenai* does not allow its conditional clause to appear in a positive form:

- (23) *gakkou-e {?ik-eba / ik-anak-ereba} ik-e-na-i.*
 school-DAT go-BA go-NEG-BA go-can-NEG-NPST
 lit. 'If you {go to / don't go to} school, it doesn't work.' (p. 44 (25))

By contrast, *-to ii* allows a negative conditional clause if the sentence expresses a wish, as in (24). This is not possible if the sentence expresses a recommendation, as in (25).

- (24) [Playing hide-and-seek]
itsumademo mitsukar-ana-i to i-i na.
 forever find-NEG-NPST TO good-NPST SFP
 'I wish they never find me.' (lit. 'If they don't find me forever, it's good.') (p. 44 (25))
- (25) *?soko-wa kikenchitai da kara ik-ana-i to i-i yo.*
 there-TOP dangerous.place COP.NPST because go-NEG-NPST TO good-NPST SFP
 'It's a very dangerous place, so you should not go.' (lit. 'If you don't go, it's good.') (p. 44 (27))

-tara ii exhibits the same behavior: a negative conditional clause is allowed only under a 'wish' interpretation:

- (26) *douse nara, hajime-kara onasake-o kake-te itadak-anakat-tara*
 after.all NARA beginning-from compassion-ACC take-CONT give-NEG-TARA
yokat-ta noni.
 good-PST CF
 'If that's what is going to happen after all, it would have been good if they had not acted out of compassion.' (p. 44 (28))
- (27) *?ik-itaku-nakat-tara, ik-anakat-tara i-i yo.*
 go-want-NEG-TARA go-NEG-TARA good-NPST SFP
 'If you don't want to go, you should not go.' (p. 45 (29))

-reba ii behaves differently from the above modals in that it allows a negative conditional clause, and this is not restricted to 'wish' interpretations:

- (28) *shi-taku-nakya, shi-nakya i-i.*
 do-want-NEG.BA do-NEG.BA good-NPST
 'If you don't want to do it, you should not do it.' (p. 45 (30a))

This means that compared with the other modals, the conditional clause of *-reba ii* still retains its own independent properties to some degree.

Third, *-to* and *-tara* allow the verbs in the conditional clause to appear in their polite forms:

- (29) *ame-ga {fur-u to / fur-imas-u to / fut-tara / fur-imashi-tara}...*
 rain-NOM fall-NPST TO fall-POL-NPST TO fall-TARA fall-POL-TARA
 'If it rains...' (p. 45 (31))

However, polite forms are not allowed in complex modals:

- (30) *sensei-ni {kii-te mi-ru to i-i / ?kii-te mi-mas-u to yoroshuu}*
 teacher-DAT ask-CONT try-NPST TO good-NPST ask-CONT try-POL-NPST TO good

- gozaimas-u*}.
be.HON-NPST
'You should ask the teacher.'
(p. 45 (32a))
- (31) *koko-de {kat-te it-tara i-i / ?kat-te ik-imashi-tara yoroshuu*
here-LOC buy-CONT go-TARA go-NPST buy-CONT go-POL-TARA good
gozaimas-u}.
be.HON-NPST
'you should buy it here.'
(p. 45 (32b))

2.2 Consequent clause

First, what we have seen above with *-nakereba naranai* regarding the impossibility of omitting the consequent clause holds for the other modals, too:

- (32) *?shigotojou-de misu-o shi-ta baai, jibun-de kaiketsusaku-o kangae-nake-reba*
work-LOC miss-ACC do-PST case self-LOC solution-ACC think-NEG-BA
ik-e-na-i. *moshikuwa tadachini joushi-ni houkoku shi-nake-reba*
go-can-NEG-NPST or immediately superior-DAT report do-NEG-BA
ik-e-na-i.
go-can-NEG-NPST
lit. 'In case one makes a mistake at work, if one doesn't think of a solution himself or if one does not report to his superior immediately, it's bad.'
(p. 45 (33a))
- (33) *?sono koto nara, minsei iin-ni tanon-de mi-ru to i-i.*
that thing NARA welfare commissioner-DAT ask-CONT try-NPST TO good-NPST
moshikuwa chokusetsu shiyakusho-ni it-te mi-ru to i-i.
or directly city.hall-DAT go-CONT try-NPST TO good-NPST
lit. 'As for that, if you go to ask the welfare commissioners, or if you go to the city hall directly, it's good.'
(p. 45 (33b))
- (34) *?gimonten-wa toshokan-de shirabe-tara i-i.* *moshikuwa sensei-ni kii-te*
question-TOP library-LOC search-TARA good-NPST or teacher-DAT ask-CONT
mi-tara i-i.
try-TARA good-NPST
'For any questions, if you go to look it up at the library, or if you go to ask your teacher, it's good.'
(p. 45 (33c))
- (35) *?kamikoppu-wa marume-te sute-reba i-i.* *moshikuwa matome-te*
paper.cup-TOP make.round-CONT throw-BA good-NPST or gather-CONT
ok-eba i-i.
prepare-BA good-NPST
'If you roll up and throw away the paper cups, or if you collect them, it's good.'
(p. 45 (33d))

Second, when the consequent clauses of *-nakereba ikenai*, *-to ii* and *-reba ii* get replaced with their positive/negative counterparts, the sentences either become unacceptable, or do not express the opposite concepts of the original modals. For instance, in contrast to *-nakereba ikenai* 'must',

-nakereba ii (lit. ‘if not...it’s good’) expresses the appropriateness or the wish of a negative event, rather than the negation of the obligation ‘must’:

- (36) *gakkou-e ik-anak-ereba i-i.*
 school-DAT go-NEG-BA good-NPST
 ‘You should not go to school.’ / ‘I wish you don’t go to school.’ (p. 45 (34a))

Similarly, in contrast with *-to ii* ‘should’, which expresses recommendations, *-to ikenai* expresses apprehension, rather than the negation of recommendations:

- (37) *gakkou-e ik-u to ik-e-na-i.*
 school-DAT go-NPST TO go-can-NEG-NPST
 ‘It will be bad to go to school.’ (p. 45 (34b))

The same can be said for *-tara ii* ‘should’. Its counterpart with a negative consequent clause expresses the appropriateness of a negative event:

- (38) *ooguchi ake-te warat-tari, mushamusha tabe-tari shi-tara ik-e-na-i*
 big.mouth open-CONT laugh-TARI munch eat-TARI do-TARA go-can-NEG-NPST
n da ne.
 FIN COP.NPST SFP
 ‘It’d bad if you laugh with your mouth wide open and munch on your food.’ (p. 45 (35))

In addition, *-reba ii* does not have a counterpart where its consequent clause appears in its negative form:

- (39) *?gakkou-e ik-eba ik-e-na-i.*
 school-DAT go-BA go-can-NEG-NPST
 ‘If you go to school, it doesn’t work.’ (p. 45 (34c))

2.3 As fossilized expressions

First, *-to ii* and *-tara ii* both do not allow the inversion of the conditional and the main clause:

- (40) a. *sensei-ni kii-te mi-ru to i-i yo.*
 teacher-DAT ask-CONT try-NPST TO good-NPST SFP
 b. *?i-i yo, sensei-ni kii-te mi-ru to.*
 good-NPST SFP teacher-DAT ask-CONT try-NPST TO
 ‘You should ask the teacher.’ (p. 47 (36a))
- (41) a. *dat-tara otsousan-ni orei nasat-tara i-i wa.*
 COP-TARA father-DAT greet do.HON-TARA good-NPST SFP
 b. *?dat-tara, i-i wa, otsousan-ni orei nasat-tara.*
 COP-TARA good-NPST SFP father-DAT greet do.HON-TARA
 ‘You should great your father.’ (p. 47 (36b))

As for inversion in *-nakereba ikenai*, the judgements are subtle, as in (42). And for *-reba ii*, inversion is possible with certain verbs, such as *-teoku* ‘to prepare’ in (43).

- (42) a. *chanto benkyou shi-nakya ik-e-na-i yo.*
 well study do-NEG.BA go-can-NEG-NPST SFP
 b. *(?)ik-e-na-i yo, chanto benkyou shi-nakya.*
 go-can-NEG-NPST SFP well study do-NEG.BA
 ‘You have to study hard.’ (p. 47 (37a))

- (43) [If the portion is too big...]
 a. *nokosh-ite ok-eba i-i des-u yo.*
 leave-CONT prepare-BA good-NPST COP.POL-NPST SFP
 b. *i-i des-u yo, nokosh-ite ok-eba.*
 good-NPST COP.POL-NPST SFP leave-CONT prepare-BA
 ‘You should just leave it.’ (p. 47 (37c))

Second, it is unnatural to insert elements into *-nakereba ikenai*, although it is more acceptable compared with the insertion of elements into *-nakereba naranai*.

- (44) *?gakkou-e ik-anak-ereba zettaini ik-e-na-i.*
 school-DAT go-NEG-BA definitely go-can-NEG-NPST
 lit. ‘If you don’t go to school, it is definitely bad.’ (p. 47 (39))

As for *-reba ii*, *-tara ii* and *-to ii*, insertion appears to be acceptable:

- (45) *sake-ga ar-ya motto i-i.*
 alcohol-NOM be-BA more good-NPST
 ‘If there’s alcohol, it will be better.’ (p. 47 (40a))
 (46) [If there’s a ceiling fan in the room...]
sore-o tsuke-ppanashi-ni su-ru to yori i-i.
 that-ACC turn.on-keep-DAT do-NPST TO more good-NPST
 ‘If you keep it turned on, it will be better.’ (p. 47 (40b))

Note that *sorede* ‘then’ can be inserted in *-reba ii*:

- (47) *aji-mo kichinto totonot-te i-reba, sorede i-i.*
 taste-ADD well arrange-CONT be-BA then good-NPST
 ‘If the taste is also in a good balance, then it’s good.’ (p. 48 (41c))

It should be mentioned that the addition of the above adverbs leads to the collapse of the fossilization of the modal expressions. For instance, it makes the inversion of the conditional and the consequent clauses easier:

- (48) *motto i-i ne, sake-ga ar-ya.*
 more good-NPST SFP alcohol-NOM be-BA
 ‘It will be better, if there’s alcohol.’ (p. 47 (40’a))
 (49) *sorede i-i, aji-mo kichinto totonot-te i-reba.*
 then good-NPST taste-ADD well arrange-CONT be-BA
 ‘Then it’s good, if the taste is also in a good balance, then it’s good.’ (p. 48 (41’c))

Finally, similarly to *-nakereba naranai*, the complex modals can co-occur with expressive ad-

verbs:

- (50) *doushitemo kat-anak-ereba ik-e-na-i.*
by.all.means win-NEG-BA go-can-NEG-NPST
'I have to win by all means.' (p. 48 (42a))
- (51) *zehi doukou su-ru to yo-i.*
definitely go.together do-NPST TO good-NPST
'I definitely should go with you.' (p. 49 (42b))
- (52) *kongo-mo zehi tsuzui-tara i-i na to omo-imas-u.*
after-ADD definitely continue-TARA good-NPST SFP C think-POL-NPST
'I think it definitely should continue in the future.' (p. 49 (42c))
- (53) *mousukoshi, semete kono miya-ga motto shikkari nasar-u toshigoro-made*
a.bit.more at.least this man-NOM more strong be-NPST age-until
soba-ni i-te age-rare-reba i-i no da ga.
next.to-DAT be-CONT give-can-BA good-NPST FIN COP.NPST but
'At least you should stay with him until he becomes a grown-up a bit more.' (p. 49 (42d))

The above adverbs do not co-occur with the conditional clauses or the consequent clauses alone, similarly to what we have shown for *-nakereba naranai*. Therefore, the reason that the adverbs in the above examples are acceptable has to do with the fossilization of the modal expressions.

3 Summary

Based on the above observation and the summary in Table 1, the modals form a continuum in terms of their degree of grammaticalization in the following order: *-nakereba naranai*, *-nakereba ikenai*, *-to ii*, *-tara ii* and *-reba ii*. It still remains to be examined where the following combinations of conditional and consequent clauses locate on this continuum:

- (54) a. {*shi-nak-ereba* / *shi-nakat-tara* / *shi-na-i to*} {*dame da* /
do-NEG-BA do-NEG-TARA do-NEG-NPST TO bad COP.NPST
komar-u / *doushiyoumonai*}
be.troubled-NPST no.other.way
'If not..., {it's bad / I'll be troubled / there's no other way}.'
- b. {*su-reba* / *shi-tara* / *su-ru to*} {*kekkou da* / *juubun da* /
do-BA do-TARA do-NPST TO not.bad COP.NPST be.sufficient COP.NPST
ooke da / *owari da*}
okay COP.NPST the.end COP.NPST
'If..., {it's not bad / it's enough / it's ok / it's the end}.'

	<i>nakereba naranai</i>	<i>nakereba ikenai</i>	<i>suru to ii</i>	<i>tara ii</i>	<i>rel</i>
co-occurrence of same conditional forms	Yes	Yes	—	Yes	Y
unacceptability of conditional adverbs	Yes	Yes	Yes	Yes	Y
no positive/negative counterparts in conditional clauses	Yes	Yes	some cases	some cases	
unacceptability of polite forms	—	—	Yes	Yes	-
impossibility of omission	Yes	Yes	Yes	Yes	Y
no positive/negative counterparts in consequents	Yes	Yes	Yes	some cases	Y
impossibility of inversion	Yes	some cases	Yes	Yes	some
impossibility of insertion of conditional and consequent clauses	Yes	some cases	some cases	some cases	some
co-occurrence with expressive adverbs	Yes	Yes	Yes	Yes	Y

Table 1: p. 49; ‘—’ for not applicable

Hasunuma, Akiko. 1993. Tara to to no jijitsuteki yoho o megutte [On the factual use of *tara* and *to*]

In Masukoka Takashi (ed.), *Nihongo no Joken Hyogen [Conditional Expressions in Japanese]*, Kuroshio.

January 23, 2022

This paper revisits and generalizes the existing observations of the factual use of *tara* and *to*. It then proposes an account based on the information-structural properties of conditionals and the distinction between narrative discourse and conversational discourse.

Existing ideas

It has been observed by Kuno (1973)¹, Inoue (1979)² etc. that factual *tara* and *to* have the following properties. First, in dialogues, when the consequent expresses an event or action that can be controlled by the subject's own intentions, *tara* and *to* are degraded. This is shown by the contrast between (1) and (2):

- (1) *yuube gohan-o {*tabe-tara / ?tabe-ru to} terebi-o mi-mash-ita.*
last.night meal-ACC eat-TARA eat-NPST TO TV-ACC see-POL-PST
'Yesterday, after I had dinner, I watched TV.' (p. 76)
- (2) *futon-ni {hait-tara / hair-u to} sonomama guuguu nemut-te shimat-ta.*
bed-DAT enter-TARA enter-NPST TO directly sound sleep-CONT RESULT-PST
'After I went into bed, I fell into a sound sleep immediately.' (p. 76)

In narrative discourse, *tara* is still subject to this constraint, whereas *to* is not:

- (3) *boku-wa terebi-o {??keshi-tara / kes-u to} tonari-ni a-ru ani-ni*
I-TOP TV-ACC turn.off-TARA turn.off-NPST TO next.to-DAT be-NPST brother-DAT
heya-ni hairi...
room-DAT enter
'After I turned off the TV, I entered my brother's room next to me...' (p. 76, based on excerpt from Haruki Murakami, *Hear the Wind Sing*)

In addition, *to* requires the antecedent and the consequent to describe states of affairs that the speaker can observe objectively at the same spot (*bamen*). This can be illustrated by the infelicity of (4):

- (4) **ie-de ne-te i-ru to, izu-de jishin-ga okot-ta.*
home-DAT sleep-CONT be-NPST TO Izu-LOC earthquake-NOM occur-PST

¹Kuno, Susumu. 1973. Tara [*Tara*]. Toki o arawasu to [Temporally-interpreted *to*]. *Nihon Bunpo Kenkyu [A Study of the Japanese Grammar]*, Taishukan.

²Inoue, K. 1979. On conditional connectives. *Nihongo no Kihonkozo ni kan suru Rironteki Jisshoteki Kenkyu [A Theoretical and Empirical Study of the Basic Structure of Japanese]*, KAKEN Grants-in-Aid for Scientific Research.

‘When I was sleeping at home, an earthquake occurred in Izu.’ (p. 77)

Inoue and Kuno did not provide an explanation of these properties.

The five uses of factual *tara* and *to*

Under the discovery use, the antecedent and the consequent describe an eventive and a stative state of affairs, respectively:

(5) Discovery (*hakken*)

mado-o {*ake-tara* / *ake-ru to*} *eegekai-ga me-no mae-ni*
 window-ACC open-TARA open-NPST TO Aegean.Sea-NOM eye-GEN front-DAT
mi-e-ta.
 see-can-PST

‘When I opened the window, I was able to see the Aegean Sea.’ (p. 78 (4))

Under the manifestational use, the antecedent describes a stative state of affairs, whereas the consequent describes an eventive one:

(6) Manifestational (*hatsugen*)

mi-age-te {*i-tara* / *i-ru to*} *sora-kara saifu-ga fut-te ki-ta.*
 see-up-CONT be-TARA be-NPST TO sky-from wallet-NOM fall-CONT come-NPST

‘When I was looking up to the sky, a wallet fell.’ (p. 78 (5))

The temporal use expresses how new states of affairs come about or how the courses of events develop:

(7) Temporal (*toki*)

yoru-ni {*nat-tara* / *nar-u to*} *mizore-wa yuki-ni kawat-ta.*
 night-DAT become-DAT become-NPST TO sleet-TOP snow-DAT become-PST

‘When the night came, the sleet turned into rain.’ (p. 78 ((6))

The responsive use expresses how the action or change of state described by the consequent comes about in response to the antecedent state of affairs:

(8) Responsive (*han’ou*)

o-keshou-o {*shi-tara* / *su-ru to*} *kanojo-wa mi-chigae-ru hodo*
 HON-make.up-ACC do-TARA do-NPST TO she-TOP see-mistake-NPST degree
kirei-ni nat-ta.
 pretty-DAT become-PST

‘When she put on make up, she became so pretty that I could barely recognize her.’ (p. 78 (7))

The continuous use describes how an action taken by an agent is followed by another. As we have seen, *tara* does not allow this use:

(9) Continuous (*renzoku*)

*kare-wa ie-ni { *kaet-tara / kaer-u to } tomodachi-ni denwa shi-ta.*
 he-TOP home-DAT return-TARA return-NPST TO friend-DAT phone do-PST

‘When he returned home, he called his friend.’

(p. 78 (8))

All five uses are common in that the antecedent state of affairs provides the basis that the consequent state of affairs gets recognized. In other words, factual *tara* and *to* express that the speaker has come to recognize some states of affairs that have recently come about.

Issue

Having identified the shared properties between the factual use of *tara* and *to*, what remains to be explained are the cases where factual *tara* and *to* come apart.

- (10) *yuube gohan-o { *tabe-tara / ?tabe-ru to } terebi-o mi-mash-ita.*
 last.night meal-ACC eat-TARA eat-NPST TO TV-ACC see-POL-PST
 ‘Yesterday, after I had dinner, I watched TV.’ (= (1) of this summary)

- (11) *kare-wa ie-ni { *kaet-tara / kaer-u to } tomodachi-ni denwa shi-ta.*
 he-TOP home-DAT return-TARA return-NPST TO friend-DAT phone do-PST
 ‘When he returned home, he called his friend.’ (= (9) of this summary)

- (12) *aru toki, sakaguchisan-ga kare-no ie-e chichi-o shibor-u tokoro-o*
 some time Mr.Sakaguchi-NOM he-GEN home-DAT milk-ACC squeeze-NPST place-ACC
mise-te kure to it-te asobi-ni { ?it-tara / ik-u to }, odoriagar-u
 show-CONT give.IMP C say-CONT play-DAT go-TARA go-NPST TO jump.up-NPST
you-ni shi-te kare-wa yorokon-da.
 C-DAT do-CONT he-TOP be.happy-PST
 ‘When Mr. Sakaguchi once came to visit him and said ‘Show me how you milk the cows!’,
 he was so excited that he almost started dancing for joy.’

(p. 79 (11), based on Ango Sakaguchi, *Wind, Light, and the Twenty-Year-Old-Me*)

- (13) *donguri-wa korokoro { ??korogat-tara / korogar-u to } ike-ni ochi-ta.*
 acorn-TOP roll.over roll.over-TARA roll.over-NPST TO pond-DAT fall-PST
 ‘As the acorn rolled over and over, it fell into the pond.’ (p. 79 (12))

- (14) *ke-no seetaa-o sentakuki-de { arat-tara / ??ara-u to } ki-rare-naku*
 hair-GEN sweater-ACC washer-LOC wash-TARA wash-NPST TO wear-can-NEG
nat-ta.
 become-PST
 ‘After I washed the sweater in the washer, it became unwearable.’ (p. 79 (13))

Solution

To explain the cases where factual *tara* and *to* come apart, this paper proposes the following two hypotheses:

- (15) Hypothesis 1: Factual *tara* is used to express a relation between the antecedent and the consequent such that, on the basis of the state of affairs described by the former, the speaker comes to recognize the state of affairs described by the latter based on her actual experience. (p. 79)
- (16) Hypothesis 2: Factual *to* is used by the speaker from the viewpoint as an external observer to describe that the antecedent state of affairs leads to the consequent state of affairs or the speaker's recognition of the consequent state of affairs. (p. 80)

There are mainly three types of conditionals where temporal *tara* is unacceptable, discussed in (i)-(iii) below. The type of conditionals where *to* is unacceptable is discussed in (iv).

(i) When the consequent expresses volitional actions

When the antecedent and the consequent express the same subject's continuous volitional actions (the continuous *renzoku* use), *tara* is unacceptable:

- (17) *yuube gohan-o { *tabe-tara / ?tabe-ru to } terebi-o mi-mash-ita.*
 last.night meal-ACC eat-TARA eat-NPST TO TV-ACC see-POL-PST
 'Yesterday, after I had dinner, I watched TV.' (=1)/(10) of this summary
- (18) *kare-wa ie-ni { *kaet-tara / kaer-u to } tomodachi-ni denwa shi-ta.*
 he-TOP home-DAT return-TARA return-NPST TO friend-DAT phone do-PST
 'When he returned home, he called his friend.' (=9)/(11) of this summary

In addition, *tara* is degraded when, under the temporal (*toki*) use, the consequent expresses the speaker's volitional action:

- (19) *daigaku-ga yasumi-ni hait-tara { ?boku / kare }-wa nimotsu-o ryukku-ni*
 university-NOM break-DAT enter-TARA I he-TOP luggage-ACC backpack-DAT
tsume, yukigutsu-o hai-te kyouto-made dekake-ta.
 add snow.boots-ACC wear-CONT Kyoto-until go.out-PST
 'When the vacation started at the university, I packed my stuff into the backpack, put on my snow boots and went to Kyoto.' (p. 80 (14))

These examples can be explained by Hypothesis 1, because the consequents of the examples do not express things that the speaker can newly come to recognize based on her actual experience. Volitional actions of the speaker are things that the speaker planned beforehand, and thus do not constitute the speaker's new knowledge when they come about. The oddness of (17) and the use of first person in (19) thus follows. The use of third person in (19) is allowed because when the antecedent describes the action of a third-person subject, it can constitute new knowledge of the speaker.

Despite having a third-person subject, (18) is odd. The reason is perhaps that the consequent action follows too naturally from the antecedent action to constitute the speaker's new knowledge. When we replace the consequent with an action that unexpectedly follows from the antecedent, the sentence becomes much more acceptable:

- (20) *kare-wa heya-ni hait-tara, nani-o omot-ta no ka, youfuku-dansu-ya*
 he-TOP room-DAT room-TARA what-ACC think-PST FIN Q clothes-closet-and
hikidashi-o tsugitsugito ake, naka-no mono-o soto-ni
 drawer-ACC one.after.another open open inside-GEN thing-ACC outside-DAT
houridashi-hajime-ta n da yo.
 pull.out-start-PST FIN COP.NPST SFP
 ‘When he returned home, as if he had suddenly remembered something, he started to pull
 out things from the closet and the drawers one after another.’ (p. 81 (15))

However, (20) would be unacceptable if we replace the third-person subject with first person. Hence, the constraint that the consequent of factual *tara* cannot describe volitional actions is most effective when the speaker is the agent of the action.

(ii) When the subject of the antecedent is not the speaker

(21) is a sentence illustrating the responsive (*han'ou*) use of *tara* and *to*. The reason that *tara* is odd in this sentence is that it describes a third-person subject's (Mr. Sakaguchi's) experience without being marked as as quotation.

- (21) *aru toki, sakaguchisan-ga kare-no ie-e chichi-o shibor-u tokoro-o*
 some time Mr.Sakaguchi-NOM he-GEN home-DAT milk-ACC squeeze-NPST place-ACC
mise-te kure to it-te asobi-ni {?it-tara / ik-u to}, odoriagar-u
 show-CONT give.IMP C say-CONT play-DAT go-TARA go-NPST TO jump.up-NPST
you-ni shi-te kare-wa yorokon-da.
 C-DAT do-CONT he-TOP be.happy-PST
 ‘When Mr. Sakaguchi once came to visit him and said ‘Show me how you milk the cows!’,
 he was so excited that he almost started dancing for joy.’ (= (12) of this summary)

When quotative *to iu* (‘it is said’) is added, *tara* becomes acceptable:

- (22) *aru toki, sakaguchisan-ga kare-no ie-e chichi-o shibor-u tokoro-o*
 some time Mr.Sakaguchi-NOM he-GEN home-DAT milk-ACC squeeze-NPST place-ACC
mise-te kure to it-te asobi-ni it-tara, odoriagar-u you-ni shi-te
 show-CONT give.IMP C say-CONT play-DAT go-TARA jump.up-NPST C-DAT do-CONT
kare-wa yorokon-da to i-u.
 he-TOP be.happy-PST C say-NPST
 ‘It was said that when Mr. Sakaguchi once came to visit him and said ‘Show me how you
 milk the cows!’, he was so excited that he almost started dancing for joy.’ (p. 75, (1'))

The contrast can again be explained by Hypothesis 1, which requires the consequent to describe a state of affairs that follows from the antecedent state of affairs and the speaker comes to recognize based on her own experience. In (22), it is not the speaker that comes to recognize the consequent state of affairs, but the third-person subject. Since the sentence describes the third-person subject's private experience, the speaker is not allowed to describe it directly without presenting it as a quotation.

(iii) When the sentence expresses non-volitional actions or changes of states

When the antecedent and the consequent describe an agent's non-volitional actions or changes of states, factual *tara* is degraded. (23) illustrates the continuous (*renzoku*) use of *tara* and *to*:

- (23) *donguri-wa korokoro* {*??korogat-tara* / *korogar-u to*} *ike-ni ochi-ta*.
 acorn-TOP roll.over roll.over-TARA roll.over-NPST TO pond-DAT fall-PST
 'As the acorn rolled over and over, it fell into the pond.' (= (13) of this summary)

Tara is degraded because the consequent state of affairs follows *too naturally* from the antecedent state of affairs; for this reason, their relation can hardly constitute new knowledge when perceived by the speaker. In contrast, *to* requires the speaker to present the relevant states of affairs objectively as an external observer, and can thus be used felicitously in (23).

(iv) When the relevant states of affairs cannot be observed at the same spot

Finally, there are cases where *tara* is acceptable, but *to* is not:

- (24) *ke-no seetaa-o sentakuki-de* {*arat-tara* / *??ara-u to*} *ki-rare-naku*
 hair-GEN sweater-ACC washer-LOC wash-TARA wash-NPST TO wear-can-NEG
nat-ta.
 become-PST
 'After I washed the sweater in the washer, it became unwearable.'
 (= (14) of this summary)
- (25) *kinou 10-kiro* {*hashit-tara* / *??hashir-u to*} *2-kiro yase-ta*.
 yesterday 10-km run-TARA run-NPST TO 2-kg lose.weight-PST
 'After I ran 10km yesterday, I lost 2kg.' (p. 84 (19))

The consequents of both (24) and (25) express states of affairs that cannot be observed at the same spot where the antecedent states of affairs obtain. In (24), it is conceivable that the fact that the sweater became unwearable did not obtain immediately after it was washed, but rather after it was dried. In (25), too, losing weight cannot be observed immediately after the speaker ran. According to Hypothesis 2, *to* describes states of affairs that the speaker can observe as an external observer. This is in conflict with what (24) and (25) convey, thus leading to the unacceptability of *to*.

When we replace the adjust the examples so that they describe states of affairs that can be observed by external observers at the same spot, *to* becomes acceptable:

- (26) *ke-no seetaa-o atsui oyu-de* *ara-u to, mirumiruuchi-ni*
 hair-GEN sweater-ACC hot boiled.water-LOC wash-NPST TO rapidly.as.I.see-DAT
chijin-de it-ta.
 shrink-CONT PST
 'When I washed the sweater with hot water, it became smaller and smaller immediately (right in front of me as I watched).' (p. 85 (13'))

Factual *to* in narrative discourse

In narrative discourse (e.g. novels), *to* can often be used to describe an agent's continuous action, and the agent can be first person, as shown in (27). This contrasts drastically with *tara*, which shows strong resistance to first-person subjects as discussed above.

- (27) *boku-wa sofaa-ni mouichido nekoron-de rajio-no toppu.footii-o kiki-nagara*
 ITOP sofa-DAT one.more.time lie.down-CONT radio-GEN Top.40-ACC listen-while
juppun-bakari bon'yari-to ten'jou-o nagame, soshite shawaa-ni hairi atsui
 10min-only aimlessly-C ceiling-ACC look then shower-DAT enter hot
yu-de teinei-ni hige-o sor-u to, kuriiningu-kara
 hot.water-LOC carefully-DAT beard-ACC shave-NPST TO cleaning-from
modot-ta bakari-no shatsu-to bamyuudaa.shootsu-o ki-ta.
 come.back-PST just-GEN shirt-and Bermuda.short-ACC wear-PST
 'I lied down in the sofa again, listened to the Top 40 from the radio, and stared at the ceiling for no reason for ten minutes, then, after I had a hot shower and shaved my beard carefully, I put on my shirt and Bermuda shorts that I have just picked up from the dry cleaner.'
 (p. 86 (21), from Haruki Murakami, *Hear the Wind Sing*)

This can be explained based on the property of narrative discourse. In narrative discourse, the narrator develops the story from the point of view of an observer who is outside the fictional world. In (27), the writer assigns two roles to the first-person *boku* 'I': the protagonist of the story, and the narrator. The narrator 'I' is able to observe the actions undertaken by the protagonist 'I' as an external observer. Hence, *to* is acceptable in this sentence.

The property of narrative discourse also explains why *to* does not require quotative markers in (28):

- (28) *aru toki, sakaguchisan-ga kare-no ie-e chichi-o shibor-u tokoro-o*
 some time Mr.Sakaguchi-NOM he-GEN home-DAT milk-ACC squeeze-NPST place-ACC
mise-te kure to it-te asobi-ni {?it-tara / ik-u to}, odoriagar-u
 show-CONT give.IMP C say-CONT play-DAT go-TARA go-NPST TO jump.up-NPST
you-ni shi-te kare-wa yorokon-da.
 C-DAT do-CONT he-TOP be.happy-PST
 'When Mr. Sakaguchi once came to visit him and said 'Show me how you milk the cows!', he was so excited that he almost started dancing for joy.'
 (= (12) of this summary)

The narrator is omniscient about the story, and thus has direct access to the protagonists' actions, emotions and private experience. Therefore, when presenting such information, the narrator does not need to use modal or hearsay markers. This is why *to* is acceptable in (28) without quotative markers. In contrast, *tara* requires the consequent to describe a state of affairs that an agent came to recognize based on his own experience, which in the case of (28) is not the speaker, but rather Mr. Sakaguchi. Therefore, quotative markers are required.

The factual and hypothetical use of *tara* and *to*

We have seen that factual *tara* and *to* express states of affairs that the speaker has just come to recognize based on state of affairs that have newly come about; hence, both the antecedent and the consequent need to express new information.

We can now understand why *tara* and *to* also have a hypothetical use. Akatsuka (1983)³ proposes that the antecedent and the consequent of (hypothetical) conditionals both describe ‘non-facts’, including (i) things that might become facts at some later point, (ii) things that are contrary to the facts, and (iii) things that have not become facts. What Akatsuka calls ‘facts’ refer to things that have been registered under the speaker’s knowledge state at the utterance time. New information that has just entered the speaker’s knowledge state counts as ‘non-facts’, and will become ‘facts’ at some later point. Although factual *tara* and *to* describe facts that have already obtained, they belong to what Akatsuka calls ‘non-facts’, since they express new information about newly obtained states of affairs. In this sense, factual *tara* and *to* share with hypothetical *tara* and *to* in terms of how the expressed information fares with the speaker’s knowledge.

³Akatsuka, N. 1983 ‘Conditionals’. *Papers in Japanese Linguistics*, vol. 9, Kurosio.

Masuoka, Takashi. 2006. *Nihongo ni okeru joken keishiki no bunka - Bun no imiteki kaiso kozo no kanten kara* [The differentiation of conditional forms in Japanese: A view from the semantic hierarchical structures of sentences]

In Takashi Masuoka (ed.), *Joken Hyogen no Taisho* [Comparisons of Conditional Expressions], pp. 31-46, Kuroshio.

January 30, 2022

This paper offers an account for the separation of the Japanese conditional form *ba* into three sub-forms: *reba*, *tara(ba)* and *nara(ba)*.

Basic properties of *reba*, *tara* and *nara*

Reba-conditionals express general causal relations between the states of affairs described by the antecedents and the consequents:

- (1) *juyou-ga fur-eba kakaku-ga agar-u.*
 need-NOM rise-REBA price-NOM go.up-NPST
 ‘If the demand rises, the price goes up.’ (p. 32 (1))

Tara-conditionals express temporal dependencies between specific states of affairs; that is, the antecedent describes a state of affairs that is not yet realized, and the consequent describes a state of affairs whose realization is dependent on the realization of the antecedent state of affairs. The speaker may be certain that the antecedent state of affairs will be realized, as in (2):

- (2) *mukou-ni tsui-tara, renraku shi-te hoshi-i.*
 there-DAT arrive-TARA contact do-CONT want-NPST
 ‘When you arrive, please let me know.’ (p. 33 (5))

The speaker may also be uncertain as to whether the antecedent state of affairs will come about. In such cases, expressions like *moshi* ‘supposedly’ and *man’ichi* ‘by any chance’ can be added:

- (3) *moshi yoi kekka-ga de-tara, sugu-ni go-houkoku shi-mas-u.*
 supposedly good result-NOM release-TARA soon-DAT HON-report do-POL-NPST
 ‘If I get any good news, I will let you know immediately.’

Tara also has a factual use, which expresses temporal dependencies between states of affairs that have already come about:

- (4) *renraku shi-te mi-tara, tantousha-ga kanchigai shi-te i-ta*
 contact do-CONT try-TARA person.in.charge-NOM misunderstand do-CONT be-PST
koto-ga wakat-ta.
 thing-NOM know-PST
 ‘When I reached out, it became clear that the person in charge misunderstood.’ (p. 33 (9))

Nara-conditionals express hypotheticality, namely that the speaker supposes that the antecedent is true, and then expresses her judgment about or attitude toward the consequent based on that

supposition.

- (5) [I thought that he would not pay for the article that I am writing, so I would write whatever I want...]

moshi genkouryou-o harat-te kure-ru no nara, kare-no hanbai kakuchou-ni
 supposedly article.fee-ACC pay-CONT give-NPST FIN NARA he-GEN sale enlarge-DAT
kyouryoku shi-you to omot-ta...
 cooperate do-VOL C think-PST

‘But if he pays me, I will help him expand his business...’ (p. 34 (14))

Analysis of the three subtypes of the *ba*-form

Masuoka (1993)¹ proposes that the separation of the *ba*-form into three sub-forms reflects the ‘conceptual levels of sentences’. The four conceptual levels of sentences are illustrated as follows:

- (6) Conceptual levels of sentences (Masuoka 1993)
- a. State of affairs (*jitai meimei*) level: expresses the types of states of affairs;
 - b. Phenomenon (*genshou*) level: expresses specific states of affairs that come about at some spatio-temporal points;
 - c. Judgment (*handan*) level: expresses interlocutors’ judgments about events;
 - d. Expression and communication (*hyougen, dentatsu*) level: expresses interlocutors’ attitudes in the process of communication.

Masuoka proposes that *reba*, *tara* and *nara* each encodes conditionalization at a different level:

- (7) Conditionalization at different conceptual levels of sentences
- a. *reba*-form: conditionalization at the state of affairs level;
 - b. *tara*-form: conditionalization at phenomenon level;
 - c. *nara*-form: conditionalization at judgment level.

Reba encodes conditionalization at the state of affairs level, because it expresses general causal relations between states of affairs. *Tara* encodes conditionalization at the phenomenon level, because it expresses temporal dependencies between specific states of affairs. *Nara* encodes conditionalization at the judgment level, because it expresses the speaker’s supposition that the antecedent be true.

However, there are two more observations that this account does not capture. First, just like *tara*, *reba* can also be used to express specific states of affairs, as exemplified in (8):

- (8) *sore-o riyoushi-te tanaka-wa nihon-ni modor-eba hakushironbun-o*
 that-ACC use do-CONT Tanaka-TOP Japan-DAT return-REBA dissertation-ACC
kak-u darou.
 write-NPST be.probable
 ‘With that, if Tanaka goes back to Japan, he will probably write his dissertation.’

¹Masuoka, Takashi. 1993. Nihongo no joken hyogen ni tsuite [On conditional expressions in Japanese]. In Takashi Masuoka (ed.), *Nihongo no Joken Hyogen [Conditional Expressions in Japanese]*, pp. 1-20, Kurosio.

(p. 37 (21))

But unlike *tara*, which does not have to express causal relations, *reba* obligatorily expresses causal relations. For instance, in (8), there is a causal relation between going back to Japan and writing a dissertation. (9) shows that *tara* can express states of affairs that are temporally dependent, but not causally dependent, on each other; *reba* is degraded in this case.

- (9) *ima tsukur-ase-te or-imas-u kara, {dekiagat-tara / ?dekiagar-eba}*
 now make-CAUS-CONT be.HON-POL-NPST because finish-TARA finish-REBA
mejiro-no kata-ni o-todoke itash-imas-u.
 Mejiro-GEN people-DAT HON-deliver do-POL-NPST
 ‘We are having people working on it now. When it’s done, we will deliver it to Mejiro.’
 (p. 37 (6), p. 38 (23))

We can further see this from the temporal use of *tara*, which does not have to involve causal dependencies between states of affairs; *reba* cannot be used in such cases:

- (10) *sapporo-ni {it-tara / *ik-eba}, nishimura-no kissaten-de takagi-san-ni*
 Sapporo-DAT go-TARA go-REBA Nishimura-GEN cafe-LOC Mr.Takagi-DAT
battari at-ta wa.
 accidentally meet-PST SFP
 ‘When I went to Sapporo, I accidentally met Mr. Takagi at the cafe at Nishimura.’
 (p. 38 (10) (24))

Second, the account also does not capture the fact that, just like *nara*, *reba* and *tara* can also be used to express hypotheticality. Both (11) and (12) are sentences where the antecedent is taken as a supposition, because it is uncertain whether the luggage will be given and whether Tatsuhisa will know:

- (11) *nimotsu-o watas-eba inochi dake-wa tasuke-te kure-ru ka.*
 luggage-ACC pass-REBA life only-TOP help-CONT give-NPST Q
 ‘I wonder whether he will at least let us live if we give him our luggage.’ (p. 37 (22))
- (12) *man’ichi tatsuhisa-ni shir-are-tara, donna koto-ni nar-u ka.*
 by.any.chance Tatsuhisa-DAT know-CAUS-TARA how thing-DAT become-NPST Q
 ‘I wonder what will happen if Tatsuhisa knows this.’ (p. 37 (8))

However, in contrast to *reba*, which obligatorily expresses causal relations, and *tara*, which obligatorily expresses temporal dependencies, *nara* is not subject to such requirements. (13) describes states of affairs that are neither temporally nor causally dependent on each other, and only *nara* is acceptable:

- (13) *kimi-ga raishuu-no kayoubi-ni {ik-u no nara / *ik-eba / *it-tara},*
 you-NOM next.week-GEN Tuesday-DAT go-NPST FIN NARA go-REBA go-TARA
boku-wa raishuu-no getsuyoubi-ni ik-u.
 I-TOP next.week-GEN Monday-DAT go-NPST
 ‘If you go next Tuesday, I will go next Monday.’
 (p. 38 (26)-(28))

To explain these observations, we refine Masuoka's (1993) account by proposing the following semantic chains that *reba*, *tara* and *nara* are associated with.

(14) Semantic chains:

- a. *reba*: general causal relations → temporal dependencies → hypotheticality
- b. *tara*: temporal dependencies → hypotheticality
- c. *nara*: hypotheticality

For *reba*, its uses to express temporal dependencies and hypotheticality are derived from its basic meaning, i.e. the meaning of expressing general causal relations. When we limit ourselves to eventive states of affairs, causal dependencies entail temporal dependencies, since causes must take place before the effects. It follows from the temporal dependencies that the antecedent describes a state of affairs that is not yet realized, or in other words, the truth value of the antecedent is not yet settled. This leads to the supposition of the truth of the antecedent, which gives rise to the use of *reba* as expressing hypotheticality.

Tara is the marked form of *ba* that contains *tari*, which expresses the completion of events. *Tara* thus explicitly expresses that the relevant states of affairs are unrealized, thus deriving its hypotheticality use. There is no requirement for causal dependencies.

Nara is the marked form of *ba* that contains the copula *nari*, and it explicitly expresses hypotheticality. It does not require causal dependencies or temporal dependencies.

Nara and to sureba

The hypothetical conditional marker *nara* can sometimes contain *to iu* 'to say' (*to* is a quotative marker, *iu* is the verb meaning 'to say'):

(15) The CEO of Fuji TV about the idea that someone might buy out one of his subsidiary companies...

baishuu su-ru to i-u no nara su-reba yo-i.
 buy.out do-NPST C say-NPST FIN NARA do-REBA good-NPST
 'If (they say) they will buy out the company, they should do it.' (p. 41 (31))

In this sentence, *to iu* makes it explicit that the relevant state of affairs is being hypothesized. This instantiates that *nara* can express hypotheticality *at the metalinguistic level*.

Apart from *nara*, another conditional form that expresses metalinguistic hypotheticality is *to sureba*. In (16), both *to sureba* and *nara* are acceptable:

(16) *moshi watashi-no bunseki-ga machigat-te i-ru {to su-reba / nara},*
 supposedly I-GEN analysis-NOM be.wrong-CONT be-NPST C do-REBA NARA
mouichido genten-ni tachimodot-te kangae-naos-u shika na-i.
 once.more origin-DAT go.back-CONT think-start.over-NPST only NEG-NPST
 'If my analysis is wrong, there's no way other than going back and thinking about it all over again.' (p. 34 (15), p. 43 (35))

To sureba is the *reba*-form of *to suru* (*to* is a quotative marker, *suru* is the light verb meaning 'to do'), which originally expresses quotes of speech or thoughts:

- (17) Although the Social Insurance Agency did not reveal who the ex-minister is, but...
ayamat-ta shori dat-ta to shi-te i-ru.
 mistake-PST process COP-PST C do-CONT be-NPST
 ‘They said that they did not deal with the matter properly. (p. 42 (33))

To suru ‘to suppose’ can also be used to signal suppositions:

- (18) *kari-ni ichi-ji-ni hajime-ru to su-ru. sou su-ru to...*
 hypothetical-DAT one-o’clock-DAT start-NPST C do-NPST that do-NPST C
 ‘Suppose that we start at 1. If we do that...’ (p. 34 (17))

Therefore, the conditional form *to sureba* expresses hypotheticality at the metalinguistic level.

Nara and *to sureba* differ in that only *nara* can be used to make suppositions about information that the speaker receives on the spot of the conversation.

- (19) *o-yasai kirai? komat-ta na. demo yasai-ga {kirai nara / *kirai*
 HON-vegetable hate be.troubled-PST SFP but vegetable-NOM hate NARA hate
da to su-reba} kudamono-o tabe-reba i-i yo.
 COP.NPST C do-REBA fruit-ACC eat-REBA good-NPST SFP
 ‘You don’t like vegetables? What should I do then...? If you don’t like vegetables, it’s ok
 for you to just eat fruits.’ (p. 43 (37), (39))

The reason that *nara* can be used in this case has to do with the base form of *nara*, namely *no da*. As argued in Masuoka (2003)², *no da* is an extended form of nominalization and comes with topicality (roughly, *X-wa Y no da* ‘As for X, it is Y’). The topicality of *no da* is preserved in *nara*, thus making it eligible for making suppositions about information received on the spot of the conversation.

²Masuoka, Takashi. 2003. Meishibun toshite no noda bun [*No da*-sentences as nominalization]. *CLAVEL*, No. 1 pp. 1-11.

Yurie Nara “Kankokugo zyookken hyoogen ‘-umyen’, ‘-ketun’, ‘-eya’: zítai no kobetusee to rearitii’ [“Korean conditional expressions ‘-umyen’, ‘-ketun’, ‘-eya’: the individuality of events and reality”]

In Takashi Masuoka (eds.) *Zyookken Hyoogen no Taisyoo* [“Comparison of conditional expressions”]

January 30, 2022, by Teruyuki Mizuno

1 Introduction

In the research history of Korean, conditionals were studied in the realm of connectives in the early 1900’s, but since 1937, they have been studied especially in relation to the conjugation of verbs.

Since 1970’s, there have been attempts to systematize conditional forms. Among those, ‘-umyen’, ‘-ketun’, ‘-eya’, ‘-edo’ have been regarded as conditional connective endings. More specifically:

- a. -umyen: general conditional expressions
- b. -ketun: general conditional expressions
- c. -eya: necessary conditional expressions
- d. -edo: concessive (converse) conditional expressions

There are the following varieties that are subsumed under the category of -umyen: -umyen, -(nun/n)tamyen, -(i)lamyen, -esstelamyen and -nulamyen.

This entry deals with -umyen, -ketun and -eya.

-umyen and -ketun are close in function to Japanese conditional connectives such as ‘-to’, ‘-ba’, ‘-tara’ and ‘nara’.

2 Necessary conditional expressions ‘-eya’

‘-ya’ is a particle that adds the meaning of ‘emphasis’:

- 1 *Ipen-eyya toykessci.*
this.time-YA alright
‘It will be fine this time.’ (there have been many trials so far)
- 2 *Cha-ya kwulleman kamyen toycyo.*
‘It is the car that should be rolled’

A sentence with ‘-eya’ is interpreted as a conditional expression in the following way:

- 3 *Kukel paakhay-ya daum pemhyangul pangcihayko peminul kemke-hayl ke aninka?*
lit. ‘It is only through grasping it that we can prevent potential

crimes and arrest the criminals, isn't it?'

(≈ 'If we didn't do it, we wouldn't be able to prevent potential crimes and arrest the criminals, right?')

A necessary condition refers to the relation in which for the consequent to be true, the antecedent must be the case.

3 -ketun and its constraints

- 11 *Hyeylilul po-ketun, nayka mannacago hantako malhay cwue.*
'If you see a helicopter, tell them that I said I want to see them.'

-ketun also has a non-conditional function. -ketun with this function is mainly used to explain reasons or current situations; -ketun with this function thus more or less corresponds to Japanese 'no da'.

4 The individuality of events

4.1 -umyen

One-case conditionals

- 17 *Ike tul-umyen, palo cenhwahay!*
'If you hear this, call me immediately!'

Multi-case conditionals

- 19 *Nitulun nwunman tt-myen ssawunya?*
'Do you guys have a fight every time you face each other?'
- 20 *Kauli toy-myen sakwaka ppalhahkey iksupnita.*
'Apples become red in autumn.'

4.2 -ketun

-ketun can only express one-case conditionals; it cannot be used for multi-case conditionals.

4.3 -*eya*

-*eya* can be used for both one-case and multi-case conditionals, but it tends to be used for multi-case conditionals. In particular, it is mostly used to refer to things that the speaker believe to be general.

5 Reality

5.1 Hypothetical conditionals

Antecedent: factual / Consequent: counterfactual This combination was not attested in Korean, which, according to Maeda (1991), exists in *nara*-conditionals in Japanese.

- 27 [Konna tokoro-e kuru no nara], [seesyo-o mot-te
such place-to come FIN NARA Bible-ACC have-GRND
kur-eba yo-katta].
come-BA good-PAST
'If I had known that I would come to such a place, I should have brought the Bible.'

To express 27 in Korean, words like 'know' must be inserted.

- 27' Ilon kosey ontako alko issesstela-myen, sengseilato kacyeol ke
kulayssney.
'If I had known that I would come to such a place, I should have brought the Bible.'

Antecedent: counterfactual / Consequent: counterfactual This combination was attested in conditionals with -*umyen* and -*eya*.

Antecedent: unrealized / Consequent: unrealized This combination was attested in conditionals with -*umyen*, -*ketun* and -*eya*. This combination concentrated on conditionals with -*ketun*.

Antecedent: factual / Consequent: unrealized This combination was only attested in conditionals with -*umyen*.

5.2 Non-hypothetical conditionals

The generic use of conditionals was attested in *-umyen* and *-eya*.

5.3 Non-conditional uses

-umyen, *-ketun* and *-eya* all have non-conditional uses.

Listing The listing use (which is attested in Japanese conditionals) is not attested in any of the conditional connectives in Korean.

Evaluative use The use as a conditional evaluative construction is attested with *-umyen* and *-eya*

6 Conclusion

	<i>-umyen</i>	<i>-ketun</i>	<i>-eya</i>
One-case	ok	ok	ok
Multi-case	ok	*	ok

	Antecedent-Consequent	<i>-umyen</i>	<i>-ketun</i>	<i>-eya</i>
Hypothetical	Factual-Counterfactual	*	*	*
	Counterfactual-Counterfactual	ok	*	ok
	Unrealized-Unrealized	ok	ok	ok
	Factual-Unrealized	ok	*	*
Non-hypothetical	Generic	ok	*	ok
Non-conditional		△	*	△

Shimoji, Sachiko. 2006. Chūgokugo no jōken hyōgen Jōkenbun ni okeru le no bunpu to imi [Conditional expressions in Chinese: The distribution and meaning of *le* in conditionals]
In Takashi Masukoka (ed.), *Jōken Hyōgen no Taishō [Comparisons of Conditional Expressions]*, pp. 83-98, Kuroshio.

February 3, 2022

This paper explores the distribution and the meaning of *le* in Chinese conditionals, with reference to Masuoka's (2006)¹ analysis of Japanese conditionals based on the notion of conceptual levels. We show that in addition to the view of *le* as an aspectual marker, it is necessary to consider *le* as an element for turns ordinary propositions into sentences that describe specific states of affairs.

Causal constructions and conditional constructions

There are three types of subordinate clauses in Chinese. First, subordinate clauses can be marked by connectives (*kanrenshi*):

- (1) *ni yaoshi bu zhenxi zhezhong chunjiede youyi, wo jiu gen bieren zuo dadang*
you if not treasure such pure friendship I JIU with others make partner
le a.
ASP SFP
As conditional: 'If you don't treasure pure friendship like ours, I'll find someone else to be my partner.'
(p. 84 (1))

Second, the subordinate and the main clauses can contain pronouns of the same form:

- (2) *ni yao duoshao, wo gei ni duoshao.*
you want how.much I give you how.much
As conditional: 'I'll give you however much you want.' (lit. If you want how much, I'll give you how much.)
(p. 84 (2))

Finally, there may be no element that signals subordination at all. (3) is ambiguous between a causal and a conditional sentence: when it is clear from the context that 'sitting in the back' is true, the sentence is interpreted as a causal sentence; when 'sitting in the back' is merely being hypothesized, the sentence is interpreted as a conditional:

- (3) *zuo zai houbianr, wo kan bu qingchu.*
sit at back I see not clearly
As causal: 'Because I'm sitting in the back, I can't see it clearly.'
As conditional: 'If I sit in the back, I won't see it clearly.'
(p. 85 (6))

¹Masuoka, Takashi. 2006. Nihongo ni okeru jōken keishiki no bunka - Bun no imiteki kaisō kōzō no kanten kara [The differentiation of conditional forms in Japanese: A view from the semantic hierarchical structures of sentences]. In Takashi Masukoka (ed.), *Jōken Hyōgen no Taishō [Comparisons of Conditional Expressions]*, pp. 31-46, Kuroshio.

As pointed out by Okochi (1967)², even in unmarked subordinated clauses, there can be cues that signal the interpretation of the sentence. As shown in (4a) vs. (4b), when the subordinate clause has a subject, the sentence receives a causal interpretation, otherwise, it receives a conditional interpretation:

- (4) a. *ta duo du shu, duo dong daoli.*
 he much read book much understand knowledge
 As causal: ‘Because he reads lots of books, he has lots of knowledge.’
 b. *duo du shu, duo dong daoli.*
 much read book much understand knowledge
 As conditional: ‘If you read more books, you will have more knowledge.’ (p. 85 (7))

Counterfactual sentences require conditional connectives and realis expressions in the antecedent clause. For instance, Chinese has two types of negation, *bu*, which negates verb phrases (Masuoka’s state of affairs level of sentences) and is irrealis, and *mei(you)*, which negates aspectual phrases (Masuoka’s phenomenon level of sentences) and is realis. Counterfactuals can be formed with the connective *ruguo* and the realis negation *meiyou*:

- (5) a. *bu xingxiu shuili, yi yu hantian jiu hui cheng zai.*
 not build water.conservancy once meet drought JIU then become disaster
 ‘If we don’t work on water conservancy projects, once the weather is dry, we’ll be in drought.’
 b. {*ruguo* / *Ø} *meiyou xingxiu shuili, yudao zheyangde hantian jiu*
 if not.PERF build water.conservancy meet such drought JIU
hui cheng zai le.
 then become disaster ASP
 ‘If we hadn’t worked on the water conservancy projects, when the weather became this dry, we would have been in drought.’ (p. 87 (9), (10))

***Le* in antecedent clauses**

The central issue of this paper concerns the distribution of *le* in conditionals and its interaction with different types of verbs. We focus on conditionals whose antecedents do not contain *le* (e.g. (6a)) and those whose antecedents contain *le* in between the verb and the object (e.g. (6b)). It is also possible for *le* to appear after the object (e.g. (6c)), but due to its low frequency, we do not discuss this type of conditionals.

- (6) a. *yaoshi zhong jiang, wo qing ni chifan.*
 if get prize I buy you eat
 ‘If I get the prize, I’ll buy you a meal.’
 b. *yaoshi zhong le jiang, wo qing ni chifan.*
 if get ASP prize I buy you eat
 ‘If I get the prize, I’ll buy you a meal.’

²Okochi, Yasunori. 1967. Fukubun ni okeru bunku no rensetsu kankei [Connections in subordinate clauses of complex sentences]. *Chūgokugogaku [Bulletin of the Chinese Language Society of Japan]*. No. 176.

- c. *yaoshi zhong jiang le, wo qing ni chifan.*
 if get prize ASP I buy you eat
 'If I get the prize, I'll buy you a meal.' (p. 88, (11))

We will look at three types of verbs, atelic verbs, telic verbs and stative verbs. They are classified as follows:

- (7) Types of verbs
 a. Eventive verbs discussed
 (i) Atelic verbs (*higenkai doushi*)
 (ii) Telic verbs (*genkaai doushi*)
 b. Stative verbs

Atelic verbs

Eventive verbs that are not accompanied by *le* in Chinese express states of affairs whose truth values are unsettled at the utterance time:

- (8) *ta chi fan.*
 he eat rice
 Future reading: 'He will eat.' or
 Habitual reading: 'He eats rice (instead of bread).' (p. 88)

In the case of atelic verbs, when the antecedent does not contain *le*, the sentence describes relations between states of affairs that hold in general, as in (9a). When the antecedent contains *le*, the sentence express dependencies between specific states of affairs, as in (9b):

- (9) a. *chi manyu, dui taier (jiu) hui you yingxiang.*
 eat eel for unborn.baby JIU will have affect
 Describing general causal relations: 'If one eats eels, it will affect one's unborn baby.'
 b. *chi le zhege manyu, dui taier (jiu) hui you yingxiang.*
 eat ASP eel for unborn.baby JIU will have affect
 'If you eat this eel, it will affect your unborn baby.' (p. 89 (15))

The contrast between (9a) and (9b) is similar to the contrast between Japanese *reba* and *tara*: on Masuoka's account, *reba* expresses general causal relations, whereas *tara* expresses temporal dependencies between specific state of affairs. Following Masuoka's proposal, *le* thus turns sentences at the state of affairs (*jitai meimei*) level into sentences expressing specific states of affairs.

When there is no temporal or causal dependency between the states of affairs described by the antecedent and the consequent, *le* does not appear, as in (10a). Otherwise, *le* appears after the verb in the antecedent, as in (10b):

- (10) a. *zhangdama, shi dianhua nin shang wo na wu shi qu.*
 Mrs.Zhang use phone you go I that room use go
 'Mrs. Zhang, if you want to use the phone, go to my room.' (p. 89 (16a))
 b. *nimen de shengyi bu shun, shi le zhe-xie qian jiu shun le.*
 you GEN business not well use ASP this-CL money JIU well ASP

‘When your business doesn’t go well, if you use this money, it will get well.’
(p. 89 (16b))

The contrast between (10a) and (10b) is similar to the contrast between Japanese *nara* and *tara*:

- (11) a. *chousan, denwa-o tsuka-u nara, watashi-no tokoro-de douzo.*
Mrs.Zhang phone-ACC use-NPST NARA I-GEN place-LOC please
‘Mrs. Zhang, if you want to use the phone, go to my place.’
b. *anata-tachi-no shoubai-ga umaku it-te i-naku-te-mo, kono*
you-PL-GEN business-NOM well go-CONT be-NEG-CONT-ADD this
okane-o tsukat-tara, junchou-ni nar-u wa yo.
money-ACC use-TARA well-DAT become-NPST SFP SFP
‘Even if your business doesn’t go well, if you use this money, it will get well.’
(p. 89 (16))

We can thus take (10) to indicate that the omission of *le* from the antecedent leads to the removal of temporal or causal dependencies from conditionals. In sum, the distinction that Japanese makes using the three conditional forms (*tara*, *reba*, *nara*) is made in Chinese by adding or removing *le* to the antecedent.

As for negation, with *bu*, the conditional expresses general causal relations; with *mei*(*you*), the sentence expresses causal relations between specific states of affairs:

- (12) {*bu / mei*} *xue hanyu, jiu bu neng zai zhongguo gongzuo.*
not not.PERF learn Chinese JIU not can at China work
‘If you don’t study Chinese, you can’t work in China.’ (p. 90 (17))

In counterfactuals, *le* in the antecedent cannot be omitted:

- (13) *yaoshi daxue shihou xue le hanyu, jiu hen rongyi zhao gongzuo le.*
if university when study ASP Chinese JIU very easy find job ASP
‘If I had studied Chinese in college, it would have been easy to find a job.’ (p. 90 (18))

The negation used in the antecedent of counterfactuals is usually *mei*(*you*), instead of *bu*, as in (14). This may have something to do with the fact that counterfactuals usually describe relations between settled, specific states of affairs.

- (14) {**bu / mei*} *xue hanyu, jiu bu neng zai zhongguo gongzuo le.*
not not.PERF learn Chinese JIU not can at China work ASP
‘If you hadn’t studied Chinese, you couldn’t have worked in China.’ (p. 90 (19))

Telic verbs

Telic verbs describe states of affairs that have endpoints themselves. Hence, when the antecedent contains telic verbs, whether the antecedent contains *le* usually does not affect the meaning of the sentence. However, there are cases where the presence and the absence of *le* makes a difference. Without *le*, the antecedent can express ‘actual’ states of affairs, i.e. states of affairs that are just about to come about, as in (15). With *le*, the sentence expresses the regular hypothetical meaning, as in (16):

- (15) *peng zhege, hen weixian!*
touch this very dangerous
‘If you touch this, it will be dangerous!’ (p. 91 (22a))
- (16) *shei yaoshi peng le ta, ta ke dui shei bu keqi.*
who if touch ASP he he rather for who not forgive
‘If anyone touches him, he will not forgive him/her.’ (p. 91 (22b))

For negation, telic verbs allow *bu* in conditional antecedents, but not *mei*. *Mei* would force the sentence to receive a causal interpretation.

- (17) {*bu* / **mei*} *kai chuanghu, wu li jiu hui hen chaoshi.*
not not.PERF open window room inside JIU will very wet
‘If I don’t open the window, the room will get very humid.’ (p. 92 (24))

In counterfactuals, when the antecedent has an eventive verb, *le* cannot be omitted, similarly to atelic verbs:

- (18) *dangshi yaoshi kai le chuanghu, wo de gaozi jiu hui bei yu linshi le.*
that.time if open ASP window I GEN draft JIU will PASS rain get.wet ASP
‘If I had opened the window at that time, the manuscript would have gotten wet because of the rain.’ (p. 92 (26))

Both *bu* and *mei* are acceptable as the negation in counterfactuals

- (19) *dangshi {bu / mei} kai chuanghu (de hua), wo de gaozi jiu bu hui bei yu linshi le.*
that.time not not.PERF open window NMLZ word I GEN draft JIU not will PASS
rain get.wet ASP
With *bu*: ‘If I hadn’t opened the window, my manuscript wouldn’t have gotten wet because of the rain.’
With *mei*: ‘If someone hadn’t opened the window, my manuscript wouldn’t have gotten wet because of the rain.’ (p. 92 (27))

With *bu*, the sentence expresses objective negation. With *mei*, the sentence refers to an opening-the-window that was brought about by the speaker’s own intention, and expresses the speaker’s remorse or regret about that action.

Stative verbs

When stative verbs are not accompanied by *le*, they express states of affairs about the actual situation. For instance, (20a) and (20b) both express states of affairs whose truth values are settled:

- (20) a. *ta you cidian.*
he have dictionary
‘He has a dictionary.’
b. *ta zhidao zhe-jian shi.*
he know this-CL matter
‘He knows this.’ (p. 93 (28))

Similarly to telic verbs, stative verbs in antecedents do not seem to be affected by the presence or the absence of *le*:

- (21) {you *xiaoxi* /you *le xiaoxi*} *wo jiu gei ni da dianhua.*
have information have ASP information I ASP give you call phone
‘If I heard anything, I’ll call you.’ (p. 93 (28))

(22) illustrates an example where the presence and the absence of *le* does make a difference. Without *le* as in (22), the sentence expresses the supposition of a state. With *le* as in (22), the sentence expresses the supposition of a change of states.

- (22) a. *wo you shenme zhixinhua dou gen ni shuo.*
I have what heart.to.heart.talk all with you say
‘If I have any personal, genuine feelings, I’ll tell you.’
b. *zhe ren yao you le xiuyang, shang le duanwen, jiu tuohaihuangu.*
this people if have ASP education get.up ASP level JIU change.totally
‘If people become educated and get to a higher level, they will become totally different.’ (p. 93 (29))

Unlike eventive verbs, in counterfactuals that contain stative verbs, *le* does not appear:

- (23) *ni dangshi yaoshi renshi wo, 400-kuaiqian jiu keyi ba ta mai xialai.*
you that.time if know me 400-yuan JIU can BA it buy down
‘If you had known me at that time, you could have bought it with 400 yuan.’ (p. 94 (30))

***Le* in consequent clauses**

The consequent clauses of counterfactuals often contain *le*:

- (24) *ruguo meiyou xingxiu shuili, yudao zheyangde hantian jiu hui cheng*
if not.PERF water.conservancy once meet such drought JIU then become
zai le.
disaster ASP
‘If we hadn’t worked on the water conservancy projects, when the weather became this dry, we would have been in drought.’ (p. 87 (10))

This observation raises two questions. First, what is the relationship between *le* in counterfactuals and *le* in other environments? Second, how does *le* in counterfactuals relate to Japanese *ta*, which also often appears in the consequents of counterfactuals?

To answer the first question, consider (25), which shows that *le* can appear in the consequents of non-counterfactual conditionals:

- (25) *nimen de shengyi bu shun, shi le zhe-xie qian jiu shun le.*
you GEN business not well use ASP this-CL money JIU well ASP
‘When your business doesn’t go well, if you use this money, it will get well.’ (p. 89 (16b))

There does not seem to be a reason to treat *le* in (24) and in (25) differently. We analyze *le* in

conditional consequents as a signal that the consequent is contrasted with its negation, i.e. in (24), that we are not in drought, and in (25), that the business dose not go well.

In contrast, Japanese *ta* in the consequents of counterfactuals have been argued to express the meaning of past, i.e. it rewinds the time to a point when the consequent is still possible (Sadanobu 2005³). Therefore, Japanese *ta* and Chinese *le* have different functions in the consequents of counterfactuals.

³Sadanobu, Toshiyuki. 2005. Nihongo ga egaku sekai no kihon kōsei [The basic structure of the world as depicted by Japanese]. The 37th meeting of Gendai Chūgokugo Kenkyūkai [Workshop of Modern Chinese].

Suzuki, Yoshikazu. 1992. Teidai no *nara* to sono shyūhen [The use of topical NARA and its relatives]. *Sonoda Journal*, vol. 26, 1–12.

May 18, 2022

In conditional sentences, the connective *nara* signals subordinate clauses:

- (1) *non-dara nor-u na. nor-u nara nom-u na.*
 drink-TARA drive-NPST PROH drive-NPST NARA drink-NPST PROH
 ‘If you drink, don’t drive. If you drive, don’t drink.’ (p. 1 (1))
- (2) A: I’m going to go shopping at the department store.
 B: *depaato-e ik-u nara, tsuideni shatsu-o kat-te ki-te*
 department.store-ACC go-NPST NARA on.the.way shirt-ACC buy-CONT come-CONT
kudasai.
 please
 ‘If you’re going to the department store, please get some shirts on your way.’ (p. 1 (2))

Apart from such uses, *nara* can also be used on nouns. For instance, Masuoka and Takubo (1992)¹ identify *nara* as one of the topic markers, in addition to *wa*. They suggest that *nara* presents issues that are raised by the addressee as the topic of the conversation:

- (3) A: Have you seen Tanaka?
 B: *Tanaka-san nara, toshokan-de benkyou shi-te-ta yo.*
 Tanaka NARA library-LOC study do-CONT-PST SFP
 lit. ‘Tanaka, he was studying at the library.’
 (Masuoka and Takubo 1992: 134 (38), cited on p. 1)

In addition, connectives like *dat-tara* (COP-TARA), *de ar-eba* (COP be-BA) and *des-u to* (COP.POL-NPST TO) also have a topical use. A similar idea has also appeared in Morita (1990)², who suggests that *nara* can have the following two types of topic-related functions:

- (4) Presenting topics
ano ko nara, mou kata-ga tsui-te i-mas-u yo.
 that girl NARA already shape-NOM get-CONT be-POL-NPST SFP
 lit. ‘That girl, we have taken care of her.’ (Morita 1990: 93, cited on p. 2)
- (5) Delimiting topics
akai hana nara, manjuushage.
 red flower NARA lycoris.radiata
 lit. ‘Red flowers, lycoris radiata (are the best).’ (Morita 1990: 93, cited on p. 2)

This paper systematizes the topical use of conditional connectives such as *nara* and *dat-tara*.

¹Masuoka, Takashi and Yukinori Takubo. 1992. *Kiso Nihongo Bunpō* [Basic Japanese Grammar]. Kurosio.

²Morita, Yoshiyuki. 1990. *Nihongogaku to Nihongo Kyōiku* [Japanese Linguistics and Japanese Education]. Bonjinsha.

1 Mikami's study and its problems

The most detailed study of topical *nara* and *dattara* so far is found in Mikami (1960)³. He raises the following points:

1. *Nara* can be used to signal a topic, and its meaning is similar to *wa*.
2. *Nara* can replace case markers like *ga*, *no*, *ni*. In addition, it can follow *ni* (i.e. *-ni nara*), *kara* (i.e. *-kara nara*) and *de* (i.e. *de nara*). These two properties are shared by *wa*.
3. But unlike *wa*, *nara* retains some properties of its conditional use.
4. Topical *nara* can be seen as the reduced form of its conditional use, and it can be 'extended' to its full conditional form. For instance, *kare nara* in (6) can be considered as a reduced form of *kare-ga yaru nara* 'if he does it'.

- (6) *kare nara, umaku yar-u darou.*
 he NARA well do-NPST be.probable
 'As for him, he will probably do it very well.' (p. 2)

5. Topical *nara* signals a topic that is under the transition from the addressee to the speaker, whereas *wa* signals a topic that already belongs to the speaker.
6. Apart from *nara*, conditional connectives that have a topical use include *dat-tara* (COP-TARA), *da to* (COP.NPST TO), *de-wa* (COP-TOP) and *de-mo* (COP-ADD).

2 Relationship with the regular conditional use

To explore topical *nara*, it is necessary to consider its difference from the regular conditional use of *nara*. We say that a sentence containing *nara* instantiates the topical use of *nara* when it satisfies the following two conditions:

- (7) a. Condition 1: The *nara* part does not allow subjects; in other words, the *nara* part does not derive from the conditional use of *nara* with the ellipsis of the subject.
 b. Conditional 2: When *nara* is replaced with *wa*, or when *nara* is omitted, the meaning of the sentence does not change significantly.

For instance, the regular conditional use of *nara* in (8a) satisfies neither of the two conditions above: the elided subject can be filled, as in (8b); also, *nara* cannot be replaced with *wa*, as in (8c).

- (8) a. *nor-u nara nom-u na.*
 drive-NPST NARA drink-NPST PROH
 'If you drive, don't drink.' (p. 5)
 b. *anata-ga jidousha-ni nor-u nara, anata-wa sake-o nom-u na.*
 you-NOM car-DAT drive-NPST NARA you-TOP alcohol-ACC drink-NPST PROH
 'If you drive a car, you don't drink alcohol.' (p. 2)

³Mikami, Akira. 1960. *Zō wa Hana ga Nagai* [As for Elephants, Their Noses are Long]. Kurosio.

- c. **nor-u-wa nom-u na.*
drive-NPST-WA drink-NPST PROH

In contrast, under the topic use such as in (6), *nara* can be replaced easily with *wa* (e.g. *kare-wa umaku yaru darou*). In addition, it is difficult to consider *kare nara* as a phrase that contains an elided subject. Hence, such uses satisfy both conditions stated in (7).

Note that the characterization of the topical *nara* given in (7) contrasts with Mikami's ideas, namely that *nara* is the reduced form of its conditional use and can be extended to its full conditional form (point 3 in Sec 1). What exactly can the full conditional form of a *nara* topic be? For instance, the *nara*-topic shown in (9B) can be paraphrased in multiple ways with a full conditional antecedent, as shown in (10):

- (9) A: Have you seen Yamada?
B: *Yamada-san nara, toshokan-ni i-mash-ita.*
Yamada NARA library-DAT be-PST
lit. 'Yamada, he was in the library.' (p. 5, (5))
- (10) a. *anata-ga i-u no-ga yamada-san nara, toshokan-ni i-mash-ita.*
you-NOM say-NPST FIN-NOM Yamada NARA library-DAT be-PST
'If it is Yamada that you're saying, he's in the library'
- b. *yamada-san-ni tsuite i-eba, toshokan-ni i-mash-ita.*
Yamada-DAT about say-BA library-DAT be-PST
'Speaking of Yamada, he was in the library.' (p. 5, (5))

It is not clear which conditional in (10) provides the most appropriate full conditional form for (9B); at least there does not seem to be a principled way to determine the choice between (10a) and (10b). Therefore, we do not adopt Mikami's characterization of topical *nara*.

3 Topical *nara* and non-topical *nara*

In canonical conditionals, the antecedents can express either the presupposition or the focus of the sentence. We propose that when following a noun phrase, *nara* also has the freedom of signaling a presupposition or a focus. This can be seen shown by the acceptability for *nara* to follow indeterminate phrases:

- (11) *yamada-san-to-mo tanaka-san-to-mo ik-itaku-nai tte it-te, ittai, dare-to*
Yamada-with-ADD Tanaka-with-ADD go-want-NEG QUOT say-CONT on.earth who-with
nara ik-u no des-u ka?
NARA go-NPST FIN COP.POL-NPST Q
'You said that you don't want to go with Yamada and Tanaka, so who on earth do you want to go with?' (p. 6 (11))
- (12) *nanji nara a-e-mas-u ka?*
what.time NARA meet-can-POL-NPST Q
'What time can you meet?' (p. 6 (12))

We propose that the distinction between signaling a presupposition and a focus constitutes the

key to determine whether *nara* is topical or not. When the phrase preceding *nara* is focused, *nara* indicates a condition that would allow the state of affairs described by the consequent to hold. In such cases, a sentence *X nara Y* express roughly the cleft sentence ‘it is X that allows Y to hold’. In particular, such uses give rise to a strong sense of contrastiveness between the conditions that allow Y to hold and those that do not. Under such uses, *nara* is not topical, even when X is constituted by a noun phrase, such as the following examples:

- (13) A: I heard that you went to France recently.
 B: *iie, chuugoku nara ik-imash-ita kedo, furansu-ni-wa it-te na-i*
 no China NARA go-POL-PST but France-DAT-TOP go-CONT NEG-NPST
des-u yo.
 POL.COP-NPST SFP
 ‘No, China, I went to, but I didn’t go to France.’ (p. 6 (8))
- (14) A: Do you have something that I can write with?
 B: *enpitsu nara ar-imas-u.*
 pencil NARA be-POL-NPST
 ‘Pencils, I have.’ (p. 6 (9))

When *nara* signals a presupposition, it is used as a topic. In such cases, the meaning expressed by *nara* is similar to ‘speaking of...’, which is a typical use of topics. Meanwhile, the topical *nara* does not give rise to a sense of contrastiveness, because it indicates that the interlocutors are interested in some information that has already been presented in the discourse, and it is not necessary to consider when certain states of affairs hold and when not. This characterization of the topical *nara* is consistent with the previous ideas by Masuoka and Takubo’s idea that *nara* presents issues that are raised by the addressee as the topic of the conversation, and also Mikami’s idea that *nara* signals a topic that is under the transition from the addressee to the speaker. In fact, such use of *nara* can also be seen when *nara* follows sentences rather than noun phrases, such as (2), repeated as follows:

- (2) A: I’m going to go shopping at the department store.
 B: *depaato-e ik-u nara, tsuideni shatsu-o kat-te ki-te*
 department.store-ACC go-NPST NARA on.the.way shirt-ACC buy-CONT come-CONT
kudasai.
 please
 ‘If you’re going to the department store, please get some shirts on your way.’ (p. 1 (2))

Because A has made an assertion about going to the department store, there is no need to consider what happens if A does not go to the department store. Hence, similarly to the topical *nara* (i.e. when it follows noun phrases), *nara* in (2) does not give rise to a sense of contrastiveness.

4 Constructions where *nara* can be used

When *wa* is used to signal a topic, such as in *zou-wa hana-ga nagai* ‘Elephants, their noses are long’ and *boku-wa unagi da* ‘I will have eel’, *wa* can be replaced by *nara* without affecting the

meaning of the sentence:

- (15) A: Let's order fish today.
 B: *sakana nara tai-ga tabe-ta-i naa.*
 fish NARA red.snapper-NOM eat-want-NPST SFP
 'Fish, I want to have red snappers.' (p. 8, (14))

Also, as pointed out by Mikami, just like *wa*, *nara* can be used to replace *ga*, *no*, *ni* and *o*. (16) is an example where *o* is replaced by *nara*:

- (16) A: Please close the window.
 B: *mado nara mou shime-mash-ita yo.*
 window NARA already close-POL-PST SFP
 'Windows, I already closed them.' (p. 8, (18))

In addition, we have also seen Mikami's observation that just like *wa*, *nara* can follow *ni*, *kara to*, *de*, *e* and *made*. (17) is an example where *nara* follows *kara*:

- (17) A: Do you know the way?
 B: *ee, watashi-mo koko-kara nara kaet-ta koto-ga ar-imas-u kara.*
 yes I-ADD here-from NARA go.back-PST thing-NOM be-POL-NPST because
 'Yes, because from here, I have gone home once.' (p. 9, (21))

Nara can also follow adverbs:

- (18) A: Can you walk?
 B: *ee, yukkuri nara aruk-e-mas-u.*
 yes slowly NARA walk-can-POL-NPST
 'Yes, slowly, I can walk.' (p. 9, (10))

However, there are some adverbs that cannot be followed by *nara*:

- (19) a. Adverbs of states:
 (i) Both *wa* and *nara* can follow: *yukkuri* 'slowly', *bonyari* 'faintly'.
 (ii) Only *wa* can follow: *ikkini* 'at a stretch, without stopping', *kippari* 'decisively, flatly', *yasuyasu to* 'without difficulty', *ukkari to* 'carelessly'.
 b. Adverbs of quantity and frequency:
 (i) Both *wa* and *nara* can follow: *sukoshi* 'a little', *tokidoki* 'sometimes', *tamani* 'occasionally', etc. (those that express a small quantity).
 (ii) Only *wa* can follow: *takusan* 'a lot', *yoku* 'often', *tappuri* 'plentifully', etc. (those that express a big amount).
 c. Adverbs of annotation can be followed by *wa* only: *kekkyoku* 'at last', *osoraku* 'probably'.
 d. Adverbs of time and location:
 (i) Both *wa* and *nara* can follow: *kinou* 'yesterday', *toukyou* 'Tokyo', *shigoto-ga sunda ato* 'after finishing work'.
 (ii) Only *wa* can follow: *X baai* 'in the occasion of X', *X toki-ni* 'at the time of X'.
 e. Adverbs of degree; *nara* cannot follow: *taihen* 'very', *motto* 'more', *kanari* 'quite';

- f. Adverbs of aspect; *nara* cannot follow: *masumasu* ‘more and more’, *youyaku* ‘finally’;
- g. Adverbs of evaluation, *nara* cannot follow: *touzen* ‘certainly’, *ainiku* ‘unfortunately’.

There are also cases where *wa* can be used to separate a predicate, such as *utsukushiku-wa nai* ‘not beautiful’. In such cases, whether *nara* is allowed varies across speakers:

- (20) A: Is that town beautiful?
 B: *iya, utsukushiku{-wa / -nara} na-i.*
 no beautiful-TOP NARA NEG-NPST
 ‘No, it is not beautiful.’ (p. 10 (34))

In addition, connectives such as *kara* ‘because’ and *ijou* ‘beyond’ and *te* (continuative verbal form) can be followed by *wa*, but not *nara*. An exception is when *te* expresses methods or states, where it can be followed by *nara*:

- (21) A: Is it true that one can sleep while standing?
 B: *suwat-te nara ne-rare-ru keredo, ikura nandemo tat-ta*
 sit-CONT NARA sleep-can-NPST but how.much whatever stand-PST
mama-de-wa ne-rare-mas-en yo.
 while-LOC-TOP sleep-can-POL-NEG SFP
 ‘Sitting, one can sleep, but whatever you say, one cannot sleep while standing.’
 (p. 10, (37))

5 Difference with *wa*

Below are some other cases where *wa*, but not *nara*, is allowed:

- (22) *kore{-wa / ?nara} odoroi-ta.*
 this-TOP NARA be.surprised-PST
 ‘This, I am surprised by it.’ (p. 10, (38))
- (23) *omae{-wa / ?nara} kaer-e.*
 you-TOP NARA go.back-IMP
 ‘You, get out.’ (p. 10, (39))
- (24) *ano hito{-wa / ?nara} dare des-u ka.*
 that person-TOP NARA who COP.POL-NPST Q
 ‘Who is that person?’ (p. 10, (40))

The reason that *nara* is not allowed in the above cases is that *nara* is a conditional form, and thus implies what happens if the supposition does not hold. In the above examples, it is very difficult to consider cases where the supposition does not hold.

Another contrast between *wa* and *nara* is shown below:

- (25) a. *yamada-san dake-wa ai-taku-na-i.*
 Yamada only-TOP meet-want-NEG-NPST

- b. *yamada-san dake nara ai-taku-na-i.*

Yamada only NARA meet-want-NEG-NPST

‘Only Yamada, I do not want to see.’

(p. 11 (41))

(25a) means ‘I’m ok with seeing anyone but Yamada’, whereas (25b) means ‘What I don’t want is to meet with only Yamada’. It might be the case that *nara* in (25b) is in fact the conditional *nara*, because the noun phrase before *nara* can be replaced with a full sentence *Yamadan-san dake de aru no nara* ‘if it is with only Yamada’.

The contrast between *wa* and *nara* can also be seen in some cases where the phrase in front of *wa/nara* describe quantities. For instance, they are exchangeable in (26), but not in (27):

- (26) A: Did you go to that meeting?

B: *maa, tokidoki {wa / nara} it-te-mash-ita.*

well sometimes WA NARA go-CONT-POL-PST

‘Well, sometimes, I went there.’

(p. 11 (44))

- (27) A: I heard that we have to go to that meeting everyday.

B: *sou, tokidoki {?wa / nara} it-te-mo yokat-ta n des-u ga,*

right sometimes WA NARA go-CONT-ADD good-PST FIN COP.POL-NPST but

mainichi-de-wa chotto ik-e-mas-en yo.

everyday-LOC-TOP a.little go-can-POL-NEG SFP

‘Right. (If it’s) Sometimes, I can go, but I can’t go there everyday.’

(p. 11 (45))

In (27), too, the contrast may have to do with the conditional use of *nara*.

Chapter 4 Hyōka no modariti no jodōshi “beki da” [The evaluative modal auxiliary *beki da*]

In Takanashi, Shino. 2007. *Hyōka no Modariti: Gendai Nihongo ni okeru Kijutsuteki Kenkyū* [Evaluative Modality: A Descriptive Study of Modern Japanese]. 91–105. Kurosio.

April 1, 2022

The expression *beki da* ‘should’ has the following characteristics:

1. It is an auxiliary that is reserved exclusively for evaluative modality.
2. It always expresses the speaker’s evaluation at the utterance time.

For the first characteristic, Japanese evaluative modals are mostly morphologically complex (e.g. ...*nakute-wa naranai* lit. ‘if..., it doesn’t work’, roughly ‘must’; ...*hou-ga ii* lit. ‘the one that...is good’, roughly ‘had better to’), and *beki da* is one of the few non-complex ones. Also, unlike the auxiliaries *mono da* and *koto da*, which have a variety of meanings including evaluative modality, *beki da* only expresses evaluative modality.

The meanings of evaluative modality can be classified into four types: the type expressing necessity and appropriateness, the type expressing lack of necessity, the type expressing permission, the type expressing lack of permission. The meaning of *beki da* belongs to the first type; specifically, it expresses that the relevant state of affairs is appropriate. For instance, when *beki da* describes a state of affairs that the hearer has not brought about, as in (1), it expresses the speaker’s advice of the action that the hearer should undertake. When the state of affairs is no longer realistic, as in (2), *beki* expresses counterfactuality, and gives rise to a connotation of regret or remorse.

- (1) A: It costs only 1000 yen to get someone to do my makeup. I want to try it.
 B: *isshou-ni ichido da kara, jibun-de su beki yo.*
 lifetime-DAT once COP.NPST because self-LOC do should SFP
 ‘It happens only once in a lifetime, so you should do it yourself.’ (p. 93 (6))
- (2) Uttered by a dairy farmer in East Hokkaido:
nousuishou-ga tettei kaimei su beki na noni.
 MAFF-NOM completely get.clear do should FIN CF.PRT
 ‘The Ministry of Agriculture, Forestry and Fisheries should have made it clear what happened.’ (p. 93 (8))

Interrogatives with *beki da* like (3) express that the speaker is not yet able to evaluate the appropriateness of the relevant state of affairs:

- (3) *dokushin-no josei nara, dare-mo-ga ichido-wa ninjin-o tabe-ru beki*
 single-GEN woman TOP who-all-NOM once-TOP carrot-ACC eat-NPST should
kadouka-o shinken-ni kangae-ru.
 whether-ACC careful-DAT think-NPST
 ‘Single women have all thought of the question whether they should eat carrots.’ (p. 94 (9))

Many contexts allow the use of *beki da*, *nakutewa ikenai* ‘must’ and *houga ii* ‘had better’:

- (4) *wakai uchi-ni iron-na hito-to {tsukia-u beki da /*
 young while-DAT various-FIN people-with associate-NPST should COP.NPST
tsukiaw-anaku-te-wa ik-e-na-i / tsukiat-ta hou-ga i-i}.
 associate-NEG-CONT-TOP go-can-NEG-NPST associate-PST direction-NOM good-NPST
 ‘One {must / should / had better to} associate with various types of people while one is young.’
 (p. 94 (12))

But in terms of their meanings, *beki da* is close with *houga ii* ‘had better’. First, *nakutewa ikenai* ‘must’ is used to express absolute evaluation. In the test of compatibility with alternative actions, *nakutewa ikenai* exhibits unacceptability:

- (5) a. *...*shi-naku-te-wa ik-e-na-i* *ga, shi-naku-te-mo i-i*.
 do-NEG-CONT-TOP go-can-NEG-NPST but do-NEG-CONT-ADD good-NPST
 ‘You must..., but you don’t have to.’
 b. ...*shi-ta hou-ga i-i* *ga, shi-naku-te-mo i-i*.
 do-PST direction-NOM good-NPST but do-NEG-CONT-ADD good-NPST
 ‘You’d better..., but you don’t have to.’
 c. (*honrai-wa*) ...*su-ru beki da* *ga, shi-naku-te-mo i-i*.
 really-TOP do-NPST should COP.NPST but do-NEG-CONT-ADD good-NPST
 ‘You (really) should..., but you don’t have to.’
 (p. 95 (13))

Second, when the necessity of the relevant state of affairs is without an exception, *beki da* and *houga ii* are not allowed:

- (6) *kuruma-o unten su-ru-ni-wa menkyo-o {*tor-u beki da / *tot-ta*
 car-ACC drive do-NPST-DAT-TOP license-ACC get-NPST should NPST.COP get-PST
hou-ga i-i / tor-anaku-te-wa ik-e-na-i}.
 direction-NOM good-NPST get-NEG-CONT-TOP go-can-NEG-NPST
 ‘To drive a car, one {must / should / had better} get a driving license.’
 (p. 95 (14))

Hence, *beki da* ‘should’ and *houga ii* ‘had better’ express appropriateness, whereas *nakutewa ikenai* ‘must’ expresses necessity.

There are also differences between *beki da* ‘should’ and *houga ii* ‘had better’. First, *houga ii* is more restricted in terms of the *wh*-words that it allows. Since *houga ii* literally expresses a comparison between two options, it can only co-occur with *dochira* and *docchi* ‘which one (of the two)’:

- (7) *dochira-no daigaku-ni hair-u {beki / hou-ga i-i} ka*
 which.one-GEN university-DAT enter-NPST should direction-NOM good-NPST Q
nayan-de i-ru.
 be.troubled-CONT be-NPST
 ‘I’m not sure which university (of the two) I should enter.’
 (p. 95 (15))
 (8) *dono daigaku-ni hair-u {beki / *hou-ga i-i} ka*
 which university-DAT enter-NPST should direction-NOM good-NPST Q
nayan-de i-ru.
 be.troubled-CONT be-NPST

‘I’m not sure which university (among these) I should enter.’ (p. 95 (16))

Second, *houga ii* often gives rise to the connotation that the failure to bring about the relevant state of affairs would lead to negative consequences; hence, it is often used to express warning, rather than advice. *Beki da* does not give rise to such connotation. Third, *beki da* often expresses that the relevant state of affairs is morally or ethically appropriate, whereas *houga ii* expresses that the relevant state of affairs will lead to positive consequences. For instance, (9a) expresses the speaker’s judgment based on what is morally appropriate, whereas (9b) does not:

- (9) *gotaimanzoku-ni un-de kure-ta oya-ni-wa rikutsunuki-ni kansha*
 healthy-DAT give.birth-CONT give-PST parent-DAT-TOP without.excuse-DAT thank
 {a. *su beki da to omo-u yo* / b. *shi-ta hou-ga i-i*}.
 do should COP.NPST C think-NPST SFP do-PST direction-NOM good-NPST
 ‘{a. I think one should / b. One had better} thank their parents without an excuse for giving
 one a healthy body.’ (p. 96 (21))

Another characteristic of *beki da* that we noted earlier is that it must express the speaker’s evaluation at the utterance time. Many other evaluative modals, such as *nakutewa ikenai* ‘must’ can also express this meaning:

- (10) A: My teeth hurts so bad today.
 B: *sore-wa taihen da. sugu haisha-ni {ik-anaku-te-wa*
 that-TOP bad COP.NPST right.away dentist-DAT go-NEG-CONT-TOP
ik-e-na-i / ik-u beki da} yo.
 go-can-NEG-NPST go-NPST should COP.NPST SFP
 ‘That’s too bad. You {must / should} go to the dentist right away.’ (p. 97 (22))

However, *nakutewa ikenai* can also express necessity and permission from an objective view, whereas *beki da* cannot, as in the contrast in (11).

- (11) *A-san-wa haisha-ni {ik-anaku-te-wa ik-e-na-i / *ik-u beki na}*
 A-TOP dentist-DAT go-NEG-CONT-TOP go-can-NEG-NPST go-NPST should FIN
node, soutai shi-mash-ita.
 because leave.early do-POL-PST
 ‘Because A {must / should} go to the dentist, he left early.’ (p. 97 (23))

Likewise, *nakutewa ikenai* cannot be replaced by *beki da* in the following context:

- (12) [Enrollment requirement]
hon koosu-no gakusei-wa zenkouki-de sanjuu tan’i-o shutoku
 this course-GEN student-TOP first.second.semester-LOC thirty credit-ACC get
shi-naku-te-wa nar-ana-i.
 do-NEG-CONT-TOP become-NEG-NPST
 ‘Students of this course must obtain thirty credits in the first and the second semester.’
 (p. 97 (24))

The inability for *beki da* to express objective evaluation stems from its lexical meaning. Since *beki*

da expresses the appropriateness of states of affairs, it has to do with the speaker's evaluation based on her subjective judgments. This property of *beki da* can also be seen when it appears in past form *beki datta* 'should have', which must express the counterfactual meaning that the relevant states of affairs were not brought about:

- (13) *uso-demo mou suokoshi tsumekon-de ok-u beki desh-ita.*
 lie-even.if more a.little squeeze-CONT prepare-NPST should COP.POL-PST
 'You should have brought more luggages, even just to pretend.' (p. 98 (26))

When negated, *beki da* requires the negation to appear outside:

- (14) a. *ik-u beki de-wa na-i.*
 go-NPST should FIN-TOP NEG-NPST
 b. **ik-ana-i beki da.*
 go-NEG-NPST should COP.NPST
 'You should not go.' (p. 94 (11))

Beki dewanai 'should not' expresses that the relevant states of affairs are inappropriate. In case the relevant state of affairs is one that has already been realized, *beki dewanai* expresses the speaker's dissatisfaction:

- (15) *hito-wa anna fuu-ni shin-u beki ja na-i des-u yo.*
 people-TOP that way-DAT die-NPST should FIN.TOP NEG-NPST COP.POL-NPST SFP
 'People should not die like that.' (p. 99 (31))

Similarly to *beki datta*, when *beki dewanai* appears in its past tense, it expresses the speaker's evaluation at the utterance time and thus requires counterfactuality:

- (16) *hontou-wa o-soushiki nante ik-u beki ja nakat-ta n des-u*
 really-TOP HON-funeral such go-NPST should FIN.TOP NEG-PST FIN COP.POL-NPST
yo.
 SFP
 'We really shouldn't have gone to the funeral.' (p. 100 (32))

In addition, *beki* can be used as a nominal modifier. As shown in (17), *beki* as a nominal modifier can be used to describe the evaluation of the appropriateness of a state of affairs, just like sentential *beki da*:

- (17) *sore-wa zehi toinaoshi-te mi-ru beki mondai de a-ru.*
 that-TOP definitely ask.again-CONT try-NPST should problem FIN be-NPST
 'It is a problem that definitely should be addressed again.' (p. 100 (33))

However, nominal *beki* has a use that its sentential counterpart does not have, i.e. the use to describe logical necessity. Under such uses, the meaning of *beki* is similar to the epistemic modal *hazu da*.

- (18) *ichi jikan mae-ni mukou-o de-ta no da kara, kare-wa mou*
 one hour before-DAT there-ACC leave-PST FIN COP.NPST because he-TOP already
*tsui-te i-ru {hazu da. / *beki da.}*
 arrive-CONT be-NPST should COP.NPST should COP.NPST

- ‘Because he left an hour ago, he should have arrived already.’ (p. 101 (5’) (35’’))
- (19) *mou tsui-te i-ru {beki / hazu no} kare-ga mada araware-na-i.*
 already arrive-CONT be-NPST should should FIN he-NOM yet appear-NEG-NPST
 lit. ‘He, who should have already arrived, has not appeared yet.’ (p. 101 (37) (37’’))

It has been argued that the epistemic use of the nominal *beki* is a remnant of the modal *beshi* in Classical Japanese (Niwa 1991, Nakahata 1999)¹. In addition to expressing the speaker’s evaluation like modern *beki da*, *beshi* can also express the speaker’s inference and possibility, i.e. it has an epistemic use. In fact, in modern Japanese, besides the nominal use of *beki*, the remnant of *beshi* can also be found in expressions like *beku shite...suru* ‘it’s natural that...’, which describes logical necessity:

- (20) *mi-te-ru to, shin-u beku shi-te shin-u ningen tte i-ru kara*
 see-CONT-NPST COND die-NPST BEKU do-CONT do-NPST people C be-NPST because
ne.
 SFP
 ‘If you look at it, there are people who die as they should (i.e. they are destined to die).’
 (p. 104 (53))

¹Niwa, Tetsuya. 1991. ‘Beki da to nakereba naranai (*beki da* and *nakereba naranai*)’. *The Bulletin of the Cultural and Natural Sciences in Osaka Gakuin University*. Vol. 23, 24. 53–72.

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